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November 6, 2026

References

Abbott et al.: A Continuous-Time Dynamic Factor Model for Intensive Longitudinal Data Arising from Mobile Health Studies **Abbott-Dempsey-NahumShani-etal-2025**

Madeline R. Abbott et al. "A Continuous-Time Dynamic Factor Model for Intensive Longitudinal Data Arising from Mobile Health Studies". In: *Psychometrika* 90.3 (June 2025), pp. 1175–1196. ISSN: 1860-0980. DOI: 10.1017/psy.2025.10023.

Abstract: Intensive longitudinal data (ILD) collected in mobile health (mHealth) studies contain rich information on the dynamics of multiple outcomes measured frequently over time. Motivated by an mHealth study in which participants self-report the intensity of many emotions multiple times per day, we describe a dynamic factor model that summarizes ILD as a low-dimensional, interpretable latent process. This model consists of (i) a measurement submodel—a factor model—that summarizes the multivariate longitudinal outcome as lower-dimensional latent variables and (ii) a structural submodel—an Ornstein-Uhlenbeck (OU) stochastic process—that captures the dynamics of the multivariate latent process in continuous time. We derive a closed-form likelihood for the marginal distribution of the outcome and the computationally-simpler sparse precision matrix for the OU process. We propose a block coordinate descent algorithm for estimation and use simulation studies to show that it has good statistical properties with ILD. Then, we use our method to analyze data from the mHealth study. We summarize the dynamics of 18 emotions using models with one, two, and three time-varying latent factors, which correspond to different behavioral science theories of emotions. We demonstrate how results can be interpreted to help improve behavioral science theories of momentary emotions, latent psychological states, and their dynamics.

Abril-Pla et al.: PyMC: A modern, and comprehensive probabilistic programming framework in Python **AbrilPla-Andreani-Carroll-etal-2023**

Oriol Abril-Pla et al. "PyMC: A modern, and comprehensive probabilistic programming framework in Python". In: *PeerJ Computer Science* 9 (Sept. 2023), e1516. ISSN: 2376-5992. DOI: 10.7717/peerj-cs.1516.

Abstract: PyMC is a probabilistic programming library for Python that provides tools for constructing and fitting Bayesian models. It offers an intuitive, readable syntax that is close to the natural syntax statisticians use to describe models. PyMC leverages the symbolic computation library PyTensor, allowing it to be compiled into a variety of computational backends, such as C, JAX, and Numba, which in turn offer access to different computational architectures including CPU, GPU, and TPU. Being a general modeling framework, PyMC supports a variety of models including generalized hierarchical linear regression and classification, time series, ordinary differential equations (ODEs), and non-parametric models such as Gaussian processes (GPs). We demonstrate PyMC's versatility and ease of use with examples spanning a range of common statistical models. Additionally, we discuss the positive role of PyMC in the development of the open-source ecosystem for probabilistic programming.

Adolf et al.: Optimal sampling rates for reliable continuous-time first-order autoregressive and vector autoregressive modeling **Adolf-Loossens-Tuerlinckx-etal-2021**

Janne K. Adolf et al. "Optimal sampling rates for reliable continuous-time first-order autoregressive and vector autoregressive modeling". In: *Psychological Methods* 26.6 (Dec. 2021), pp. 701–718. ISSN: 1082-989X. DOI: 10.1037/met0000398.

Abstract: Autoregressive and vector autoregressive models are a driving force in current psychological research. In affect research they are, for instance, frequently used to formalize affective processes and estimate affective dynamics. Discrete-time model variants are most commonly used, but continuous-time formulations are gaining popularity, because they can handle data from longi-

tudinal studies in which the sampling rate varies within the study period, and yield results that can be compared across data sets from studies with different sampling rates. However, whether and how the sampling rate affects the quality with which such continuous-time models can be estimated, has largely been ignored in the literature. In the present article, we show how the sampling rate affects the estimation reliability (i.e., the standard errors of the parameter estimators, with smaller values indicating higher reliability) of continuous-time autoregressive and vector autoregressive models. Moreover, we determine which sampling rates are optimal in the sense that they lead to standard errors of minimal size (subject to the assumption that the models are correct). Our results are based on the theories of optimal design and maximum likelihood estimation. We illustrate them making use of data from the COGITO Study. We formulate recommendations for study planning, and elaborate on strengths and limitations of our approach.

Alter et al.: Determining negligible associations in regression **Alter-Counsell-2023**

Udi Alter and Alyssa Counsell. "Determining negligible associations in regression". In: *The Quantitative Methods for Psychology* 19.1 (2023), pp. 59–83. DOI: 10.20982/tqmp.19.1.p059.

Anderson: A confidence interval for the difference between standardized regression coefficients **Anderson-2024**

Samantha F. Anderson. "A confidence interval for the difference between standardized regression coefficients". In: *Multivariate Behavioral Research* 59.4 (July 2024), pp. 758–780. ISSN: 1532-7906. DOI: 10.1080/00273171.2024.2318784.

Antonakis et al.: On ignoring the random effects assumption in multilevel models: Review, critique, and recommendations **Antonakis-Bastardoz-Ronkko-2021**

John Antonakis, Nicolas Bastardoz, and Mikko Rönkkö. "On ignoring the random effects assump-

tion in multilevel models: Review, critique, and recommendations". In: *Organizational Research Methods* 24.2 (Apr. 2021), pp. 443–483. ISSN: 1552-7425. DOI: 10.1177/1094428119877457.

Abstract: Entities such as individuals, teams, or organizations can vary systematically from one another. Researchers typically model such data using multilevel models, assuming that the random effects are uncorrelated with the regressors. Violating this testable assumption, which is often ignored, creates an endogeneity problem thus preventing causal interpretations. Focusing on two-level models, we explain how researchers can avoid this problem by including cluster means of the Level 1 explanatory variables as controls; we explain this point conceptually and with a large-scale simulation. We further show why the common practice of centering the predictor variables is mostly unnecessary. Moreover, to examine the state of the science, we reviewed 204 randomly drawn articles from macro and micro organizational science and applied psychology journals, finding that only 106 articles—with a slightly higher proportion from macro-oriented fields—properly deal with the random effects assumption. Alarmingly, most models also failed on the usual exogeneity requirement of the regressors, leaving only 25 mostly macro-level articles that potentially reported trustworthy multilevel estimates. We offer a set of practical recommendations for researchers to model multilevel data appropriately.

Ash et al.: Sensitivity, specificity, and tolerability of the BACTrack Skyn compared to other alcohol monitoring approaches among young adults in a field-based setting

Ash-Gueorguieva-Barnett-etal-2022

Garrett I. Ash et al. "Sensitivity, specificity, and tolerability of the BACTrack Skyn compared to other alcohol monitoring approaches among young adults in a field-based setting". In: *Alcoholism: Clinical and Experimental Research* 46.5 (May 2022), pp. 783–796. ISSN: 1530-0277. DOI: 10.1111/acer.14804.

Abstract: Background: There is a need for novel alcohol biosensors that are accurate, able to detect alcohol concentration close in time to consumption, and feasible and acceptable for many clinical and

research applications. We evaluated the field accuracy and tolerability of novel (BACTrack Skyn) and established (Alcohol Monitoring Systems SCRAM CAM) alcohol biosensors. Methods: The sensor and diary data were collected in a larger study of a biofeedback intervention and compared observationally in the present sub-study. Participants (high-risk drinkers, 40% female; median age 21) wore both Skyn and SCRAM CAM sensors for 1-6 days and were instructed to drink as usual. Data from the first cohort of participants ($N = 27$; 101 person-days) were used to find threshold values of transdermal alcohol that classified each day as meeting or not meeting defined levels of drinking (heavy, above-moderate, any). These values were used to develop scoring metrics that were subsequently tested using the second cohort ($N = 20$; 57 person-days). Data from both biosensors were compared to mobile diary self-report to evaluate sensitivity and specificity in relation to a priori standards established in the literature. Results: Skyn classification rules for Cohort #1 within 3 months of device shipment showed excellent sensitivity for heavy drinking (94%) and exceeded expectations for above-moderate and any drinking (78% and 69%, respectively), while specificity met expectations (91%). However, classification worsened when Cohort #1 devices $\geq$ 3 months from shipment were tested (area under curve for receiver operator characteristic 0.87 vs. 0.79) and the derived classification threshold when applied to Cohort #2 was inadequately specific (70%). Skyn tolerability metrics were excellent and exceeded the SCRAM CAM ($p \leq 0.001$). Conclusions: Skyn tolerability was favorable and accuracy rules were internally derivable but did not yield useful scoring metrics going forward across device lots and months of usage.

Baek et al.: Meta-analysis of single-case experimental design using multilevel modeling

Baek-Luo-Lam-2023

Eunkyeng Baek, Wen Luo, and Kwok Hap Lam. "Meta-analysis of single-case experimental design using multilevel modeling". In: *Behavior Modification* 47.6 (Jan. 2023), pp. 1546–1573. ISSN: 1552-4167. DOI: 10.1177/01454455221144034.

Abstract: Multilevel modeling (MLM) is an approach for meta-analyzing single-case experimental designs (SCED). In this paper, we provide a step-by-step guideline for using the MLM to meta-

analyze SCED time-series data. The MLM approach is first presented using a basic three-level model, then gradually extended to represent more realistic situations of SCED data, such as modeling a time variable, moderators representing different design types and multiple outcomes, and heterogeneous within-case variance. The presented approach is then illustrated using real SCED data. Practical recommendations using the MLM approach are also provided for applied researchers based on the current methodological literature. Available free and commercial software programs to meta-analyze SCED data are also introduced, along with several hands-on software codes for applied researchers to implement their own studies. Potential advantages and limitations of using the MLM approach to meta-analyzing SCED are discussed.

Bakk et al.: Relating latent class membership to external variables: An overview

Bakk-Kuha-2021

Zsuzsa Bakk and Jouni Kuha. "Relating latent class membership to external variables: An overview". In: *British Journal of Mathematical and Statistical Psychology* 74.2 (May 2021), pp. 340–362. ISSN: 2044-8317. DOI: 10.1111/bmsp.12227.

Abstract: In this article we provide an overview of existing approaches for relating latent class membership to external variables of interest. We extend on the work of Nylund-Gibson et al. (Structural Equation Modeling: A Multidisciplinary Journal, 2019, 26, 967), who summarize models with distal outcomes by providing an overview of most recommended modeling options for models with covariates and larger models with multiple latent variables as well. We exemplify the modeling approaches using data from the General Social Survey for a model with a distal outcome where underlying model assumptions are violated, and a model with multiple latent variables. We discuss software availability and provide example syntax for the real data examples in Latent GOLD.

Berli et al.: Understanding Between-Person Interventions With Time-Intensive Longitudinal Outcome Data: Longitudinal Mediation Analyses **Berli-Inauen-Stadler-etal-2021**

Corina Berli et al. "Understanding Between-Person Interventions With Time-Intensive Longitudinal Outcome Data: Longitudinal Mediation Analyses". In: *Annals of Behavioral Medicine* 55.5 (May 2021), pp. 476–488. ISSN: 1532-4796. DOI: 10.1093/abm/kaaa066.

Abstract: Mediation analysis is commonly used to investigate how interventions influence health outcomes, but conventional approaches often give limited attention to how mediation processes unfold over time. This article presents a framework for studying the effects of a between-person intervention on repeatedly measured mediators and outcomes. The framework consists of five conceptual and analytic steps that examine intervention effects on the outcome, intervention effects on the mediator, associations between mediator and outcome, the resulting mediated process over time, and selection of an appropriate statistical model. Two health behavior intervention examples illustrate how longitudinal multilevel structural equation models can represent changes in the magnitudes of direct and indirect effects across time. The authors emphasize that the temporal dynamics of mediation should be treated as an important component of causal analyses of interventions.

Boker et al.: Separating long-term equilibrium adaptation from short-term self-regulation dynamics using latent differential equations **Boker-Daniel-Orzek-2024**

Steven M. Boker, Katharine E. Daniel, and Jannik Orzek. "Separating long-term equilibrium adaptation from short-term self-regulation dynamics using latent differential equations". In: *Multivariate Behavioral Research* 59.6 (Nov. 2024), pp. 1177–1187. ISSN: 1532-7906. DOI: 10.1080/00273171.2023.2228302.

Abstract: Self-regulatory systems may change simultaneously at different timescales, with short-term fluctuations occurring around an equilibrium that itself changes gradually over longer periods. The authors propose a model that combines a second-order linear differential equation for short-

timescale regulation with a first-order linear differential equation for long-timescale adaptation of the equilibrium. The model also permits individual differences to moderate parameters governing short-term dynamics. A simulation study indicates that the model can recover reasonable parameter estimates even when the short and long timescales substantially overlap. An empirical application to emotional well-being among recently bereaved widows illustrates how the approach can distinguish short-term regulation from longer-term adaptation.

Bolt et al.: Educating the future generation of researchers: A cross-disciplinary survey of trends in analysis methods **Bolt-Nomi-Bzdok-etal-2021**

Taylor Bolt et al. "Educating the future generation of researchers: A cross-disciplinary survey of trends in analysis methods". In: *PLOS Biology* 19.7 (July 2021). Ed. by Ulrich Dirnagl, e3001313. ISSN: 1545-7885. DOI: 10.1371/journal.pbio.3001313.

Abstract: Methods for data analysis in the biomedical, life, and social (BLS) sciences are developing at a rapid pace. At the same time, there is increasing concern that education in quantitative methods is failing to adequately prepare students for contemporary research. These trends have led to calls for educational reform to undergraduate and graduate quantitative research method curricula. We argue that such reform should be based on data-driven insights into within- and cross-disciplinary use of analytic methods. Our survey of peer-reviewed literature analyzed approximately 1.3 million openly available research articles to monitor the cross-disciplinary mentions of analytic methods in the past decade. We applied data-driven text mining analyses to the "Methods" and "Results" sections of a large subset of this corpus to identify trends in analytic method mentions shared across disciplines, as well as those unique to each discipline. We found that the t test, analysis of variance (ANOVA), linear regression, chi-squared test, and other classical statistical methods have been and remain the most mentioned analytic methods in biomedical, life science, and social science research articles. However, mentions of these methods have declined as a percentage of the published literature between 2009 and 2020. On the other hand, multivariate statistical and machine learning approaches, such as artificial neural networks (ANNs), have seen a significant increase in the total

share of scientific publications. We also found unique groupings of analytic methods associated with each BLS science discipline, such as the use of structural equation modeling (SEM) in psychology, survival models in oncology, and manifold learning in ecology. We discuss the implications of these findings for education in statistics and research methods, as well as within- and cross-disciplinary collaboration.

Borenstein: Avoiding common mistakes in meta-analysis: Understanding the distinct roles of Q, I-squared, tau-squared, and the prediction interval in reporting heterogeneity **Borenstein-2024**

Michael Borenstein. "Avoiding common mistakes in meta-analysis: Understanding the distinct roles of Q, I-squared, tau-squared, and the prediction interval in reporting heterogeneity". In: *Research Synthesis Methods* 15.2 (Mar. 2024), pp. 354–368. ISSN: 1759-2887. DOI: 10.1002/jrsm.1678.

Abstract: In any meta-analysis, it is critically important to report the dispersion in effects as well as the mean effect. If an intervention has a moderate clinical impact on average we also need to know if the impact is moderate for all relevant populations, or if it varies from trivial in some to major in others. Or indeed, if the intervention is beneficial in some cases but harmful in others. Researchers typically report a series of statistics such as the Q -value, the p -value, and I 2 , which are intended to address this issue. Often, they use these statistics to classify the heterogeneity as being low, moderate, or high and then use these classifications when considering the potential utility of the intervention. While this practice is ubiquitous, it is nevertheless incorrect. The statistics mentioned above do not actually tell us how much the effect size varies. Classifications of heterogeneity based on these statistics are uninformative at best, and often misleading. My goal in this paper is to explain what these statistics do tell us, and that none of them tells us how much the effect size varies. Then I will introduce the prediction interval, the statistic that does tell us how much the effect size varies, and that addresses the question we have in mind when we ask about heterogeneity. This paper is adapted from a chapter in "Common Mistakes in Meta-Analysis and How to Avoid Them." A free PDF of the book is available at https://www.Meta-Analysis.com/rsm .

Borghuis et al.: Longitudinal associations between trait neuroticism and negative daily experiences in adolescence **Borghuis-Bleidorn-Sijtsma-etal-2020**

Jeroen Borghuis et al. "Longitudinal associations between trait neuroticism and negative daily experiences in adolescence". In: *Journal of Personality and Social Psychology* 118.2 (Feb. 2020), pp. 348–363. ISSN: 0022-3514. DOI: 10.1037/pspp0000233.

Abstract: It is well established that trait neuroticism bears strong links with negative affect and interpersonal problems. The goal of this study was to examine the longitudinal associations between neuroticism and daily experiences of negative affect and interpersonal problems during the developmentally important period of adolescence. Dutch adolescents and their best friends ($N = 1{,}046$) completed up to 6 yearly personality trait questionnaires and up to 15 between-year assessment bursts between the ages of 13 and 18. During each assessment burst, participants reported on 5 consecutive days about their experiences of negative affect and interpersonal conflict with their mother and their best friend. We estimated a series of multilevel random-intercept cross-lagged panel models to differentiate covariance at the level of constant between-person differences from dynamic processes that occurred within persons. At the level of constant between-person differences, higher neuroticism was associated with more negative daily experiences. At the within-person level, yearly changes in neuroticism were bidirectionally and positively associated with yearly changes in daily negative affect. The most parsimonious, best fitting models did not contain a random intercept for daily conflict with friend and adolescents' contingency between daily experiences of conflict with mother and negative affect. Rank-order differences in these variables were positively associated with subsequent within-person changes in neuroticism. We discuss these results with regard to endogenous versus dynamic theories of personality development and the value of using a differentiated statistical approach.

L. F. Bringmann: Person-specific networks in psychopathology: Past, present, and future **Bringmann-2021**

Laura F. Bringmann. "Person-specific networks in psychopathology: Past, present, and future". In: *Current Opinion in Psychology* 41 (Oct. 2021), pp. 59–64. ISSN: 2352-250X. DOI: 10.1016/j.copsyc.2021.03.004.

Abstract: In the psychological network approach, mental disorders such as major depressive disorder are conceptualized as networks. The network approach focuses on the symptom structure or the connections between symptoms instead of the severity (i.e., mean level) of a symptom. To infer a person-specific network for a patient, time-series data are needed. By far the most common model to statistically model the person-specific interactions between symptoms or momentary states has been the vector autoregressive (VAR) model. Although the VAR model helps to bring psychological network theory into clinical research and closer to clinical practice, several discrepancies arise when we map the psychological network theory onto the VAR-based network models. These challenges and possible solutions are discussed in this review.

Brose et al.: Integrating state dynamics and trait change: A tutorial using the example of stress reactivity and change in well-being **Brose-Neubauer-Schmiedek-2022**

Annette Brose, Andreas Benjamin Neubauer, and Florian Schmiedek. "Integrating state dynamics and trait change: A tutorial using the example of stress reactivity and change in well-being". In: *European Journal of Personality* 36.2 (Mar. 2022), pp. 180–199. ISSN: 1099-0984. DOI: 10.1177/08902070211014055.

Abstract: Theories of trait change increasingly emphasize the possibility that repeated state experiences and state dynamics contribute to longer-term changes in individual differences. Research linking processes such as daily stress reactivity to changes in well-being has often relied on a two-step approach in which state dynamics and trait change are modeled separately. This tutorial presents one-step procedures for estimating state dynamics and longer-term change simultaneously

within a multilevel structural equation modeling framework. Using stress reactivity and change in well-being as an example, the authors describe the advantages of jointly modeling within-person dynamics and between-person differences in change and provide practical guidance for substantive researchers interested in connecting short-term processes with longer-term development.

Campbell: Equivalence testing for linear regression **Campbell-2024**

Harlan Campbell. "Equivalence testing for linear regression". In: *Psychological Methods* 29.1 (Feb. 2024), pp. 88–98. ISSN: 1939-1463. DOI: 10.1037/met0000596.

Abstract: We introduce equivalence testing procedures for linear regression analyses. Such tests can be very useful for confirming the lack of a meaningful association between a continuous outcome and a continuous or binary predictor. Specifically, we propose an equivalence test for unstandardized regression coefficients and an equivalence test for semipartial correlation coefficients. We review how to define valid hypotheses, how to calculate p values, and how these tests compare to an alternative Bayesian approach with applications to examples in the literature. (PsycInfo Database Record (c) 2024 APA, all rights reserved).

Man Chen et al.: Multilevel meta-analysis of single-case experimental designs using robust variance estimation. **Chen-Pustejovsky-2024**

Man Chen and James E. Pustejovsky. "Multilevel meta-analysis of single-case experimental designs using robust variance estimation." In: *Psychological Methods* 29.3 (June 2024), pp. 537–560. ISSN: 1082-989X. DOI: 10.1037/met0000510.

Abstract: Single-case experimental designs (SCEDs) are used to study the effects of interventions on the behavior of individual cases, by making comparisons between repeated measurements of an outcome under different conditions. In research areas where SCEDs are prevalent, there is a need for methods to synthesize results across multiple studies. One approach to synthesis uses a multilevel meta-analysis (MLMA) model to describe the distribution of effect sizes across studies and across

cases within studies. However, MLMA relies on having accurate sampling variances of effect size estimates for each case, which may not be possible due to auto-correlation in the raw data series. One possible solution is to combine MLMA with robust variance estimation (RVE), which provides valid assessments of uncertainty even if the sampling variances of effect size estimates are inaccurate. Another possible solution is to forgo MLMA and use simpler, ordinary least squares (OLS) methods with RVE. This study evaluates the performance of effect size estimators and methods of synthesizing SCEDs in the presence of auto-correlation, for several different effect size metrics, via a Monte Carlo simulation designed to emulate the features of real data series. Results demonstrate that the MLMA model with RVE performs properly in terms of bias, accuracy, and confidence interval coverage for estimating overall average log response ratios. The OLS estimator corrected with RVE performs the best in estimating overall average Tau effect sizes. None of the available methods perform adequately for meta-analysis of within-case standardized mean differences.

Meng Chen et al.: A person- and time-varying vector autoregressive model to capture interactive infant-mother head movement dynamics **Chen-Chow-Hammal-etal-2021**

Meng Chen et al. "A person- and time-varying vector autoregressive model to capture interactive infant-mother head movement dynamics". In: *Multivariate Behavioral Research* 56.5 (Sept. 2021), pp. 739–767. ISSN: 1532-7906. DOI: 10.1080/00273171.2020.1762065.

Abstract: Head movement is an important but often overlooked component of emotion and social interaction. Examination of regularity and differences in head movements of infant-mother dyads over time and across dyads can shed light on whether and how mothers and infants alter their dynamics over the course of an interaction to adapt to each others. One way to study these emergent differences in dynamics is to allow parameters that govern the patterns of interactions to change over time, and according to person- and dyad-specific characteristics. Using two estimation approaches to implement variations of a vector-autoregressive model with time-varying coefficients, we investigated the dynamics of automatically-tracked head movements in mothers and infants during the Face-Face/Still-Face Procedure (SFP) with 24 infant-mother dyads. The

first approach requires specification of a confirmatory model for the time-varying parameters as part of a state-space model, whereas the second approach handles the time-varying parameters in a semi-parametric ("mostly" model-free) fashion within a generalized additive modeling framework. Results suggested that infant-mother head movement dynamics varied in time both within and across episodes of the SFP, and varied based on infants' subsequently-assessed attachment security. Code for implementing the time-varying vector-autoregressive model using two R packages, dynr and mgcv, is provided.

M. W.-L. Cheung: Synthesizing indirect effects in mediation models with meta-analytic methods **Cheung-2022**

Mike W.-L. Cheung. "Synthesizing indirect effects in mediation models with meta-analytic methods". In: *Alcohol and Alcoholism* 57.1 (Jan. 2022), pp. 5–15. DOI: 10.1093/alcalc/agab044.

Abstract: Aims: A mediator is a variable that explains the underlying mechanism between an independent variable and a dependent variable. The indirect effect indicates the effect from the predictor to the outcome variable via the mediator. In contrast, the direct effect represents the predictor's effort on the outcome variable after controlling for the mediator. Methods: A single study rarely provides enough evidence to answer research questions in a particular domain. Replications are generally recommended as the gold standard to conduct scientific research. When a sufficient number of studies have been conducted addressing similar research questions, a meta-analysis can be used to synthesize those studies' findings. Results: The main objective of this paper is to introduce two frameworks to integrating studies using mediation analysis. The first framework involves calculating standardized indirect effects and direct effects and conducting a multivariate meta-analysis on those effect sizes. The second one uses meta-analytic structural equation modeling to synthesize correlation matrices and fit mediation models on the average correlation matrix. We illustrate these procedures on a real dataset using the R statistical platform. Conclusion: This paper closes with some further directions for future studies.

S. F. Cheung et al.: Improving an old way to measure moderation effect in standardized units. **Cheung-Cheung-Lau-etal-2022**

Shu Fai Cheung, Sing-Hang Cheung, et al. "Improving an old way to measure moderation effect in standardized units." In: *Health Psychology* 41.7 (July 2022), pp. 502–505. ISSN: 0278-6133. DOI: 10.1037/hea0001188.

Abstract: Moderation effects in multiple regression, tested usually by the inclusion of a product term, are frequently investigated in health psychology. However, several issues in presenting the moderation effects in standardized units and their associated confidence intervals are commonly observed. While an old method had been proposed to standardize variables in moderated regression before fitting a moderated regression model, this method was rarely used due to inconvenience and even when used, the confidence intervals derived were biased. Here, we attempt to solve these two problems by providing a tool to conveniently conduct standardization in moderated regression without the step of standardizing the variables beforehand and to accurately form the nonparametric bootstrapping confidence intervals for this standardized measure of moderation effects. Health psychology researchers are now equipped with a tool that can be used to report and interpret standardized moderation effects correctly.

S. F. Cheung et al.: FINDOUT: Using either SPSS commands or graphical user interface to identify influential cases in structural equation modeling in AMOS **Cheung-Pesigan-2023a**

Shu Fai Cheung and Ivan Jacob Agaloos Pesigan. "FINDOUT: Using either SPSS commands or graphical user interface to identify influential cases in structural equation modeling in AMOS". In: *Multivariate Behavioral Research* 58.5 (Jan. 2023), pp. 964–968. DOI: 10.1080/00273171.2022.2148089.

Abstract: The results in a structural equation modeling (SEM) analysis can be influenced by just a few observations, called influential cases. Tools have been developed for users of R to identify

them. However, similar tools are not available for AMOS, which is also a popular SEM software package. We introduce the FINDOUT toolset, a group of SPSS extension commands, and an AMOS plugin, to identify influential cases and examine how these cases influence the results. The SPSS commands can be used either as syntax commands or as custom dialogs from pull-down menus, and the AMOS plugin can be run from AMOS pull-down menu. We believe these tools can help researchers to examine the robustness of their findings to influential cases.

S. F. Cheung et al.: semlbci: An R package for forming likelihood-based confidence intervals for parameter estimates, correlations, indirect effects, and other derived parameters **Cheung-Pesigan-2023b**

Shu Fai Cheung and Ivan Jacob Agaloos Pesigan. "semlbci: An R package for forming likelihood-based confidence intervals for parameter estimates, correlations, indirect effects, and other derived parameters". In: *Structural Equation Modeling: A Multidisciplinary Journal* 30.6 (May 2023), pp. 985–999. DOI: 10.1080/10705511.2023.2183860.

Abstract: There are three common types of confidence interval (CI) in structural equation modeling (SEM): Wald-type CI, bootstrapping CI, and likelihood-based CI (LBCI). LBCI has the following advantages: (1) it has better coverage probabilities and Type I error rate compared to Wald-type CI when the sample size is finite; (2) it correctly tests the null hypothesis of a parameter based on likelihood ratio chi-square difference test; (3) it is less computationally intensive than bootstrapping CI; and (4) it is invariant to transformations. However, LBCI is not available in many popular SEM software packages. We developed an R package, semlbci, for forming LBCI for parameters in models fitted by lavaan, a popular open-source SEM package, such that researchers have more options in forming CIs for parameters in SEM. The package supports both unstandardized and standardized estimates, derived parameters such as indirect effect, multisample models, and the robust LBCI proposed by Falk.

S. F. Cheung et al.: DIY bootstrapping: Getting the nonparametric bootstrap confidence interval in SPSS for any statistics or function of statistics (when this bootstrapping is appropriate) **Cheung-Pesigan-Vong-2023**

Shu Fai Cheung, Ivan Jacob Agaloos Pesigan, and Weng Ngai Vong. "DIY bootstrapping: Getting the nonparametric bootstrap confidence interval in SPSS for any statistics or function of statistics (when this bootstrapping is appropriate)". In: *Behavior Research Methods* 55.2 (2023), pp. 474–490. DOI: 10.3758/s13428-022-01808-5.

Abstract: Researchers can generate bootstrap confidence intervals for some statistics in SPSS using the BOOTSTRAP command. However, this command can only be applied to selected procedures, and only to selected statistics in these procedures. We developed an extension command and prepared some sample syntax files based on existing approaches from the Internet to illustrate how researchers can (a) generate a large number of nonparametric bootstrap samples, (b) do desired analysis on all these samples, and (c) form the bootstrap confidence intervals for selected statistics using the OMS commands. We developed these tools to help researchers apply nonparametric bootstrapping to any statistics for which this method is appropriate, including statistics derived from other statistics, such as standardized effect size measures computed from the t test results. We also discussed how researchers can extend the tools for other statistics and scenarios they encounter.

Cho et al.: Multilevel latent differential structural equation model with short time series and time-varying covariates: A comparison of frequentist and Bayesian estimators **Cho-Chow-Marini-etal-2024**

Young Won Cho et al. "Multilevel latent differential structural equation model with short time series and time-varying covariates: A comparison of frequentist and Bayesian estimators". In: *Multivariate Behavioral Research* 59.5 (May 2024), pp. 934–956. ISSN: 1532-7906. DOI: 10.1080/00273171.2024.2347959.

Abstract: Continuous-time modeling using differential equations is a promising technique to model change processes with longitudinal data. Among ways to fit this model, the Latent Differential Structural Equation Modeling (LDSEM) approach defines latent derivative variables within a structural equation modeling (SEM) framework, thereby allowing researchers to leverage advantages of the SEM framework for model building, estimation, inference, and comparison purposes. Still, a few issues remain unresolved, including performance of multilevel variations of the LDSEM under short time lengths (e.g., 14 time points), particularly when coupled multivariate processes and time-varying covariates are involved. Additionally, the possibility of using Bayesian estimation to facilitate the estimation of multilevel LDSEM (M-LDSEM) models with complex and higher-dimensional random effect structures has not been investigated. We present a series of Monte Carlo simulations to evaluate three possible approaches to fitting M-LDSEM, including: frequentist single-level and two-level robust estimators and Bayesian two-level estimator. Our findings suggested that the Bayesian approach outperformed other frequentist approaches. The effects of time-varying covariates are well recovered, and coupling parameters are the least biased especially using higher-order derivative information with the Bayesian estimator. Finally, an empirical example is provided to show the applicability of the approach.

Courtney et al.: Testing affect regulation models of drinking prior to and after drinking initiation using ecological momentary assessment **Courtney-Russell-2021**

Jimikaye B. Courtney and Michael A. Russell. "Testing affect regulation models of drinking prior to and after drinking initiation using ecological momentary assessment". In: *Psychology of Addictive Behaviors* 35.5 (Aug. 2021), pp. 597–608. ISSN: 0893-164X. DOI: 10.1037/adb0000763.

Abstract: Objective: Affect regulation models of drinking state that affect motivates and reinforces drinking. Few studies have been able to elucidate the timing of these associations in natural settings. We tested positive affect (PA) and negative affect (NA) as predictors of drinking behavior, both prior to and during drinking episodes, and whether drinking predicted changes in affect during episodes. Method: Two hundred twenty two regularly drinking young adults (21-29 years, 84% un-

dergraduates), completed an ecological momentary assessment (EMA) protocol for five consecutive 24-hr periods stretching across 6 days (Wednesday-Monday). Participants provided PA and NA reports three times daily and every half hour during drinking episodes. Alcohol consumption reports were provided each morning and every half hour during drinking episodes. Results: Multi-level models showed that greater pre-drinking PA predicted higher odds of drinking and greater number of drinks consumed. Pre-drinking NA did not predict same day odds of drinking or drinks consumed. Episode-level results revealed different associations for PA and NA with drinking. Current PA did not predict drinks consumed over the next half hour; however, increased drinking was associated with greater increases in PA over the next half hour. Higher NA predicted fewer drinks consumed in the next half hour and higher odds of the end of a drinking episode; however, increased drinking was not associated with changes in NA. Conclusions: PA increased following drinking during episodes. Our results suggest that a focus on PA prior to episodes and a focus on NA during episodes may interrupt processes leading to heavy drinking, and may therefore aid prevention efforts.

Declercq et al.: Multilevel Meta-Analysis of Individual Participant Data of Single-Case Experimental Designs: One-Stage versus Two-Stage Methods

Declercq-Jamshidi-FernandezCastilla-etal-2022

Lies Declercq et al. "Multilevel Meta-Analysis of Individual Participant Data of Single-Case Experimental Designs: One-Stage versus Two-Stage Methods". In: *Multivariate Behavioral Research* 57.2-3 (May 2022), pp. 298–317. ISSN: 1532-7906. DOI: 10.1080/00273171.2020.1822148.

Abstract: To conduct a multilevel meta-analysis of multiple single-case experimental design (SCED) studies, the individual participant data (IPD) can be analyzed in one or two stages. In the one-stage approach, a multilevel model is estimated based on the raw data. In the two-stage approach, an effect size is calculated for each participant and these effect sizes and their sampling variances are subsequently combined to estimate a meta-analytic multilevel model. The multilevel model in the two-stage approach has fewer parameters to estimate, in exchange for the reduction of information of the raw data to effect sizes. In this paper we explore how the one-stage and two-stage IPD

approaches can be applied in the context of meta-analysis of single-case designs. Both approaches are compared for several single-case designs of increasing complexity. Through a simulation study we show that the two-stage approach obtains better convergence rates for more complex models, but that model estimation does not necessarily converge at a faster speed. The point estimates of the fixed effects are unbiased for both approaches across all models, as such confirming results from methodological research on IPD meta-analysis of group-comparison designs. In light of these results, we discuss the implementation of both methods in R.

Dedecker et al.: Asymptotic confidence interval for R^2 in multiple linear regression **Dedecker-Guedj-Taupin-2025**

Jérôme Dedecker, Odelia Guedj, and Marie-Luce Taupin. "Asymptotic confidence interval for R^2 in multiple linear regression". In: *Statistics* 59.1 (Jan. 2025), pp. 1–36. ISSN: 0233-1888. DOI: 10.1080/02331888.2024.2428978.

DeMartini et al.: Dynamic structural equation modeling of the relationship between alcohol habit and drinking variability **DeMartini-Gueorguieva-Taylor-etal-2022**

Kelly S. DeMartini et al. "Dynamic structural equation modeling of the relationship between alcohol habit and drinking variability". In: *Drug and Alcohol Dependence* 233 (Apr. 2022), p. 109202. ISSN: 0376-8716. DOI: 10.1016/j.drugalcdep.2021.109202.

Abstract: Background: A hyper-engaged habit system may be common in alcohol use disorders (AUDs). Regarding drinking patterns, habit may be expressed as higher levels of drinking autoregression, where previous day drinking is correlated with next day drinking. This study utilized dynamic structural equation models (DSEM) with intensive longitudinal data to understand whether alcohol habit relates to drinking autoregression and variable levels of alcohol consumption. Methods: Participants were adult drinkers ($N = 313$) who completed baseline self-report assessments of past 30-day alcohol consumption and alcohol habit. Alcohol habit was measured by the Self Report

Habit Index (SRHI). Thirty-day coding of the Timeline Followback assessed total daily drinking and any daily heavy drinking. Results: The DSEM model for daily drinking found a weak but significant autoregressive data structure. Alcohol habit was related to increased mean drinking but did not strengthen the autoregressive effect of drinks per day. Higher alcohol habit was associated with higher levels of drinks per day person-specific variability. This pattern was replicated with the DSEM model for heavy drinking. Alcohol habit did not impact the autoregressive effect of heavy drinking but was associated with higher levels of heavy drinking. Conclusions: While both drinks per day and heavy drinking showed a significant autoregressive structure, evidence of alcohol habit did not strengthen this effect. Alcohol habit did impact drinking variability; higher alcohol habit is associated with greater levels of drinking variability and higher mean drinking. Strategies to regulate drinking variability, including heavier drinking occasions, could target AUD habit.

Didier et al.: Signal processing and machine learning with transdermal alcohol concentration to predict natural environment alcohol consumption.

Didier-King-Polley-etal-2024

Nathan A. Didier et al. "Signal processing and machine learning with transdermal alcohol concentration to predict natural environment alcohol consumption." In: *Experimental and Clinical Psychopharmacology* 32.2 (Apr. 2024), pp. 245–254. ISSN: 1064-1297. DOI: 10.1037/pha0000683.

Abstract: Wrist-worn alcohol biosensors continuously and discreetly record transdermal alcohol concentration (TAC) and may allow alcohol researchers to monitor alcohol consumption in participants' natural environments. However, the field lacks established methods for signal processing and detecting alcohol events using these devices. We developed software that streamlines analysis of raw data (TAC, temperature, and motion) from a wrist-worn alcohol biosensor (BACtrack Skyn) through a signal processing and machine learning pipeline: biologically implausible skin surface temperature readings (¡ 28C) were screened for potential device removal and TAC artifacts were corrected, features that describe TAC (e.g., rise duration) were calculated and used to train models (random forest and logistic regression) that predict self-reported alcohol consumption, and model

performances were measured and summarized in autogenerated reports. The software was tested using 60 Skyn data sets recorded during 30 alcohol drinking episodes and 30 nonalcohol drinking episodes. Participants (N = 36; 13 with alcohol use disorder) wore the Skyn during one alcohol drinking episode and one nonalcohol drinking episode in their natural environment. In terms of distinguishing alcohol from nonalcohol drinking, correcting artifacts in the data resulted in 10% improvement in model accuracy relative to using raw data. Random forest and logistic regression models were both accurate, correctly predicting 97% (58/60; AUC-ROCs = 0.98, 0.96) of episodes. Area under TAC curve, rise duration of TAC curve, and peak TAC were the most important features for predictive accuracy. With promising model performance, this protocol will enhance the efficiency and reliability of TAC sensors for future alcohol monitoring research.

Don et al.: Training in mindfulness or loving-kindness meditation is associated with lower variability in social connectedness across time

Don-VanCappellen-Fredrickson-2022

Brian P. Don, Patty Van Cappellen, and Barbara L. Fredrickson. "Training in mindfulness or loving-kindness meditation is associated with lower variability in social connectedness across time". In: *Mindfulness* 13.5 (Apr. 2022), pp. 1173–1184. ISSN: 1868-8535. DOI: 10.1007/s12671-022-01856-0.

Abstract: Meditation research has generally focused on average levels of social outcomes, although fluctuations in social functioning may also be important for well-being. Using data from a randomized study of 224 mid-life adults, this study examined changes in the variability of daily social connectedness during mindfulness or loving-kindness meditation training. Participants reported social connectedness and emotions daily for two weeks before training, six weeks during training, and three weeks afterward. Participants in both meditation conditions showed reduced variability in social connectedness during the intervention even after accounting for average connectedness. Reductions in variability of both positive and negative emotions partially accounted for the decline in social connectedness variability. The findings suggest that meditation training may be associated

not only with levels of social connectedness but also with greater stability in social experience over time.

Dora et al.: The daily association between affect and alcohol use: A meta-analysis of individual participant data **Dora-Piccirillo-Foster-etal-2023**

Jonas Dora et al. "The daily association between affect and alcohol use: A meta-analysis of individual participant data". In: *Psychological Bulletin* 149.1-2 (Jan. 2023), pp. 1–24. ISSN: 0033-2909. DOI: 10.1037/bul0000387.

Abstract: Influential psychological theories hypothesize that people consume alcohol in response to the experience of both negative and positive emotions. Despite two decades of daily diary and ecological momentary assessment research, it remains unclear whether people consume more alcohol on days they experience higher negative and positive affects in everyday life. In this preregistered meta-analysis, we synthesized the evidence for these daily associations between affect and alcohol use. We included individual participant data from 69 studies (N = 12,394), which used daily and momentary surveys to assess the affect and the number of alcoholic drinks consumed. Results indicate that people are not more likely to drink on days they experience high negative affect but are more likely to drink and drink heavily on days high in positive affect. People self-reporting a motivational tendency to drink-to-cope and drink-to-enhance consumed more alcohol but not on days they experienced higher negative and positive affects. Results were robust across different operationalizations of affect, study designs, study populations, and individual characteristics. These findings challenge the long-held belief that people drink more alcohol following increase in negative affect. Integrating these findings under different theoretical models and limitations of this field of research, we collectively propose an agenda for future research to explore open questions surrounding affect and alcohol use.

Driver: Inference with cross-lagged effects—Problems in time **Driver-2025**

Charles C. Driver. "Inference with cross-lagged effects—Problems in time". In: *Psychological Methods* 30.1 (Feb. 2025), pp. 174–202. ISSN: 1082-989X. DOI: 10.1037/met0000665.

Abstract: The interpretation of cross-effects from vector autoregressive models to infer structure and causality among constructs is widespread and sometimes problematic. I describe problems in the interpretation of cross-effects when processes that are thought to fluctuate continuously in time are, as is typically done, modeled as changing only in discrete steps (as in e.g., structural equation modeling)–zeroes in a discrete-time temporal matrix do not necessarily correspond to zero effects in the underlying continuous processes, and vice versa. This has implications for the common case when the presence or absence of cross-effects is used for inference about underlying causal processes. I demonstrate these problems via simulation, and also show that when an underlying set of processes are continuous in time, even relatively few direct causal links can result in much denser temporal effect matrices in discrete-time. I demonstrate one solution to these issues, namely parameterizing the system as a stochastic differential equation and focusing inference on the continuous-time temporal effects. I follow this with some discussion of issues regarding the switch to continuous-time, specifically regularization, appropriate measurement time lag, and model order. An empirical example using intensive longitudinal data highlights some of the complexities of applying such approaches to real data, particularly with respect to model specification, examining misspecification, and parameter interpretation.

Elmer et al.: Modeling categorical time-to-event data: The example of social interaction dynamics captured with event-contingent experience sampling methods. **Elmer-vanDuijn-Ram-etal-2025**

Timon Elmer et al. "Modeling categorical time-to-event data: The example of social interaction dynamics captured with event-contingent experience sampling methods." In: *Psychological Methods* 30.6 (Dec. 2025), pp. 1345–1363. ISSN: 1082-989X. DOI: 10.1037/met0000598.

Abstract: The depth of information collected in participants' daily lives with active (e.g., experience sampling surveys) and passive (e.g., smartphone sensors) ambulatory measurement methods is immense. When measuring participants' behaviors in daily life, the timing of particular events—such as social interactions—is often recorded. These data facilitate the investigation of new types of research questions about the timing of those events, including whether individuals' affective state is associated with the rate of social interactions (binary event occurrence) and what types of social interactions are likely to occur (multicategory event occurrences, e.g., interactions with friends or family). Although survival analysis methods have been used to analyze time-to-event data in longitudinal settings for several decades, these methods have not yet been incorporated into ambulatory assessment research. This article illustrates how multilevel and multistate survival analysis methods can be used to model the social interaction dynamics captured in intensive longitudinal data, specifically *when individuals exhibit particular categories of behavior*. We provide an introduction to these models and a tutorial on how the timing and type of social interactions can be modeled using the R statistical programming language. Using event-contingent reports ($N = 150$, $N_{\text{events}} = 64,112$) obtained in an ambulatory study of interpersonal interactions, we further exemplify an empirical application case. In sum, this article demonstrates how survival models can advance the understanding of (social interaction) dynamics that unfold in daily life.

Feinn et al.: Individual differences in affect dynamics and alcohol-related outcomes

Feinn-Armeli-Tennen-2023

Richard Feinn, Stephen Armeli, and Howard Tennen. "Individual differences in affect dynamics and alcohol-related outcomes". In: *Substance Use & Misuse* 58.8 (Apr. 2023), pp. 967–974. ISSN: 1532-2491. DOI: 10.1080/10826084.2023.2201829.

Abstract: Background: To examine whether individual differences in intensive longitudinal data-derived affective dynamics (i.e. positive and negative affect variability and inertia and positive affect-negative affect bipolarity) - posited to be indicative of emotion dysregulation - are uniquely related to drinking level and affect-regulation drinking motives after controlling for mean levels

of affective states. Method: We used a large sample of college student drinkers (N = 1640, 54% women) who reported on their affective states, drinking levels and drinking motives daily for 30 days using a web-based daily diary. We then calculated from the daily data positive and negative affect variability, inertia, affect bipolarity and mean levels of affect and used these as predictors of average drinking level and affect-regulation drinking motives (assessed using both retrospective and daily reporting methods). Results: Findings from dynamic structural equation models indicated that mean levels of affect were uniquely related to drinking motives, but not drinking level. Few dynamic affect predictors were uniquely related to outcomes in the predicted direction after controlling for mean affect levels. Conclusion: Our results add to the inconsistent literature regarding the associations between affective dynamics and alcohol-related outcomes, suggesting that any effects of these indicators, after controlling for mean affect levels, might be more complex than can be detected with simple linear models.

Fisher et al.: A square-root second-order extended Kalman filtering approach for estimating smoothly time-varying parameters **Fisher-Chow-Molenaar-etal-2022**

Zachary F. Fisher, Sy-Miin Chow, et al. "A square-root second-order extended Kalman filtering approach for estimating smoothly time-varying parameters". In: *Multivariate Behavioral Research* 57.1 (Jan. 2022), pp. 134–152. ISSN: 1532-7906. DOI: 10.1080/00273171.2020.1815513.

Abstract: Researchers collecting intensive longitudinal data (ILD) are increasingly looking to model psychological processes, such as emotional dynamics, that organize and adapt across time in complex and meaningful ways. This is also the case for researchers looking to characterize the impact of an intervention on individual behavior. To be useful, statistical models must be capable of characterizing these processes as complex, time-dependent phenomenon, otherwise only a fraction of the system dynamics will be recovered. In this paper we introduce a Square-Root Second-Order Extended Kalman Filtering approach for estimating smoothly time-varying parameters. This approach is capable of handling dynamic factor models where the relations between variables underlying the processes of interest change in a manner that may be difficult to specify in advance.

We examine the performance of our approach in a Monte Carlo simulation and show the proposed algorithm accurately recovers the unobserved states in the case of a bivariate dynamic factor model with time-varying dynamics and treatment effects. Furthermore, we illustrate the utility of our approach in characterizing the time-varying effect of a meditation intervention on day-to-day emotional experiences.

Fisher et al.: Penalized estimation and forecasting of multiple subject intensive longitudinal data **Fisher-Kim-Fredrickson-etal-2022**

Zachary F. Fisher, Younghoon Kim, et al. "Penalized estimation and forecasting of multiple subject intensive longitudinal data". In: *Psychometrika* 87.2 (June 2022), pp. 403–431. ISSN: 1860-0980. DOI: 10.1007/s11336-021-09825-7.

Abstract: Intensive longitudinal data (ILD) is an increasingly common data type in the social and behavioral sciences. Despite the many benefits these data provide, little work has been dedicated to realize the potential such data hold for forecasting dynamic processes at the individual level. To address this gap in the literature, we present the multi-VAR framework , a novel methodological approach allowing for penalized estimation of ILD collected from multiple individuals. Importantly, our approach estimates models for all individuals simultaneously and is capable of adaptively adjusting to the amount of heterogeneity present across individual dynamic processes. To accomplish this, we propose a novel proximal gradient descent algorithm for solving the multi-VAR problem and prove the consistency of the recovered transition matrices. We evaluate the forecasting performance of our method in comparison with a number of benchmark methods and provide an illustrative example involving the day-to-day emotional experiences of 16 individuals over an 11-week period.

Fredrickson et al.: Same-day, cross-day, and upward spiral relations between positive affect and positive health behaviours **Fredrickson-Arizmendi-VanCappellen-2021**

Barbara L. Fredrickson, Cara Arizmendi, and Patty Van Cappellen. "Same-day, cross-day, and

upward spiral relations between positive affect and positive health behaviours". In: *Psychology & Health* 36.4 (Apr. 2021), pp. 444–460. ISSN: 1476-8321. DOI: 10.1080/08870446.2020.1778696.

Abstract: This study examined same-day and cross-day relations between positive affect and positive health behaviours, including physical activity, fruit and vegetable consumption, and meditation. The analyses used intensive daily data from midlife adults participating in a randomized meditation study. The authors examined whether positive affect and health behaviours were associated within the same day, whether either predicted the other on the following day, and whether repeated associations were consistent with an upward spiral process. Results indicated substantial same-day associations between positive affect and several health behaviours, whereas cross-day associations were more selective. The findings highlight the importance of distinguishing contemporaneous from prospective relations when evaluating reciprocal links between positive affect and health behaviour.

Fridberg et al.: Examining features of transdermal alcohol biosensor readings: A promising approach with implications for research and intervention

Fridberg-Wang-Porges-2022

Daniel J. Fridberg, Yan Wang, and Eric Porges. "Examining features of transdermal alcohol biosensor readings: A promising approach with implications for research and intervention". In: *Alcoholism: Clinical and Experimental Research* 46.4 (Feb. 2022), pp. 514–516. ISSN: 1530-0277. DOI: 10.1111/acer.14794.

Georgeson et al.: A sensitivity analysis for temporal bias in cross-sectional mediation

Georgeson-AlvarezBartolo-MacKinnon-2025

A. R. Georgeson, Diana Alvarez-Bartolo, and David P. MacKinnon. "A sensitivity analysis for temporal bias in cross-sectional mediation". In: *Psychological Methods* 30.6 (Dec. 2025), pp. 1326–1344. ISSN: 1082-989X. DOI: 10.1037/met0000628.

Abstract: For over three decades, methodologists have cautioned against the use of cross-sectional mediation analyses because they yield biased parameter estimates. Yet, cross-sectional mediation models persist in practice and sometimes represent the only analytic option. We propose a sensitivity analysis procedure to encourage a more principled use of cross-sectional mediation analysis, drawing inspiration from Gollob and Reichardt (1987, 1991). The procedure is based on the two-wave longitudinal mediation model and uses phantom variables for the baseline data. After a researcher provides ranges of possible values for cross-lagged, autoregressive, and baseline Y and M correlations among the phantom and observed variables, they can use the sensitivity analysis to identify longitudinal conditions in which conclusions from a cross-sectional model would differ most from a longitudinal model. To support the procedure, we first show that differences in sign and effect size of the b-path occur most often when the cross-sectional effect size of the b-path is small and the cross-lagged and the autoregressive correlations are equal or similar in magnitude. We then apply the procedure to cross-sectional analyses from real studies and compare the sensitivity analysis results to actual results from a longitudinal mediation analysis. While no statistical procedure can replace longitudinal data, these examples demonstrate that the sensitivity analysis can recover the effect that was actually observed in the longitudinal data if provided with the correct input information. Implications of the routine application of sensitivity analysis to temporal bias are discussed. R code for the procedure is provided in the online supplementary materials.

Gistelinck et al.: Multilevel autoregressive models for longitudinal dyadic data

Gistelinck-Loeys-2020

Fien Gistelinck and Tom Loeys. "Multilevel autoregressive models for longitudinal dyadic data". In: *TPM - Testing, Psychometrics, Methodology in Applied Psychology* 27 (2020), pp. 433–452. ISSN: 1972-6325. DOI: 10.4473/TPM27.3.7.

Abstract: In social and behavioral science, dyadic research has become more and more popular. In case of cross-sectional dyadic data, one can apply the actor-partner interdependence model (APIM). When dyads are measured repeatedly over time, applied researchers are often hesitant to analyze

such data due to the statistical complexity. In this paper, we introduce a user-friendly Shiny-application, called the LDDinSEM-application. The app automatically fits the lagged dependent actor-partner interdependence model (LD-APIM), a multilevel autoregressive model extension of the APIM within the structural equation modeling (SEM) framework. The application allows the researcher to investigate the effects of an antecedent on an outcome, given the previous outcome. We illustrate the app using an empirical example assessing the actor and partner effects of positive relationship feelings on next day's intimacy in heterosexual couples.

Goldberg et al.: Does mindfulness practice promote psychological functioning or is it the other way around? A daily diary study **Goldberg-Hanley-Baldwin-etal-2020**

Simon B. Goldberg et al. "Does mindfulness practice promote psychological functioning or is it the other way around? A daily diary study". In: *Psychotherapy* 57.3 (Sept. 2020), pp. 310–322. DOI: 10.1037/pst0000286.

Abstract: Although mindfulness-based interventions are widely used, the relationship between practice dosage and psychological functioning remains unclear. Twenty-five participants completed daily assessments throughout a 12-week Mindfulness-Oriented Recovery Enhancement intervention. Analyses separated whether meditation occurred from the amount of time practiced and examined positive affect, negative affect, and state mindfulness. Practice and psychological outcomes were related within the same day, but practice did not prospectively predict next-day outcomes. Instead, greater negative affect predicted less next-day practice, whereas greater mindfulness predicted more next-day practice. Additional analyses indicated that practice altered the continuity of daily affect. The findings suggest that psychological functioning may influence subsequent meditation behavior and that the association may be reciprocal rather than simply reflecting benefits produced by greater practice time.

Gonzales et al.: What effect sizes should researchers report for multiple regression under non-normal data? Gonzales-Li-2024

Ian Gonzales and Johnson Ching-Hong Li. "What effect sizes should researchers report for multiple regression under non-normal data?" In: *Communications in Statistics - Simulation and Computation* 53.6 (June 2024), pp. 2786–2804. ISSN: 0361-0918. DOI: 10.1080/03610918.2022.2091778.

Grayson et al.: The impact of stress on personality state expressions Grayson-Harari-Matz-2026

Samantha J. Grayson, Gabriella M. Harari, and Sandra C. Matz. "The impact of stress on personality state expressions". In: *Communications Psychology* 4.1 (Mar. 2026). ISSN: 2731-9121. DOI: 10.1038/s44271-026-00438-3.

Abstract: Who we are at any given moment depends not only on our psychological dispositions but also the situational forces we experience. Here, we explore how stress—an intrapsychic situational force—impacts momentary personality expression. In Study 1 ($N = 792$), we experimentally induced stress, showing that the experience of stress is causally related to lower levels of state Openness, Conscientiousness, Extraversion, Agreeableness and higher levels of state Neuroticism. In Study 2 ($N = 713$; 17,853 observations), we used experience sampling to capture naturally occurring stress and show that stress is associated with lower levels of state Extraversion and Agreeableness, and higher levels of state Neuroticism and Openness in everyday situations. Notably, across both studies, the associations between stress and personality state expressions are largely robust when accounting for momentary affect. Together, our work highlights stress as a distinct and dynamic driver of personality state variability in everyday life.

Groot et al.: Checking the inventory: Illustrating different methods for individual participant data meta-analytic structural equation modeling Groot-Kan-Jak-2024

Lennert J. Groot, Kees-Jan Kan, and Suzanne Jak. "Checking the inventory: Illustrating different

methods for individual participant data meta-analytic structural equation modeling". In: *Research Synthesis Methods* 15.6 (Aug. 2024), pp. 872–895. ISSN: 1759-2887. DOI: 10.1002/jrsm.1735.

Abstract: Researchers may have at their disposal the raw data of the studies they wish to meta-analyze. The goal of this study is to identify, illustrate, and compare a range of possible analysis options for researchers to whom raw data are available, wanting to fit a structural equation model (SEM) to these data. This study illustrates techniques that directly analyze the raw data, such as multilevel and multigroup SEM, and techniques based on summary statistics, such as correlation-based meta-analytical structural equation modeling (MASEM), discussing differences in procedures, capabilities, and outcomes. This is done by analyzing a previously published collection of datasets using open source software. A path model reflecting the theory of planned behavior is fitted to these datasets using different techniques involving SEM. Apart from differences in handling of missing data, the ability to include study-level moderators, and conceptualization of heterogeneity, results show differences in parameter estimates and standard errors across methods. Further research is needed to properly formulate guidelines for applied researchers looking to conduct individual participant data MASEM.

Gu et al.: A simple analytic approach to regression with or without interaction terms: Integrating the covariance structure model and the delta method

Gu-Punjatewakupt-Sattayathamrongthian-etal-2026

Fei Gu et al. "A simple analytic approach to regression with or without interaction terms: Integrating the covariance structure model and the delta method". In: *Psychometrika* 91.4 (Sept. 2026), pp. 1426–1478. ISSN: 1860-0980. DOI: 10.1017/psy.2026.10107.

Abstract: Several textbooks give the incorrect formula to calculate the standard error estimates for standardized regression coefficients. As a remedy, two analytic methods have been developed: (1) the delta method and (2) the covariance structure modeling method. However, neither method is applicable to compute the standard error estimates for unstandardized regression coefficients of

products of Z -scores. In the literature, a nonparametric bootstrap procedure is advocated to test the significance of unstandardized regression coefficients of products of Z -scores. In this article, we propose a simple analytic approach that can produce the standard error estimates not only for standardized regression coefficients when interaction terms are not included, but also for unstandardized regression coefficients when interaction terms of Z -scores are included. Two numeric examples are used to compare our analytic approach with the existing methods, and simulation studies are conducted to further evaluate the performances of regular regression, our analytic approach, and the nonparametric bootstrap procedure at finite sample sizes. It is found that (1) regular regression performs well only when the variances of predictor variables are small, (2) our analytic approach performs well at the sample size of 200 or larger, and (3) the nonparametric bootstrap procedure performs (almost) perfectly in all conditions.

Gunn et al.: Characterising patterns of alcohol use among heavy drinkers: A cluster analysis utilising alcohol biosensor data **Gunn-Steingrimsson-Merrill-etal-2021**

Rachel L. Gunn et al. "Characterising patterns of alcohol use among heavy drinkers: A cluster analysis utilising alcohol biosensor data". In: *Drug and Alcohol Review* 40.7 (May 2021), pp. 1155–1164. ISSN: 1465-3362. DOI: 10.1111/dar.13306.

Abstract: Introduction: Previous research has predominately relied on person-level or single characteristics of drinking episodes to characterise patterns of drinking that may confer risk. This research often relies on self-report measures. Advancements in wearable alcohol biosensors provide a multifaceted objective measure of drinking. The current study aimed to characterise drinking episodes using data derived from a wearable alcohol biosensor. Methods: Participants ($n = 45$) were adult heavy drinkers who wore the Secure Continuous Remote Alcohol Monitoring (SCRAM) bracelet and reported on their drinking behaviours. Cluster analysis was used to evaluate unique combinations of alcohol episode characteristics. Associations between clusters and self-reported person and event-level factors were also examined in univariable and multivariable models. Results: Results suggested three unique clusters: Cluster 1 (most common, slowest rate of rise to and decline

from peak), Cluster 2 (highest peak transdermal alcohol concentration and area under the curve) and Cluster 3 (fastest rate of decline from peak). Univariable analyses distinguished Cluster 1 as having fewer self-reported drinks and fewer episodes that occurred on weekends relative to Cluster 2. The effect for number of drinks remained in multivariable analyses. Discussion and Conclusions: This is the first study to characterise drinking patterns at the event-level using objective data. Results suggest that it is possible to distinguish drinking episodes based on several characteristics derived from wearable alcohol biosensors. This examination lays the groundwork for future studies to characterise patterns of drinking and their association with consequences of drinking behaviour.

Hallquist et al.: Problems with centrality measures in psychopathology symptom networks: Why network psychometrics cannot escape psychometric theory

Hallquist-Wright-Molenaar-2021

Michael N. Hallquist, Aidan G. C. Wright, and Peter C. M. Molenaar. "Problems with centrality measures in psychopathology symptom networks: Why network psychometrics cannot escape psychometric theory". In: *Multivariate Behavioral Research* 56.2 (Mar. 2021), pp. 199–223. ISSN: 1532-7906. DOI: 10.1080/00273171.2019.1640103.

Abstract: Understanding patterns of symptom co-occurrence is one of the most difficult challenges in psychopathology research. Do symptoms co-occur because of a latent factor, or might they directly and causally influence one another? Motivated by such questions, there has been a surge of interest in network analyses that emphasize the putatively direct role symptoms play in influencing each other. In this critical paper, we highlight conceptual and statistical problems with using centrality measures in cross-sectional networks. In particular, common network analyses assume that there are no unmodeled latent variables that confound symptom co-occurrence. The traditions of clinical taxonomy and test development in psychometric theory, however, greatly increase the possibility that latent variables exist in symptom data. In simulations that include latent variables, we demonstrate that closeness and betweenness are vulnerable to spurious covariance among symptoms that connect subgraphs (e.g., diagnoses). We further show that strength is redundant

with factor loading in several cases. Finally, if a symptom reflects multiple latent causes, centrality metrics reflect a weighted combination, undermining their interpretability in empirical data. Our results suggest that it is essential for network psychometric approaches to examine the evidence for latent variables prior to analyzing or interpreting patterns at the symptom level. Failing to do so risks identifying spurious relationships or failing to detect causally important effects. Altogether, we argue that centrality measures do not provide solid ground for understanding the structure of psychopathology when latent confounding exists.

Hamaker: Analysis of intensive longitudinal data: Putting psychological processes in perspective **Hamaker-2025**

Ellen L. Hamaker. "Analysis of intensive longitudinal data: Putting psychological processes in perspective". In: *Annual Review of Clinical Psychology* 21.1 (May 2025), pp. 379–405. ISSN: 1548-5951. DOI: 10.1146/annurev-clinpsy-081423-022947.

Abstract: Research based on intensive longitudinal data (ILD)—consisting of many repeated measures from one or multiple individuals—is rapidly gaining popularity in psychological science. To appreciate the unique potential of ILD research for clinical psychology, this review begins by examining how our three traditional research approaches fall short when the goal is to investigate processes. It then explores how the analysis of ILD can be used to study a process as it unfolds within a specific person over time but also to study average process features or individual differences therein. By emphasizing the alignment between research questions, data collection, and analytical strategies, the potential of ILD research is further highlighted. It is argued that for future progress it is essential to integrate machine learning and causal inference methods with statistical techniques for ILD and to become more explicit about timescales, time frames, and dynamics in psychological theories.

Hamaker et al.: The fixed versus random effects debate and how it relates to centering in multilevel modeling **Hamaker-Muthen-2020**

Ellen L. Hamaker and Bengt Muthén. "The fixed versus random effects debate and how it relates to centering in multilevel modeling". In: *Psychological Methods* 25.3 (June 2020), pp. 365–379. ISSN: 1082-989X. DOI: 10.1037/met0000239.

Abstract: In many disciplines researchers use longitudinal panel data to investigate the potentially causal relationship between 2 variables. However, the conventions and concerns vary widely across disciplines. Here we focus on 2 concerns, that is: (a) the concern about random effects versus fixed effects, which is central in the (micro)econometrics/sociology literature; and (b) the concern about grand mean versus group (or person) mean centering, which is central in the multilevel literature associated with disciplines like psychology and educational sciences. We show that these 2 concerns are actually addressing the same underlying issue. We discuss diverse modeling methods based on either multilevel regression modeling with the data in long format, or structural equation modeling with the data in wide format, and compare these approaches with simulated data. We extend the multilevel model with random slopes and discuss the consequences of this. Subsequently, we provide guidelines on how to choose between the diverse modeling options. We illustrate the use of these guidelines with an empirical example based on intensive longitudinal data, in which we consider both a time-varying and a time-invariant covariate.

Haslbeck et al.: Recovering within-person dynamics from psychological time series **Haslbeck-Ryan-2022**

Jonas M. B. Haslbeck and Oisin Ryan. "Recovering within-person dynamics from psychological time series". In: *Multivariate Behavioral Research* 57.5 (Sept. 2022), pp. 735–766. ISSN: 1532-7906. DOI: 10.1080/00273171.2021.1896353.

Abstract: Idiographic modeling is rapidly gaining popularity, promising to tap into the within-person dynamics underlying psychological phenomena. To gain theoretical understanding of these

dynamics, we need to make inferences from time series models about the underlying system. Such inferences are subject to two challenges: first, time series models will arguably always be misspecified, meaning it is unclear how to make inferences to the underlying system; and second, the sampling frequency must be sufficient to capture the dynamics of interest. We discuss both problems with the following approach: we specify a toy model for emotion dynamics as the true system, generate time series data from it, and then try to recover that system with the most popular time series analysis tools. We show that making straightforward inferences from time series models about an underlying system is difficult. We also show that if the sampling frequency is insufficient, the dynamics of interest cannot be recovered. However, we also show that global characteristics of the system can be recovered reliably. We conclude by discussing the consequences of our findings for idiographic modeling and suggest a modeling methodology that goes beyond fitting time series models alone and puts formal theories at the center of theory development.

Hecht et al.: Continuous-time modeling in prevention research: An illustration

Hecht-Voelkle-2021

Martin Hecht and Manuel C. Voelkle. "Continuous-time modeling in prevention research: An illustration". In: *International Journal of Behavioral Development* 45.1 (Jan. 2021), pp. 19–27. ISSN: 1464-0651. DOI: 10.1177/0165025419885026.

Abstract: The analysis of cross-lagged relationships is a popular approach in prevention research to explore the dynamics between constructs over time. However, a limitation of commonly used cross-lagged models is the requirement of equally spaced measurement occasions that prevents the usage of flexible longitudinal designs and complicates cross-study comparisons. Continuous-time modeling overcomes these limitations. In this article, we illustrate the use of continuous-time models using Bayesian and frequentist approaches to model estimation. As an empirical example, we study the dynamic interplay of physical activity and health, a classic research topic in prevention science, using data from the "Midlife in the United States (MIDUS 2): Daily Stress Project, 2004-2009."

To help prevention researchers in adopting the approach, we provide annotated R scripts and a simulated data set based on the results from analyzing the MIDUS 2 data.

Hecht et al.: A computationally more efficient Bayesian approach for estimating continuous-time models **Hecht-Zitzmann-2020**

Martin Hecht and Steffen Zitzmann. "A computationally more efficient Bayesian approach for estimating continuous-time models". In: *Structural Equation Modeling: A Multidisciplinary Journal* 27.6 (Mar. 2020), pp. 829–840. ISSN: 1532-8007. DOI: 10.1080/10705511.2020.1719107.

Abstract: Continuous-time modeling is gaining in popularity as more and more intensive longitudinal data need to be analyzed. Current Bayesian software implementations of continuous-time models suffer from rather high, inadequate run times. Therefore, we apply a model reformulation approach to reduce run time. In a simulation study, we investigate the estimation quality and run time gain. We then illustrate our optimized Bayesian continuous-time model estimation and compare it to established continuous-time modeling software using an empirical example. Parameter estimates and inference statistics were very comparable, while run times were very different. Our approach reduces the run times for Bayesian estimations of continuous-time models from hours to minutes.

Hecht et al.: Exploring the unfolding of dynamic effects with continuous-time models: Recommendations concerning statistical power to detect peak cross-lagged effects **Hecht-Zitzmann-2021b**

Martin Hecht and Steffen Zitzmann. "Exploring the unfolding of dynamic effects with continuous-time models: Recommendations concerning statistical power to detect peak cross-lagged effects". In: *Structural Equation Modeling: A Multidisciplinary Journal* 28.6 (May 2021), pp. 894–902. ISSN: 1532-8007. DOI: 10.1080/10705511.2021.1914627.

Abstract: Cross-lagged panel models have been commonly applied to investigate the dynamic interplay of variables. In such discrete-time models, the size of the cross-lagged effects depends on the length of the time interval between the measurement occasions. Continuous-time modeling allows to explore this interval dependence of cross-lagged effects and thus to identify the maximal "peak" cross-lagged effects. To detect these peak effects, sufficient statistical power is needed. Based on results from a simulation study, we employed machine learning algorithms to identify a highly accurate prediction model. Results are incorporated into a Shiny App (available at https://psychtools.shinyapps.io/ ContinuousTimePowerCalculation) for easy power calculations. Although limitations apply, our results might be helpful for study planning.

Hecht et al.: Sample size recommendations for continuous-time models: Compensating shorter time series with larger numbers of persons and vice versa

Hecht-Zitzmann-2021a

Martin Hecht and Steffen Zitzmann. "Sample size recommendations for continuous-time models: Compensating shorter time series with larger numbers of persons and vice versa". In: *Structural Equation Modeling: A Multidisciplinary Journal* 28.2 (Mar. 2021), pp. 229–236. ISSN: 1532-8007. DOI: 10.1080/10705511.2020.1779069.

Abstract: Autoregressive modeling has traditionally been concerned with time-series data from one unit (N = 1). For short time series (T ¡ 50), estimation performance problems are well studied and documented. Fortunately, in psychological and social science research, besides T, another source of information is often available for model estimation, that is, the persons (N ¿ 1). In this work, we illustrate the N/T compensation effect: With an increasing number of persons N at constant T, the model estimation performance increases, and vice versa, with an increasing number of time points T at constant N, the performance increases as well. Based on these observations, we develop sample size recommendations in the form of easily accessible N/T heatmaps for two popular autoregressive continuous-time models.

Hisler et al.: Neuroticism as the intensity, reactivity, and variability in day-to-day affect **Hisler-Krizan-DeHart-etal-2020**

Garrett C. Hisler et al. "Neuroticism as the intensity, reactivity, and variability in day-to-day affect". In: *Journal of Research in Personality* 87 (Aug. 2020), p. 103964. ISSN: 0092-6566. DOI: 10.1016/j.jrp.2020.103964.

Abstract: Neuroticism has been linked to typical levels of affect, affect reactivity to negative events, and variability in affect over time. However, the intercorrelations among these characteristics make it unclear whether neuroticism reflects unique variance in each of these aspects of emotional life. Data from two daily-diary samples revealed that neuroticism was associated with average levels and variability of positive and negative affect and reactivity of negative affect to stressors, but was only uniquely related to mean levels of positive and negative affect. Findings highlight the substantial overlap in affect indices, suggesting that mean levels of affect, at the very least, are at the core of neuroticism, and reveal the need for further research using more nuanced approaches.

Hoekstra et al.: Heterogeneity in individual network analysis: Reality or illusion? **Hoekstra-Epskamp-Borsboom-2023**

Ria H. A. Hoekstra, Sacha Epskamp, and Denny Borsboom. "Heterogeneity in individual network analysis: Reality or illusion?" In: *Multivariate Behavioral Research* 58.4 (July 2023), pp. 762–786. ISSN: 1532-7906. DOI: 10.1080/00273171.2022.2128020.

Abstract: The use of idiographic research techniques has gained popularity within psychological research and network analysis in particular. Idiographic research has been proposed as a promising avenue for future research, with differences between idiographic results highlighting evidence for radical heterogeneity. However, in the quest to address the individual in psychology, some classic statistical problems, such as those arising from sampling variation and power limitations, should not be overlooked. This article aims to determine to what extent current tools to compare idiographic networks are suited to disentangle true from illusory heterogeneity in the presence of sampling

error. To this end, we investigate the performance of tools to inspect heterogeneity (visual inspection, comparison of centrality measures, investigating standard deviations of random effects, and GIMME) through simulations. Results show that power limitations hamper the validity of conclusions regarding heterogeneity and that the power required to assess heterogeneity adequately is often not realized in current research practice. Of the tools investigated, inspecting standard deviations of random effects and GIMME proved the most suited. However, all tools evaluated leave the door wide open to misinterpret all observed variability in terms of individual differences. Hence, the current paper calls for caution in the use and interpretation of new time-series techniques when it comes to heterogeneity.

Horstmann et al.: Assessing personality states: What to consider when constructing personality state measures **Horstmann-Ziegler-2020**

Kai T. Horstmann and Matthias Ziegler. "Assessing personality states: What to consider when constructing personality state measures". In: *European Journal of Personality* 34.6 (Dec. 2020), pp. 1037–1059. ISSN: 1099-0984. DOI: 10.1002/per.2266.

Abstract: Repeated assessments of personality states in daily diary or experience sampling studies have become a more and more common tool in the psychologist's toolbox. However, and contrary to the widely available literature on personality traits, no best practices for the development of personality state measures exist, and personality state measures have been developed in many different ways. To address this, we first define what a personality state is and discuss important components. On the basis of this, we define what a personality state measure is and suggest a general guideline for the development of such measures. Following the ABC of test construction can then guide the strategy for obtaining validity and reliability evidence: (A) What is the construct being measured? (B) What is the intended purpose of the measure? And (C) What is the targeted population of persons and situations? We then conclude with an example by developing an initial item pool for the assessment of conscientiousness personality states.

Hunter: State space mixture modeling: Finding people with similar patterns of change **Hunter-2024**

Michael D. Hunter. "State space mixture modeling: Finding people with similar patterns of change". In: *Multivariate Behavioral Research* 59.6 (Nov. 2024), pp. 1253–1269. ISSN: 1532-7906. DOI: 10.1080/00273171.2023.2261224.

Abstract: Increasingly, behavioral scientists encounter data where several individuals were measured on multiple variables over numerous occasions. Many current methods combine these data, assuming all individuals are randomly equivalent. An extreme alternative assumes no one is randomly equivalent. We propose state space mixture modeling as one possible compromise. State space mixture modeling assumes that unknown groups of people exist who share the same parameters of a state space model, and simultaneously estimates both the state space parameters and group membership. The goal is to find people that are undergoing similar change processes over time. The present work demonstrates state space mixture modeling on a simulated data set, and summarizes the results from a large simulation study. The illustration shows how the analysis is conducted, whereas the simulation provides evidence of its general validity and applicability. In the simulation study, sample size had the greatest influence on parameter estimation and the dimension of the change process had the greatest impact on correctly grouping people together, likely due to the distinctiveness of their patterns of change. State space mixture modeling offers one of the best-performing methods for simultaneously drawing conclusions about individual change processes while also analyzing multiple people.

Hunter et al.: What ergodicity means for you **Hunter-Fisher-Geier-2024**

Michael D. Hunter, Zachary F. Fisher, and Charles F. Geier. "What ergodicity means for you". In: *Developmental Cognitive Neuroscience* 68 (Aug. 2024), p. 101406. ISSN: 1878-9293. DOI: 10.1016/j.dcn.2024.101406.

Abstract: This paper explores the relation between within-person and between-person research designs using the concept of ergodicity from statistical mechanics in physics. We demonstrate the consequences of ergodicity using several real data examples from previously published studies. We then create several simulated examples that illustrate the independence of within-person processes from between-person differences, and pair these examples with analytic results that reinforce our conclusions. Finally, we discuss the plausibility of ergodicity being the general rule rather than the exception for social and behavioral processes, address common arguments against heeding the implications of ergodicity for behavioral research, and offer several possible solutions.

Ion et al.: Events-affect-personality: A daily diary investigation of the mediating effects of affect on the events-personality relationship **Ion-Georgescu-Iliescu-etal-2025**

Andrei Ion et al. "Events-affect-personality: A daily diary investigation of the mediating effects of affect on the events-personality relationship". In: *Psychological Reports* 128.3 (June 2025), pp. 1861–1886. ISSN: 1558-691X. DOI: 10.1177/00332941231175363.

Abstract: Our 10-day diary investigation anchored in dynamic personality theories, such as Whole Trait Theory examined (a) whether within-person variability in two broad personality traits Extraversion and Neuroticism is consistently predicted by daily events, (b) whether positive and negative affect, respectively partly mediate this relationship and (c) the lagged relationships between events, and next day variations in affect and personality. Results revealed that personality exhibited significant within-person variability, that positive and negative affect partly mediate the relationship between events and personality, affect accounting for up to 60% of the effects of events on personality. Additionally, we identified that event-affect congruency was accountable for larger effects compared to event-affect non-congruency.

Jak et al.: Meta-analytic structural equation modeling with moderating effects on SEM parameters **Jak-Cheung-2020**

Suzanne Jak and Mike W.-L. Cheung. "Meta-analytic structural equation modeling with moderating effects on SEM parameters". In: *Psychological Methods* 25.4 (Aug. 2020), pp. 430–455. ISSN: 1082-989X. DOI: 10.1037/met0000245.

Abstract: Meta-analytic structural equation modeling (MASEM) is an increasingly popular meta-analytic technique that combines the strengths of meta-analysis and structural equation modeling. MASEM facilitates the evaluation of complete theoretical models (e.g., path models or factor analytic models), accounts for sampling covariance between effect sizes, and provides measures of overall fit of the hypothesized model on meta-analytic data. We propose a novel MASEM method, one-stage MASEM, which is better suitable to explain study-level heterogeneity than existing methods. One-stage MASEM allows researchers to incorporate continuous or categorical moderators into the MASEM, in which any parameter in the structural equation model (e.g., path coefficients and factor loadings) can be modeled by the moderator variable, while the method does not require complete data for the primary studies included in the meta-analysis. We illustrate the new method on two real data sets, evaluate its empirical performance via a computer simulation study, and provide user-friendly R-functions and annotated syntax to assist researchers in applying one-stage MASEM. We close the article by presenting several future research directions.

Kang et al.: Accurate confidence and Bayesian interval estimation for non-centrality parameters and effect size indices **Kang-Jones-Armstrong-etal-2023**

Kaidi Kang et al. "Accurate confidence and Bayesian interval estimation for non-centrality parameters and effect size indices". In: *Psychometrika* 88.1 (Mar. 2023), pp. 253–273. ISSN: 1860-0980. DOI: 10.1007/s11336-022-09899-x.

Abstract: Reporting effect size index estimates with their confidence intervals (CIs) can be an excellent way to simultaneously communicate the strength and precision of the observed evidence.

We recently proposed a robust effect size index (RESI) that is advantageous over common indices because it's widely applicable to different types of data. Here, we use statistical theory and simulations to develop and evaluate RESI estimators and confidence/credible intervals that rely on different covariance estimators. Our results show (1) counter to intuition, the randomness of covariates reduces coverage for Chi-squared and F CIs; (2) when the variance of the estimators is estimated, the non-central Chi-squared and F CIs using the parametric and robust RESI estimators fail to cover the true effect size at the nominal level. Using the robust estimator along with the proposed nonparametric bootstrap or Bayesian (credible) intervals provides valid inference for the RESI, even when model assumptions may be violated. This work forms a unified effect size reporting procedure, such that effect sizes with confidence/credible intervals can be easily reported in an analysis of variance (ANOVA) table format.

Kraiss et al.: Idiographic bidirectional associations of stressfulness of events and negative affect in daily life as indicators for mental health: An experience sampling study

Kraiss-Vaessen-Klooster-2024

Jannis T. Kraiss, Thomas Vaessen, and Peter M. ten Klooster. "Idiographic bidirectional associations of stressfulness of events and negative affect in daily life as indicators for mental health: An experience sampling study". In: *Stress and Health* 40.5 (May 2024). ISSN: 1532-2998. DOI: 10.1002/smi.3433.

Abstract: Evidence suggests that complex micro-dynamics occurring in daily life underly the development of mental distress. We aimed to (1) study the cross-lagged association between stressful events and negative affect (NA), (2) show that there is substantial between-person variability in idiographic associations and (3) show that idiographic associations are indicative of mental health. Experience sampling study assessing perceived stressfulness of events (PSE) and NA four times per day for 2 weeks in a non-clinical convenience sample ($N = 70$, mean age = 22.9, 61% female, 69% German). Bivariate vector autoregressive model implemented in dynamic structural equation modelling to model the associations between stressful events and NA and obtain idiographic asso-

ciations. Stressfulness of events and NA were significantly reciprocally associated. Autocorrelations and cross-lagged associations from PSE to NA showed substantial variability and were significantly related with trait measures of depression, anxiety, well-being, and perceived stress. Contrary to expectations, cross-lagged associations from NA to stressfulness of events were not related to trait mental health. The approach outlined in this article is useful for studying idiographic dynamics in daily life. The findings increase our understanding of micro-dynamics underlying mental health and individual differences in these processes.

Kuiper et al.: Meta-analysis of lagged regression models: A continuous-time approach

Kuiper-Ryan-2020

Rebecca M. Kuiper and Oisin Ryan. "Meta-analysis of lagged regression models: A continuous-time approach". In: *Structural Equation Modeling: A Multidisciplinary Journal* 27.3 (May 2020), pp. 396–413. ISSN: 1532-8007. DOI: 10.1080/10705511.2019.1652613.

Abstract: In science, the gold standard for evidence is an empirical result which is consistent across multiple studies. Meta-analysis techniques allow researchers to combine the results of different studies. Due to the increasing availability of longitudinal data, studying lagged effects is increasingly popular also in meta-analytic studies. However, in current practice, little attention is paid to the unique challenges of meta-analyzing these lagged effects. Namely, it is well known that lagged effects estimates change depending on the time that elapses between measurement waves. This means that studies that use different uniform time intervals between observations (e.g., 1 hour vs 3 hours or 1 month vs 2 months) can come to very different parameter estimates, and seemingly contradictory conclusions, about the same underlying process. In this article, we introduce, describe, and illustrate a new meta-analysis method (CTmeta) which assumes an underlying continuous-time process, and compare it with current practice.

Kury et al.: Understanding the affective and mental health outcomes of meditation interventions: The role of individual differences in self-compassion

Kury-Fredrickson-VanCappellen-etal-2026

Manuella B. Kury et al. "Understanding the affective and mental health outcomes of meditation interventions: The role of individual differences in self-compassion". In: *Mindfulness* 17 (May 2026), pp. 1497–1513. DOI: 10.1007/s12671-026-02803-z.

Abstract: Meditation interventions can benefit emotional, social, and mental health, but their effects vary across individuals. This preregistered study tested whether baseline self-compassion moderated responses to mindfulness meditation and loving-kindness meditation. Data came from 217 adults who completed six weeks of meditation training and reported emotions, social connectedness, and depressive symptoms. Participants with lower initial self-compassion generally experienced larger reductions in guilt, shame, and depression and greater increases in social connectedness across the intervention. Increases in self-compassion mediated reductions in depression among participants who began with lower self-compassion. Baseline self-compassion did not consistently moderate associations between meditation on a specific day and that day's outcomes. The results suggest that people lower in self-compassion may derive especially strong longer-term benefits from meditation training.

Lai et al.: Two-stage path analysis with definition variables: An alternative framework to account for measurement error

Lai-Hsiao-2022

Mark H. C. Lai and Yu-Yu Hsiao. "Two-stage path analysis with definition variables: An alternative framework to account for measurement error". In: *Psychological Methods* 27.4 (Aug. 2022), pp. 568–588. ISSN: 1082-989X. DOI: 10.1037/met0000410.

Abstract: When estimating path coefficients among psychological constructs measured with error, structural equation modeling (SEM), which simultaneously estimates the measurement and structural parameters, is generally regarded as the gold standard. In practice, however, researchers

usually first compute composite scores or factor scores, and use those as observed variables in a path analysis, for purposes of simplifying the model or avoiding model convergence issues. Whereas recent approaches, such as reliability adjustment methods and factor score regression, has been proposed to mitigate the bias induced by ignoring measurement error in composite/factor scores with continuous indicators, those approaches are not yet applicable to models with categorical indicators. In this article, we introduce the two-stage path analysis (2S-PA) with definition variables as a general framework for path modeling to handle categorical indicators, in which estimation of factor scores and path coefficients are separated. It thus allows for different estimation methods in the measurement and the structural path models and easier diagnoses of violations of model assumptions. We conducted three simulation studies, ranging from latent regression to mediation analysis with categorical indicators, and showed that 2S-PA generally produced similar estimates to those using SEM in large samples, but gave better convergence rates, less standard error bias, and better control of Type I error rates in small samples. We illustrate 2S-PA using data from a national data set, and show how researchers can implement it in Mplus and OpenMx. Possible extensions and future directions of 2S-PA are discussed.

Le et al.: Temporal bidirectional dynamics of momentary stress and affect: Dynamic structural equation modelling of ecological momentary assessments

Le-Sun-Yap-etal-2026

Flora Le et al. "Temporal bidirectional dynamics of momentary stress and affect: Dynamic structural equation modelling of ecological momentary assessments". In: *Journal of Affective Disorders* 396 (Mar. 2026), p. 120810. ISSN: 0165-0327. DOI: 10.1016/j.jad.2025.120810.

Abstract: The process model of emotion regulation highlights affect's ebb and flow in daily life in response to external events or internal processes, such as stress. Collectively, daily stress and affect can shape daily experiences and influence long-term health. Understanding the dynamics of the stress-affect relationship requires examining intensity (average level), inertia (autoregression), and variability (residual variances), yet few studies simultaneously consider both arousal (high vs. low,

denoted as H vs. L) and valence (positive vs. negative, denoted as PA vs. NA) dimensions of affect. Participants ($N = 424$; 72.4% women, $M_{\text{age}} = 22.79 \pm 5.35$) came from three longitudinal ecological momentary assessment studies (completed between 2017 and 2020) across 7–15 days. Dynamic structural equation modelling tested the dynamics between stress and affect across valence and arousal, and their bidirectional relationships. Results showed that individuals with higher stress intensity experienced higher HNA and LNA and, to a lesser extent, lower HPA and LPA intensity. Individuals with higher stress intensity also had higher overall stress variability and less stress in response to PA. Higher average HNA was associated with higher stress variability, whereas higher average LNA was associated with higher affect variability. Additionally, higher HNA inertia was associated with more stress in response to HNA and less HNA reactivity to stress. Our findings support the bidirectional and dynamic nature of stress-affect relationships and the value of incorporating nuanced aspects of stress and affect into future research across populations and clinical intervention frameworks.

Le Bourdonnec et al.: Continuous-time mediation analysis for repeatedly measured mediators and outcomes **LeBourdonnec-Valeri-ProustLima-2025**

Kateline Le Bourdonnec, Linda Valeri, and Cécile Proust-Lima. "Continuous-time mediation analysis for repeatedly measured mediators and outcomes". In: *Biometrics* 81.2 (Apr. 2025). ISSN: 1541-0420. DOI: 10.1093/biomtc/ujaf062.

Abstract: Mediation analysis aims to decipher the underlying causal mechanisms between an exposure, an outcome, and intermediate variables called mediators. Initially developed for a fixed-time mediator and outcome, it has been extended to the framework of longitudinal data by discretizing the assessment times of the mediator and outcome. Yet, processes in play in longitudinal studies are usually defined in continuous time and measured at irregular and subject-specific visits. This is the case in dementia research when cerebral and cognitive changes measured at planned visits in cohorts are of interest. We thus propose a methodology to estimate the causal mechanisms between a time-fixed exposure (X), a mediator process ($\mathcal{M}_t$), and an outcome process ($\mathcal{Y}_t$), both measured

repeatedly over time in the presence of a time-dependent confounding process ($\mathcal{L}_t$). We consider 2 types of causal estimands: natural effects and path-specific effects. We provide identifiability assumptions, and we employ a multivariate mixed model based on differential equations for their estimation. The performances of the method are assessed in simulations, and the method is illustrated in 2 real-world examples motivated by the 3C cerebral aging study to assess (1) the effect of educational level on functional dependency through depressive symptomatology and cognitive functioning and (2) the effect of a genetic factor on cognitive functioning potentially mediated by vascular brain lesions and confounded by neurodegeneration.

D. Lee et al.: Do I even need a multilevel model? Negligible effect significance testing for intraclass correlation coefficients **Lee-Hancock-2026**

Dayeon Lee and Gregory R. Hancock. "Do I even need a multilevel model? Negligible effect significance testing for intraclass correlation coefficients". In: *Journal of Educational and Behavioral Statistics* (Aug. 2026). ISSN: 1935-1054. DOI: 10.3102/10769986261478956.

Abstract: In applied research, the decision to utilize multilevel modeling (MLM) is commonly guided by comparing a point estimate of the unconditional intraclass correlation coefficient (ICC) against some recommended threshold (e.g., 0.05). This naïve approach, however, fails to account for sampling uncertainty and provides no formal inferential justification for a decision. The current work introduces negligible effect significance testing (or equivalence testing) for the ICC, proposing a framework that performs this test using a pivotal quantity of the ICC based on the F statistic. Furthermore, we offer guidance to help researchers define "negligible" that utilizes the design effect to characterize a tolerable level of variance inflation. R code illustrating the proposed procedure is provided.

S. A. W. Lee et al.: From the individual to the group: Using idiographic analyses and two-stage random effects meta-analysis to obtain population level inferences for within-person processes **Lee-Gates-2024**

Sandra A. W. Lee and Kathleen M. Gates. "From the individual to the group: Using idiographic analyses and two-stage random effects meta-analysis to obtain population level inferences for within-person processes". In: *Multivariate Behavioral Research* 59.6 (Nov. 2024), pp. 1220–1239. ISSN: 1532-7906. DOI: 10.1080/00273171.2023.2229310.

Abstract: In psychology, the use of portable technology and wearable devices to ease participant burden in data collection is on the rise. This creates increased interest in collecting real-time or near real-time data from individuals within their natural environments. As a result, vast amounts of observational time series data are generated. Often, motivation for collecting this data hinges on understanding within-person processes that underlie psychological phenomena. Motivated by the body of Dr. Peter Molenaar's life work calling for analytical approaches that consider potential heterogeneity and non-ergodicity, the focus of this paper is on using idiographic analyses to generate population inferences for within-person processes. Meta-analysis techniques using one-stage and two-stage random effects meta-analysis as implemented in single-case experimental designs are presented. The case for preferring a two-stage approach for meta-analysis of single-subject observational time series data is made and demonstrated using an empirical example. This provides a novel implementation of the methodology as prior implementations focus on applications to short time series with experimental designs. Inspired by Dr. Molenaar's work, we describe how an approach, two-stage random effects meta-analysis (2SRE-MA), aligns with recent calls to consider idiographic approaches when making population-level inferences regarding within-person processes.

S. A. Lee et al.: The mediating role of daily affect dynamics in bidirectional and longitudinal associations between depressive symptoms and inflammation

Lee-GrahamEngeland-Lee-etal-2026

Sun Ah Lee et al. "The mediating role of daily affect dynamics in bidirectional and longitudinal associations between depressive symptoms and inflammation". In: *Biopsychosocial Science and Medicine* 88.6 (May 2026), pp. 521–534. ISSN: 2998-8756. DOI: 10.1097/psy.0000000000001494.

Abstract: Objectives: Extensive research has established relationships between depression and inflammation, yet greater attention is needed to elucidate mechanisms through which these processes unfold across multiple levels of analysis and time scales. This cohort panel study evaluated the role of daily affect dynamics in bidirectional and longitudinal associations between depressive symptoms and inflammation. Methods: Using data from the second and third waves of the Midlife in the United States (MIDUS) study, the sample included 563 adults (M_{age} = 52.4; 57% female; 84% White). During the Biomarker Project, depressive symptoms were assessed at a single time point using the CESD-20, and the inflammatory markers were collected including CRP and four cytokines (IL-6, IL-8, TNFα, IL-10). During the Daily Diary Project, daily affect and stress were measured over eight consecutive days. Two affect dynamics indicators were specified: affective variability and reactivity to daily stressors. Multilevel structural equation models simultaneously tested autoregressive and cross-lagged associations between depressive symptoms and inflammation at the person-level, and day-level mediated paths involving daily affect dynamics. Results: Results revealed that affective variability and reactivity mediated the autoregressive paths of depressive symptoms across two waves. Positive affective variability mediated the pathway from heightened baseline depressive symptoms to higher CRP levels at follow-up. Negative affective reactivity to daily stressors mediated the pathway from higher baseline CRP to subsequent depressive symptoms. Conclusions: Findings suggest daily affective processes as potential mechanisms linking depressive symptoms and inflammation, underscoring the significance of promoting affective stability and adaptive stress responses in everyday life.

Levi et al.: The daily dose-response hypothesis of mindfulness meditation practice: An experience sampling study **Levi-Shoham-Amir-etal-2021**

Karin Levi et al. "The daily dose-response hypothesis of mindfulness meditation practice: An experience sampling study". In: *Psychosomatic Medicine* 83.6 (July 2021), pp. 624–630. DOI: 10.1097/PSY.0000000000000912.

Abstract: This study examined whether the amount of daily mindfulness meditation practice predicts temporally proximal outcomes during an intervention. Eighty-two meditation-naive adults completed intensive assessments of home practice, state mindfulness, decentering, emotional valence, and arousal during a 21-day mindfulness-based intervention. Total accumulated practice did not predict the assessed outcomes at postintervention. At the daily level, however, more practice was associated with greater same-day mindfulness, decentering, and positive emotional valence, but not arousal. These associations did not persist into the following day. The results support a daily dose-response process in which the benefits associated with home practice are reliable but short-lived rather than cumulative.

Y. Li et al.: Multiple imputation with factor scores: A practical approach for handling simultaneous missingness across items in longitudinal designs **Li-Oravecz-Ji-etal-2025**

Yanling Li, Zita Oravecz, Linying Ji, et al. "Multiple imputation with factor scores: A practical approach for handling simultaneous missingness across items in longitudinal designs". In: *Multivariate Behavioral Research* 60.1 (Jan. 2025), pp. 61–89. ISSN: 1532-7906. DOI: 10.1080/00273171.2024.2371816.

Abstract: Missingness in intensive longitudinal data triggered by latent factors constitute one type of nonignorable missingness that can generate simultaneous missingness across multiple items on each measurement occasion. To address this issue, we propose a multiple imputation (MI) strategy called MI-FS, which incorporates factor scores, lag/lead variables, and missing data indicators into the imputation model. In the context of process factor analysis (PFA), we conducted a Monte

Carlo simulation study to compare the performance of MI-FS to listwise deletion (LD), MI with manifest variables (MI-MV, which implements MI on both dependent variables and covariates), and partial MI with MVs (PMI-MV, which implements MI on covariates and handles missing dependent variables via full-information maximum likelihood) under different conditions. Across conditions, we found MI-based methods overall outperformed the LD; the MI-FS approach yielded lower root mean square errors (RMSEs) and higher coverage rates for auto-regression (AR) parameters compared to MI-MV; and the PMI-MV and MI-MV approaches yielded higher coverage rates for most parameters except AR parameters compared to MI-FS. These approaches were also compared using an empirical example investigating the relationships between negative affect and perceived stress over time. Recommendations on when and how to incorporate factor scores into MI processes were discussed.

Y. Li et al.: Bayesian forecasting with a regime-switching zero-inflated multilevel poisson regression model: An application to adolescent alcohol use with spatial covariates

Li-Oravecz-Zhou-etal-2022

Yanling Li, Zita Oravecz, Shuai Zhou, et al. "Bayesian forecasting with a regime-switching zero-inflated multilevel poisson regression model: An application to adolescent alcohol use with spatial covariates". In: *Psychometrika* 87.2 (Jan. 2022), pp. 376–402. DOI: 10.1007/s11336-021-09831-9.

Abstract: In this paper, we present and evaluate a novel Bayesian regime-switching zero-inflated multilevel Poisson (RS-ZIMLP) regression model for forecasting alcohol use dynamics. The model partitions individuals' data into two phases, known as regimes, with: (1) a zero-inflation regime that is used to accommodate high instances of zeros (non-drinking) and (2) a multilevel Poisson regression regime in which variations in individuals' log-transformed average rates of alcohol use are captured by means of an autoregressive process with exogenous predictors and a person-specific intercept. The times at which individuals are in each regime are unknown, but may be estimated from the data. We assume that the regime indicator follows a first-order Markov process as related to exogenous predictors of interest. The forecast performance of the proposed model was evaluated using a Monte Carlo simulation study and further demonstrated using substance use and spatial

covariate data from the Colorado Online Twin Study (CoTwins). Results showed that the proposed model yielded better forecast performance compared to a baseline model which predicted all cases as non-drinking and a reduced ZIMLP model without the RS structure, as indicated by higher AUC (the area under the receiver operating characteristic (ROC) curve) scores, and lower mean absolute errors (MAEs) and root-mean-square errors (RMSEs). The improvements in forecast performance were even more pronounced when we limited the comparisons to participants who showed at least one instance of transition to drinking.

Y. Li et al.: A growth of hierarchical autoregression model for capturing individual differences in changes of dynamic characteristics of psychological processes

Li-Williams-Muth-etal-2025

Yanling Li, Lindy Williams, et al. "A growth of hierarchical autoregression model for capturing individual differences in changes of dynamic characteristics of psychological processes". In: *Structural Equation Modeling: A Multidisciplinary Journal* 32.2 (Mar. 2025), pp. 237–250. ISSN: 1532-8007. DOI: 10.1080/10705511.2024.2402328.

Abstract: Intensive longitudinal studies sometimes include multiple measurement phases or bursts, making it possible to investigate psychological change at several timescales. Existing combinations of growth curve and autoregressive models do not always make full use of known phase structure when modeling such data. The authors propose a Bayesian growth of hierarchical autoregression model that integrates autoregressive processes with growth curves to describe changes in multiple dynamic characteristics, including baseline levels, variability, and regulation, while also estimating individual differences in these changes. The model is applied to ecological momentary assessment data collected across a mobile health intervention. Results for meaning in life suggested decreasing fluctuations and increasing regulation over time, although the intervention itself did not provide a credible additional benefit. The approach provides a unified framework for examining developmental changes in psychological dynamics across individuals and study phases.

Y. Li et al.: Fitting multilevel vector autoregressive models in Stan, JAGS, and Mplus

Li-Wood-Ji-etal-2022

Yanling Li, Julie Wood, et al. "Fitting multilevel vector autoregressive models in Stan, JAGS, and Mplus". In: *Structural Equation Modeling: A Multidisciplinary Journal* 29.3 (May 2022), pp. 452–475. DOI: 10.1080/10705511.2021.1911657.

Abstract: The influx of intensive longitudinal data creates a pressing need for complex modeling tools that help enrich our understanding of how individuals change over time. Multilevel vector autoregressive (mlVAR) models allow for simultaneous evaluations of reciprocal linkages between dynamic processes and individual differences, and have gained increased recognition in recent years. High-dimensional and other complex variations of mlVAR models, though often computationally intractable in the frequentist framework, can be readily handled using Markov chain Monte Carlo techniques in a Bayesian framework. However, researchers in social science fields may be unfamiliar with ways to capitalize on recent developments in Bayesian software programs. In this paper, we provide step-by-step illustrations and comparisons of options to fit Bayesian mlVAR models using Stan, JAGS and Mplus, supplemented with a Monte Carlo simulation study. An empirical example is used to demonstrate the utility of mlVAR models in studying intra- and inter-individual variations in affective dynamics.

S. Liu et al.: On the use of empirical Bayes estimates as measures of individual traits

Liu-Kuppens-Bringmann-2021

Siwei Liu, Peter Kuppens, and Laura Bringmann. "On the use of empirical Bayes estimates as measures of individual traits". In: *Assessment* 28.3 (Apr. 2021), pp. 845–857. ISSN: 1552-3489. DOI: 10.1177/1073191119885019.

Abstract: Empirical Bayes (EB) estimates of the random effects in multilevel models represent how individuals deviate from the population averages and are often extracted to detect outliers or used

as predictors in follow-up analysis. However, little research has examined whether EB estimates are indeed reliable and valid measures of individual traits. In this article, we use statistical theory and simulated data to show that EB estimates are biased toward zero, a phenomenon known as "shrinkage." The degree of shrinkage and reliability of EB estimates depend on a number of factors, including Level-1 residual variance, Level-1 predictor variance, Level-2 random effects variance, and number of within-person observations. As a result, EB estimates may not be ideal for detecting outliers, and they produce biased regression coefficients when used as predictors. We illustrate these issues using an empirical data set on emotion regulation and neuroticism.

S. Liu et al.: Treating random effects as observed versus latent predictors: The bias-variance tradeoff in small samples **Liu-Rhemtulla-2022**

Siwei Liu and Mijke Rhemtulla. "Treating random effects as observed versus latent predictors: The bias-variance tradeoff in small samples". In: *British Journal of Mathematical and Statistical Psychology* 75.1 (Feb. 2022), pp. 158–181. ISSN: 2044-8317. DOI: 10.1111/bmsp.12253.

Abstract: Random effects in longitudinal multilevel models represent individuals' deviations from population means and are indicators of individual differences. Researchers are often interested in examining how these random effects predict outcome variables that vary across individuals. This can be done via a two-step approach in which empirical Bayes (EB) estimates of the random effects are extracted and then treated as observed predictor variables in follow-up regression analyses. This approach ignores the unreliability of EB estimates, leading to underestimation of regression coefficients. As such, previous studies have recommended a multilevel structural equation modeling (ML-SEM) approach that treats random effects as latent variables. The current study uses simulation and empirical data to show that a bias-variance tradeoff exists when selecting between the two approaches. ML-SEM produces generally unbiased regression coefficient estimates but also larger standard errors, which can lead to lower power than the two-step approach. Implications of the results for model selection and alternative solutions are discussed.

Lohmann et al.: A primer on continuous-time modeling in educational research: An exemplary application of a continuous-time latent curve model with structured residuals (CT-LCM-SR) to PISA data **Lohmann-Zitzmann-Voelkle-etal-2022**

Julian F. Lohmann et al. "A primer on continuous-time modeling in educational research: An exemplary application of a continuous-time latent curve model with structured residuals (CT-LCM-SR) to PISA data". In: *Large-scale Assessments in Education* 10.1 (July 2022). ISSN: 2196-0739. DOI: 10.1186/s40536-022-00126-8.

Abstract: One major challenge of longitudinal data analysis is to find an appropriate statistical model that corresponds to the theory of change and the research questions at hand. In the present article, we argue that continuous-time models are well suited to study the continuously developing constructs of primary interest in the education sciences and outline key advantages of using this type of model. Furthermore, we propose the continuous-time latent curve model with structured residuals (CT-LCM-SR) as a suitable model for many research questions in the education sciences. The CT-LCM-SR combines growth and dynamic modeling and thus provides descriptions of both trends and process dynamics. We illustrate the application of the CT-LCM-SR with data from PISA reading literacy assessment of 2000 to 2018 and provide a tutorial and annotated code for setting up the CT-LCM-SR model.

Loossens et al.: The Affective Ising Model: A computational account of human affect dynamics **Loossens-Mestdagh-Dejonckheere-etal-2020**

Tim Loossens, Merijn Mestdagh, et al. "The Affective Ising Model: A computational account of human affect dynamics". In: *PLOS Computational Biology* 16.5 (May 2020). Ed. by Jacopo Grilli, e1007860. ISSN: 1553-7358. DOI: 10.1371/journal.pcbi.1007860.

Abstract: The human affect system is responsible for producing the positive and negative feelings that color and guide our lives. At the same time, when disrupted, its workings lie at the basis of the occurrence of mood disorder. Understanding the functioning and dynamics of the affect

system is therefore crucial to understand the feelings that people experience on a daily basis, their dynamics across time, and how they can become dysregulated in mood disorder. In this paper, a nonlinear stochastic model for the dynamics of positive and negative affect is proposed called the Affective Ising Model (AIM). It incorporates principles of statistical mechanics, is inspired by neurophysiological and behavioral evidence about auto-excitation and mutual inhibition of the positive and negative affect dimensions, and is intended to better explain empirical phenomena such as skewness, multimodality, and non-linear relations of positive and negative affect. The AIM is applied to two large experience sampling studies on the occurrence of positive and negative affect in daily life in both normality and mood disorder. It is examined to what extent the model is able to reproduce the aforementioned non-Gaussian features observed in the data, using two sightly different continuous-time vector autoregressive (VAR) models as benchmarks. The predictive performance of the models is also compared by means of leave-one-out cross-validation. The results indicate that the AIM is better at reproducing non-Gaussian features while their performance is comparable for strictly Gaussian features. The predictive performance of the AIM is also shown to be better for the majority of the affect time series. The potential and limitations of the AIM as a computational model approximating the workings of the human affect system are discussed.

Loossens et al.: A comparison of continuous and discrete time modeling of affective processes in terms of predictive accuracy **Loossens-Tuerlinckx-Verdonck-2021**

Tim Loossens, Francis Tuerlinckx, and Stijn Verdonck. "A comparison of continuous and discrete time modeling of affective processes in terms of predictive accuracy". In: *Scientific Reports* 11.1 (Mar. 2021). ISSN: 2045-2322. DOI: 10.1038/s41598-021-85320-4.

Abstract: Intra-individual processes are thought to continuously unfold across time. For equally spaced time intervals, the discrete-time lag-1 vector autoregressive (VAR(1)) model and the continuous-time Ornstein-Uhlenbeck (OU) model are equivalent. It is expected that by taking into account the unequal spacings of the time intervals in real data between observations will lead to an advantage for the OU in terms of predictive accuracy. In this paper, this is claim is being investigated by

comparing the predictive accuracy of the OU model to that of the VAR(1) model on typical ESM data obtained in the context of affect research. It is shown that the VAR(1) model outperforms the OU model for the majority of the time series, even though time intervals in the data are unequally spaced. Accounting for measurement error does not change the result. Deleting large abrupt changes on short time intervals (that may be caused by externally driven events) does however lead to a significant improvement for the OU model. This suggests that processes in psychology may be continuously evolving, but that there are factors, like external events, which can disrupt the continuous flow.

J. Luo et al.: The stressful personality: A meta-analytical review of the relation between personality and stress **Luo-Zhang-Cao-etal-2023**

Jing Luo et al. "The stressful personality: A meta-analytical review of the relation between personality and stress". In: *Personality and Social Psychology Review* 27.2 (May 2023), pp. 128–194. ISSN: 1532-7957. DOI: 10.1177/10888683221104002.

Abstract: The current study presented the first meta-analytic review on the associations between the Big Five personality traits and stress measured under different conceptualizations (stressor exposure, psychological and physiological stress responses) using a total of 1,575 effect sizes drawn from 298 samples. Overall, neuroticism was found to be positively related to stress, whereas extraversion, agreeableness, conscientiousness, and openness were negatively linked to stress. When stress assessed under different conceptualizations was tested, only neuroticism, agreeableness, and conscientiousness were related to stressor exposure. All of the Big Five personality traits were significantly associated with psychological stress perception, whereas the five personality traits showed weak to null associations with physiological stress response. Further moderation analyses suggested that the associations between personality traits and stress under different conceptualizations were also contingent upon different characteristics of stress, sample, study design, and measures. The results supported the important role of personality traits in individual differences in stress.

Mader et al.: Emotional (in)stability: Neuroticism is associated with increased variability in negative emotion after all **Mader-Arslan-Schmukle-etal-2023**

Nina Mader et al. "Emotional (in)stability: Neuroticism is associated with increased variability in negative emotion after all". In: *Proceedings of the National Academy of Sciences* 120.23 (May 2023). ISSN: 1091-6490. DOI: 10.1073/pnas.2212154120.

Abstract: The personality trait neuroticism is tightly linked to mental health, and neurotic people experience stronger negative emotions in everyday life. But, do their negative emotions also show greater fluctuation? This commonsensical notion was recently questioned by [Kalokerinos et al. Proc Natl Acad Sci USA 112, 15838-15843 (2020)], who suggested that the associations found in previous studies were spurious. Less neurotic people often report very low levels of negative emotion, which is usually measured with bounded rating scales. Therefore, they often pick the lowest possible response option, which severely constrains the amount of emotional variability that can be observed in principle. Applying a multistep statistical procedure that is supposed to correct for this dependency, [Kalokerinos et al. Proc Natl Acad Sci USA 112, 15838-15843 (2020)] no longer found an association between neuroticism and emotional variability. However, like other common approaches for controlling for undesirable effects due to bounded scales, this method is opaque with respect to the assumed mechanism of data generation and might not result in a successful correction. We thus suggest an alternative approach that a) takes into account that emotional states outside of the scale bounds can occur and b) models associations between neuroticism and both the mean and variability of emotion in a single step with the help of Bayesian censored location-scale models. Simulations supported this model over alternative approaches. We analyzed 13 longitudinal datasets (2,518 individuals and 11,170 measurements in total) and found clear evidence that more neurotic people experience greater variability in negative emotion.

Manthey et al.: What are the economic costs to society attributable to alcohol use? A systematic review and modelling study **Manthey-Hassan-Carr-etal-2021**

Jakob Manthey et al. "What are the economic costs to society attributable to alcohol use? A systematic review and modelling study". In: *PharmacoEconomics* 39.7 (May 2021), pp. 809–822. ISSN: 1179-2027. DOI: 10.1007/s40273-021-01031-8.

Abstract: Background: Alcohol-attributable costs to society are captured by cost-of-illness studies, however estimates are often not comparable, e.g. due to the omission of relevant cost components. In this contribution we (1) summarize the societal costs attributable to alcohol use, and (2) estimate the total costs under the assumption that all cost components are considered. Methods: A systematic review and meta-analyses were conducted for studies reporting costs from alcohol consumption for the years 2000 and later, using the EMBASE and MEDLINE databases. Cost estimates were converted into 2019 international dollars (Int$) per adult and into percentage of gross domestic product (GDP). For each study, weights were calculated to correct for the exclusion of cost indicators. Results: Of 1708 studies identified, 29 were included, and the mean costs of alcohol use amounted to 817.6 Int$ per adult (95% confidence interval [CI] 601.8-1033.4), equivalent to 1.5% of the GDP (95% CI 1.2-1.7%). Adjusting for omission of cost components, the economic costs of alcohol consumption were estimated to amount to 1306 Int$ per adult (95% CI 873-1738), or 2.6% (95% CI 2.0-3.1%) of the GDP. About one-third of costs (38.8%) were incurred through direct costs, while the majority of costs were due to losses in productivity (61.2%). Discussion: The identified cost studies were mainly conducted in high-income settings, with high heterogeneity in the employed methodology. Accounting for some methodological variations, our findings demonstrate that alcohol use continues to incur a high level of cost to many societies.

McKendrick et al.: Drug-induced conditioned place preference and its practical use in substance use disorder research **McKendrick-Graziane-2020**

Greer McKendrick and Nicholas M. Graziane. "Drug-induced conditioned place preference and its

practical use in substance use disorder research". In: *Frontiers in Behavioral Neuroscience* 14 (Sept. 2020). ISSN: 1662-5153. DOI: 10.3389/fnbeh.2020.582147.

Abstract: The conditioned place preference (CPP) paradigm is a well-established model utilized to study the role of context associations in reward-related behaviors, including both natural rewards and drugs of abuse. In this review article, we discuss the basic history, various uses, and considerations that are tied to this technique. There are many potential takeaway implications of this model, including negative affective states, conditioned drug effects, memory, and motivation, which are all considered here. We also discuss the neurobiology of CPP including relevant brain regions, molecular signaling cascades, and neuromodulatory systems. We further examine some of our prior findings and how they integrate CPP with self-administration paradigms. Overall, by describing the fundamentals of CPP, findings from the past few decades, and implications of using CPP as a research paradigm, we have endeavored to support the case that the CPP method is specifically advantageous for studying the role of a form of Pavlovian learning that associates drug use with the surrounding environment.

McNeish et al.: A primer on two-level dynamic structural equation models for intensive longitudinal data in Mplus **McNeish-Hamaker-2020**

Daniel McNeish and Ellen L. Hamaker. "A primer on two-level dynamic structural equation models for intensive longitudinal data in Mplus". In: *Psychological Methods* 25.5 (Oct. 2020), pp. 610–635. DOI: 10.1037/met0000250.

Abstract: Technological advances have led to an increase in intensive longitudinal data and the statistical literature on modeling such data is rapidly expanding, as are software capabilities. Common methods in this area are related to time-series analysis, a framework that historically has received little exposure in psychology. There is a scarcity of psychology-based resources introducing the basic ideas of time-series analysis, especially for data sets featuring multiple people. We begin with basics of N = 1 time-series analysis and build up to complex dynamic structural equation

models available in the newest release of Mplus Version 8. The goal is to provide readers with a basic conceptual understanding of common models, template code, and result interpretation. We provide short descriptions of some advanced issues, but our main priority is to supply readers with a solid knowledge base so that the more advanced literature on the topic is more readily digestible to a larger group of researchers.

McNeish et al.: Intensive longitudinal mediation in Mplus McNeish-MacKinnon-2025

Daniel McNeish and David P. MacKinnon. "Intensive longitudinal mediation in Mplus". In: *Psychological Methods* 30.2 (Apr. 2025), pp. 393–415. ISSN: 1082-989X. DOI: 10.1037/met0000536.

Abstract: Much of the existing longitudinal mediation literature focuses on panel data where relatively few repeated measures are collected over a relatively broad timespan. However, technological advances in data collection (e.g., smartphones, wearables) have led to a proliferation of short duration, densely collected longitudinal data in behavioral research. These intensive longitudinal data differ in structure and focus relative to traditionally collected panel data. As a result, existing methodological resources do not necessarily extend to nuances present in the recent influx of intensive longitudinal data and designs. In this tutorial, we first cover potential limitations of traditional longitudinal mediation models to accommodate unique characteristics of intensive longitudinal data. Then, we discuss how recently developed dynamic structural equation models (DSEMs) may be well-suited for mediation modeling with intensive longitudinal data and can overcome some of the limitations associated with traditional approaches. We describe four increasingly complex intensive longitudinal mediation models: (a) stationary models where the indirect effect is constant over time and people, (b) person-specific models where the indirect effect varies across people, (c) dynamic models where the indirect effect varies across time, and (d) cross-classified models where the indirect effect varies across both time and people. We apply each model to a running example featuring a mobile health intervention designed to improve health behavior of individuals with binge eating disorder. In each example, we provide annotated Mplus code and interpretation of the output

to guide empirical researchers through mediation modeling with this increasingly popular type of longitudinal data.

McNeish et al.: Thinking twice about sum scores **McNeish-Wolf-2020**

Daniel McNeish and Melissa Gordon Wolf. "Thinking twice about sum scores". In: *Behavior Research Methods* 52.6 (Apr. 2020), pp. 2287–2305. ISSN: 1554-3528. DOI: 10.3758/s13428-020-01398-0.

Abstract: A common way to form scores from multiple-item scales is to sum responses of all items. Though sum scoring is often contrasted with factor analysis as a competing method, we review how factor analysis and sum scoring both fall under the larger umbrella of latent variable models, with sum scoring being a constrained version of a factor analysis. Despite similarities, reporting of psychometric properties for sum scored or factor analyzed scales are quite different. Further, if researchers use factor analysis to validate a scale but subsequently sum score the scale, this employs a model that differs from validation model. By framing sum scoring within a latent variable framework, our goal is to raise awareness that (a) sum scoring requires rather strict constraints, (b) imposing these constraints requires the same type of justification as any other latent variable model, and (c) sum scoring corresponds to a statistical model and is not a model-free arithmetic calculation. We discuss how unjustified sum scoring can have adverse effects on validity, reliability, and qualitative classification from sum score cut-offs. We also discuss considerations for how to use scale scores in subsequent analyses and how these choices can alter conclusions. The general goal is to encourage researchers to more critically evaluate how they obtain, justify, and use multiple-item scale scores.

Mey et al.: Increases of negative affect following daily hassles are not moderated by neuroticism: An ecological momentary assessment study

Mey-Chmitorz-Kurth-etal-2020

Lara Kristin Mey et al. "Increases of negative affect following daily hassles are not moderated by neuroticism: An ecological momentary assessment study". In: *Stress and Health* 36.5 (June 2020), pp. 615–628. ISSN: 1532-2998. DOI: 10.1002/smi.2964.

Abstract: The occurrence of daily hassles is associated with increased subsequent levels of negative affect. Neuroticism has been found to exacerbate this effect. So far, most research used single-item measures for the assessment of daily hassles or relied on daily diary studies. This study aimed to examine the interrelations of daily hassles, negative affect reactivity, and neuroticism in daily life employing an extensive inventory of daily hassles. Seventy participants (18-30 years; $M = 23.9$ years, 59% female) completed a 4-week smartphone-based ecological momentary assessment study reporting the occurrence and perceived strain of daily hassles as well as negative affect at five semi-random signals between 9 a.m. and 8 p.m. Multilevel analyses revealed significant associations between elevated levels of negative affect and higher cumulative daily hassle strain ratings per signal in concurrent and time-lagged analyses. Contrary to our expectations, there was no moderation by neuroticism on these associations. The results suggest that daily hassles can accumulate in their impact on mood in daily life and exert a prolonged effect on negative affect. The absence of a significant moderation by neuroticism may be interpreted in the light of methodological specifics of this study.

Moeyaert et al.: Meta-analysis of single-case research via multilevel models: Fundamental concepts and methodological considerations **Moeyaert-Manolov-Rodabaugh-2020**

Mariola Moeyaert, Rumen Manolov, and Emily Rodabaugh. "Meta-analysis of single-case research via multilevel models: Fundamental concepts and methodological considerations". In: *Behavior Modification* 44.2 (Mar. 2020), pp. 265–295. ISSN: 1552-4167. DOI: 10.1177/0145445518806867.

Abstract: Multilevel modeling is an approach that can be used to summarize single-case experimental design (SCED) data. Multilevel models were developed to analyze hierarchical structured data with units at a lower level nested within higher level units. SCEDs use time series data collected from multiple cases (or subjects) within a study that allow researchers to investigate intervention effectiveness at the individual level and also to investigate how these individual intervention effects change over time. There is an increased interest in the field regarding how SCEDs can be used to establish an evidence base for interventions by synthesizing data from a series of intervention studies. Although using multilevel models to meta-analyze SCED studies is promising, application is often hampered by being potentially excessively technical. First, this article provides an accessible description and overview of the potential of multilevel meta-analysis to combine SCED data. Second, a summary of the methodological evidence on the performance of multilevel models for meta-analysis is provided, which is useful given that such evidence is currently scattered over multiple technical articles in the literature. Third, the actual steps to perform a multilevel meta-analysis are outlined in a brief practical guide. Fourth, a suggestion for integrating the quantitative results with a visual representation is provided.

Mulder: Power analysis for the random intercept cross-lagged panel model using the powRICLPM R-package **Mulder-2023**

Jeroen D. Mulder. "Power analysis for the random intercept cross-lagged panel model using the powRICLPM R-package". In: *Structural Equation Modeling: A Multidisciplinary Journal* 30.4 (July 2023), pp. 645–658. ISSN: 1532-8007. DOI: 10.1080/10705511.2022.2122467.

Abstract: The random intercept cross-lagged panel model (RI-CLPM) is a popular model among psychologists for studying reciprocal effects in longitudinal panel data. Although various texts and software packages have been published concerning power analyses for structural equation models (SEM) generally, none have proposed a power analysis strategy that is tailored to the particularities of the RI-CLPM. This can be problematic because mismatches between the power analysis design, the model, and reality, can negatively impact the validity of the recommended sample size and

number of repeated measures. As power analyses play an increasingly important role in the preparation phase of research projects, an RI-CLPM-specific strategy for the design of a power analysis is detailed, and implemented in the R-package powRICLPM. This paper focuses on the (basic) bivariate RI-CLPM, and extensions to include constraints over time, measurement error (leading to the stable trait autoregressive trait state model), non-normal data, and bounded estimation.

Mulder et al.: Three extensions of the random intercept cross-lagged panel model

Mulder-Hamaker-2021

Jeroen D. Mulder and Ellen L. Hamaker. "Three extensions of the random intercept cross-lagged panel model". In: *Structural Equation Modeling: A Multidisciplinary Journal* 28.4 (July 2021), pp. 638–648. ISSN: 1532-8007. DOI: 10.1080/10705511.2020.1784738.

Abstract: The random intercept cross-lagged panel model (RI-CLPM) is rapidly gaining popularity in psychology and related fields as a structural equation modeling (SEM) approach to longitudinal data. It decomposes observed scores into within-unit dynamics and stable, between-unit differences. This paper discusses three extensions of the RI-CLPM that researchers may be interested in, but are unsure of how to accomplish: (a) including stable, person-level characteristics as predictors and/or outcomes; (b) specifying a multiple-group version; and (c) including multiple indicators. For each extension, we discuss which models need to be run in order to investigate underlying assumptions, and we demonstrate the various modeling options using a motivating example. We provide fully annotated code for lavaan (R-package) and Mplus on an accompanying website.

B. O. Muthén et al.: Latent transition analysis with random intercepts (RI-LTA)

Muthen-Asparouhov-2022

Bengt O. Muthén and Tihomir Asparouhov. "Latent transition analysis with random intercepts (RI-LTA)". In: *Psychological Methods* 27.1 (Feb. 2022), pp. 1–16. ISSN: 1082-989X. DOI: 10.1037/met0000370.

Abstract: This article demonstrates that the regular LTA model is unnecessarily restrictive and that an alternative model is readily available that typically fits the data much better, leads to better estimates of the transition probabilities, and extracts new information from the data. By allowing random intercept variation in the model, between-subject variation is separated from the within-subject latent class transitions over time allowing a clearer interpretation of the data. Analysis of two examples from the literature demonstrates the advantages of random intercept LTA. Model variations include Mover-Stayer analysis, measurement invariance analysis, and analysis with covariates.

Nestler: Modeling intraindividual variability in growth with measurement burst designs **Nestler-2021**

Steffen Nestler. "Modeling intraindividual variability in growth with measurement burst designs". In: *Structural Equation Modeling: A Multidisciplinary Journal* 28.1 (Jan. 2021), pp. 28–39. ISSN: 1532-8007. DOI: 10.1080/10705511.2020.1757455.

Abstract: Technical advancements have made it easier for researchers to conduct studies in which single-measure panel studies are combined with experience sampling studies. This measurement burst design allows applied researchers to answer a number of interesting questions, such as whether the relationships between variables change over bursts. Here we suggest a heterogeneous variability growth model that researchers can use to examine growth across mean levels and across intraindividual variability. Furthermore, interindividual differences in the two types of growth and their relations can be investigated. After introducing the model, we show how the model's parameters can be estimated with a Maximum Likelihood estimator and a Bayesian approach. Furthermore, we illustrate the approach using both fictitious and real data examples and we present the results of a simulation study.

Norman et al.: Combining transdermal and breath alcohol assessments, real-time drink logs and retrospective self-reports to measure alcohol consumption and intoxication across a multi-day music festival **Norman-Peacock-Ferguson-etal-2021**

Thomas Norman et al. "Combining transdermal and breath alcohol assessments, real-time drink logs and retrospective self-reports to measure alcohol consumption and intoxication across a multi-day music festival". In: *Drug and Alcohol Review* 40.7 (Nov. 2021), pp. 1112–1121. ISSN: 1465-3362. DOI: 10.1111/dar.13215.

Abstract: Introduction and Aims: Comprehensively investigating alcohol-related behaviours in the context of a dynamic multi-day alcohol-licensed event is important for understanding and minimising patron risk. We aimed to assess the measurement utility of implementing a multi-dimensional alcohol assessment battery using biometric data collection, real-time drink logs and retrospective self-report measures over the course of a 4-day music festival. Methods: Fourteen adults participated ($n = 7$ male, mean age 21.9 years). Breath and transdermal alcohol concentration (BrAC and TAC, respectively) were measured using breathalysers and transdermal alcohol bracelets. A real-time drink log was completed via smartphones on initiating each drink, and a retrospective questionnaire was administered up to twice daily throughout the event (6 timepoints total). Results: While almost all participants (92.9%) logged significantly fewer drinks in real-time than they retrospectively reported via the twice-daily questionnaires, logs provided important contextual information including the types of drinks consumed and drinking intensity. Compared to BrAC, TAC provided a better understanding of the time course of intoxication, indicating highest alcohol consumption outside of static BrAC assessment windows. However, BrAC provided a better assessment of present state: all participants were 0.00% BrAC at departure despite over two-fifths (42.9%) of the sample's last TAC reading exceeding 0.00%. Conclusions: As standalone assessments, each method possessed limitations. As a combined battery, they were successfully administered simultaneously, resulting in a more comprehensive overview of alcohol consumption/intoxication over the prolonged drinking session. However, the marked burden of simultaneous administration should be considered, and measures should be chosen judiciously based on research needs.

Nüst et al.: The Rockerverse: Packages and applications for containerisation with R **Nust-Eddelbuettel-Bennett-etal-2020**

Daniel Nüst et al. "The Rockerverse: Packages and applications for containerisation with R". In: *The R Journal* 12.1 (2020), p. 437. DOI: 10.32614/rj-2020-007.

Abstract: The Rocker Project provides widely used Docker images for R across different application scenarios. This article surveys downstream projects that build upon the Rocker Project images and presents the current state of R packages for managing Docker images and controlling containers. These use cases cover diverse topics such as package development, reproducible research, collaborative work, cloud-based data processing, and production deployment of services. The variety of applications demonstrates the power of the Rocker Project specifically and containerisation in general. Across the diverse ways to use containers, we identified common themes: reproducible environments, scalability and efficiency, and portability across clouds. We conclude that the current growth and diversification of use cases is likely to continue its positive impact, but see the need for consolidating the Rockerverse ecosystem of packages, developing common practices for applications, and exploring alternative containerisation software.

Oh et al.: Measurement model misspecification in dynamic structural equation models: Power, reliability, and other considerations **Oh-Hunter-Chow-2025**

Hyungeun Oh, Michael D. Hunter, and Sy-Miin Chow. "Measurement model misspecification in dynamic structural equation models: Power, reliability, and other considerations". In: *Structural Equation Modeling: A Multidisciplinary Journal* 32.3 (Feb. 2025), pp. 511–528. ISSN: 1532-8007. DOI: 10.1080/10705511.2025.2452884.

Abstract: Dynamic Structural Equation Models (DSEMs) integrate multilevel modeling, time series analysis, and structural equation modeling within a Bayesian estimation framework, offering a versatile tool for analyzing intensive longitudinal data (ILD). However, the impact of measurement

structure misspecification in DSEMs, especially under varying reliability conditions and model complexities, remains underexplored. Our Monte Carlo simulation revealed that omitting measurement errors when present led to severe biases in dynamic parameters regardless of reliability conditions, though power remained high. Increasing the number of participants and time points ameliorated but did not eliminate all biases. A single-indicator DSEMs with a measurement structure using composite scores showed similar performance to multiple indicators DSEMs. Empirical applications showed discrepancies in dynamic parameters based on the number of indicators and measurement structures used. Leveraging these findings, we provide design recommendations, functions for extending reliability indices from single-indicator to multiple-indicator models, and guidelines for power evaluations under different reliability conditions.

Ong et al.: Positive affect dynamics **Ong-Dejonckheere-Ram-2025**

Anthony D. Ong, Egon Dejonckheere, and Nilam Ram. "Positive affect dynamics". In: *Current Directions in Psychological Science* 34.5 (Oct. 2025), pp. 301–308. DOI: 10.1177/09637214251339454.

Abstract: Positive affect is central to well-being and contributes to psychological and physical functioning. This review synthesizes research on how positive affect unfolds within people in everyday settings and across multiple timescales. It considers variability, instability, inertia, and reactivity and examines their associations with adaptive functioning and health. Dynamic approaches may improve prediction beyond conventional trait- and state-based summaries, but their use is complicated by difficulties in measurement, modeling, and integration across temporal resolutions and contexts. The authors outline methodological priorities for improving the precision and interpretation of positive affect dynamics and discuss applications for promoting health and resilience.

Orth et al.: Testing prospective effects in longitudinal research: Comparing seven competing cross-lagged models **Orth-Clark-Donnellan-etal-2021**

Ulrich Orth, D. Angus Clark, et al. "Testing prospective effects in longitudinal research: Comparing

seven competing cross-lagged models". In: *Journal of Personality and Social Psychology* 120.4 (Apr. 2021), pp. 1013–1034. ISSN: 0022-3514. DOI: 10.1037/pspp0000358.

Abstract: In virtually all areas of psychology, the question of whether a particular construct has a prospective effect on another is of fundamental importance. For decades, the cross-lagged panel model (CLPM) has been the model of choice for addressing this question. However, CLPMs have recently been critiqued, and numerous alternative models have been proposed. Using the association between low self-esteem and depression as a case study, we examined the behavior of seven competing longitudinal models in 10 samples, each with at least four waves of data and sample sizes ranging from 326 to 8,259. The models were compared in terms of convergence, fit statistics, and consistency of parameter estimates. The traditional CLPM and the random intercepts cross-lagged panel model (RI-CLPM) converged in every sample, whereas the other models frequently failed to converge or did not converge properly. The RI-CLPM exhibited better model fit than the CLPM, whereas the CLPM produced more consistent cross-lagged effects (both across and within samples) than the RI-CLPM. We discuss the models from a conceptual perspective, emphasizing that the models test conceptually distinct psychological and developmental processes, and we address the implications of the empirical findings with regard to model selection. Moreover, we provide practical recommendations for researchers interested in testing prospective associations between constructs and suggest using the CLPM when focused on between-person effects and the RI-CLPM when focused on within-person effects.

Orth et al.: Effect size guidelines for cross-lagged effects Orth-Meier-Buhler-etal-2024

Ulrich Orth, Laurenz L. Meier, et al. "Effect size guidelines for cross-lagged effects". In: *Psychological Methods* 29.2 (Apr. 2024), pp. 421–433. ISSN: 1082-989X. DOI: 10.1037/met0000499.

J. H. Orzek et al.: Regularized continuous time structural equation models: A network perspective **Orzek-Voelkle-2023**

Jannik H. Orzek and Manuel C. Voelkle. "Regularized continuous time structural equation models: A network perspective". In: *Psychological Methods* 28.6 (Dec. 2023), pp. 1286–1320. ISSN: 1082-989X. DOI: 10.1037/met0000550.

Abstract: Regularized continuous time structural equation models are proposed to address two recent challenges in longitudinal research: Unequally spaced measurement occasions and high model complexity. Unequally spaced measurement occasions are part of most longitudinal studies, sometimes intentionally (e.g., in experience sampling methods) sometimes unintentionally (e.g., due to missing data). Yet, prominent dynamic models, such as the autoregressive cross-lagged model, assume equally spaced measurement occasions. If this assumption is violated parameter estimates can be biased, potentially leading to false conclusions. Continuous time structural equation models (CTSEM) resolve this problem by taking the exact time point of a measurement into account. This allows for any arbitrary measurement scheme. We combine CTSEM with LASSO and adaptive LASSO regularization. Such regularization techniques are especially promising for the increasingly complex models in psychological research, the most prominent example being network models with often dozens or hundreds of parameters. Here, LASSO regularization can reduce the risk of overfitting and simplify the model interpretation. In this article we highlight unique challenges in regularizing continuous time dynamic models, such as standardization or the optimization of the objective function, and offer different solutions. Our approach is implemented in the R (R Core Team, 2022) package regCtsem. We demonstrate the use of regCtsem in a simulation study, showing that the proposed regularization improves the parameter estimates, especially in small samples. The approach correctly eliminates true-zero parameters while retaining true-nonzero parameters. We present two empirical examples and end with a discussion on current limitations and future research directions.

Ou et al.: Estimation of nonlinear mixed-effects continuous-time models using the continuous-discrete extended Kalman filter **Ou-Hunter-Lu-etal-2023**

Lu Ou et al. "Estimation of nonlinear mixed-effects continuous-time models using the continuous-discrete extended Kalman filter". In: *British Journal of Mathematical and Statistical Psychology* 76.3 (Sept. 2023), pp. 462–490. ISSN: 2044-8317. DOI: 10.1111/bmsp.12318.

Abstract: Many intensive longitudinal measurements are collected at irregularly spaced time intervals, and involve complex, possibly nonlinear and heterogeneous patterns of change. Effective modelling of such change processes requires continuous-time differential equation models that may be nonlinear and include mixed effects in the parameters. One approach of fitting such models is to define random effect variables as additional latent variables in a stochastic differential equation (SDE) model of choice, and use estimation algorithms designed for fitting SDE models, such as the continuous-discrete extended Kalman filter (CDEKF) approach implemented in the dynr R package, to estimate the random effect variables as latent variables. However, this approach's efficacy and identification constraints in handling mixed-effects SDE models have not been investigated. In the current study, we analytically inspect the identification constraints of using the CDEKF approach to fit nonlinear mixed-effects SDE models; extend a published model of emotions to a nonlinear mixed-effects SDE model as an example, and fit it to a set of irregularly spaced ecological momentary assessment data; and evaluate the feasibility of the proposed approach to fit the model through a Monte Carlo simulation study. Results show that the proposed approach produces reasonable parameter and standard error estimates when some identification constraint is met. We address the effects of sample size, process noise variance, and data spacing conditions on estimation results.

Park et al.: Subgrouping with chain graphical VAR models **Park-Chow-Epskamp-etal-2024**

Jonathan J. Park, Sy-Miin Chow, Sacha Epskamp, et al. "Subgrouping with chain graphical VAR

models". In: *Multivariate Behavioral Research* 59.3 (Feb. 2024), pp. 543–565. ISSN: 1532-7906. DOI: 10.1080/00273171.2023.2289058.

Abstract: Recent years have seen the emergence of an "idio-thetic" class of methods to bridge the gap between nomothetic and idiographic inference. These methods describe nomothetic trends in idiographic processes by pooling intraindividual information across individuals to inform group-level inference or vice versa. The current work introduces a novel "idio-thetic" model: the subgrouped chain graphical vector autoregression (scGVAR). The scGVAR is unique in its ability to identify subgroups of individuals who share common dynamic network structures in both lag(1) and contemporaneous effects. Results from Monte Carlo simulations indicate that the scGVAR shows promise over similar approaches when clusters of individuals differ in their contemporaneous dynamics and in showing increased sensitivity in detecting nuanced group differences while keeping Type-I error rates low. In contrast, a competing approach-the Alternating Least Squares VAR (ALS VAR) performs well when groups were separated by larger distances. Further considerations are provided regarding applications of the ALS VAR and scGVAR on real data and the strengths and limitations of both methods.

Park et al.: Affect and personality: Ramifications of modeling (non-)directionality in dynamic network models **Park-Chow-Fisher-etal-2020**

Jonathan J. Park, Sy-Miin Chow, Zachary F. Fisher, et al. "Affect and personality: Ramifications of modeling (non-)directionality in dynamic network models". In: *European Journal of Psychological Assessment* 36.6 (Nov. 2020), pp. 1009–1023. ISSN: 2151-2426. DOI: 10.1027/1015-5759/a000612.

Abstract: The use of dynamic network models has grown in recent years. These models allow researchers to capture both lagged and contemporaneous effects in longitudinal data typically as variations, reformulations, or extensions of the standard vector autoregressive (VAR) models. To date, many of these dynamic networks have not been explicitly compared to one another. We compare three popular dynamic network approaches—GIMME, uSEM, and LASSO gVAR—in

terms of their differences in modeling assumptions, estimation procedures, statistical properties based on a Monte Carlo simulation, and implications for affect and personality researchers. We found that all three dynamic network approaches provided yielded group-level empirical results in partial support of affect and personality theories. However, individual-level results revealed a great deal of heterogeneity across approaches and participants. Reasons for discrepancies are discussed alongside these approaches' respective strengths and limitations.

Park et al.: On subgrouping continuous processes in discrete time

Park-Fisher-Chow-etal-2023

Jonathan J. Park, Zachary F. Fisher, Sy-Miin Chow, et al. "On subgrouping continuous processes in discrete time". In: *Multivariate Behavioral Research* 58.1 (Jan. 2023), pp. 154–155. ISSN: 1532-7906. DOI: 10.1080/00273171.2022.2160957.

Park et al.: Evaluating discrete time methods for subgrouping continuous processes

Park-Fisher-Chow-etal-2024

Jonathan J. Park, Zachary F. Fisher, Sy-Miin Chow, et al. "Evaluating discrete time methods for subgrouping continuous processes". In: *Multivariate Behavioral Research* 59.6 (Nov. 2024), pp. 1240–1252. ISSN: 1532-7906. DOI: 10.1080/00273171.2023.2235685.

Abstract: Rapid developments over the last several decades have brought increased focus and attention to the role of time scales and heterogeneity in the modeling of human processes. To address these emerging questions, subgrouping methods developed in the discrete-time framework—such as the vector autoregression (VAR)—have undergone widespread development to identify shared nomothetic trends from idiographic modeling results. Given the dependence of VAR-based parameters on the measurement intervals of the data, we sought to clarify the strengths and limitations of these methods in recovering subgroup dynamics under different measurement intervals. Building on the work of Molenaar and collaborators for subgrouping individual time-series by means of

the subgrouped chain graphical VAR (scgVAR) and the subgrouping option in the group iterative multiple model estimation (S-GIMME), we present results from a Monte Carlo study aimed at addressing the implications of identifying subgroups using these discrete-time methods when applied to continuous-time data. Results indicate that discrete-time subgrouping methods perform well at recovering true subgroups when the measurement intervals are large enough to capture the full range of a system's dynamics, either via lagged or contemporaneous effects. Further implications and limitations are discussed therein.

Park et al.: Unsupervised model construction in continuous-time

Park-Fisher-Hunter-etal-2025

Jonathan J. Park, Zachary F. Fisher, Michael D. Hunter, et al. "Unsupervised model construction in continuous-time". In: *Structural Equation Modeling: A Multidisciplinary Journal* 32.3 (May 2025), pp. 377–399. ISSN: 1532-8007. DOI: 10.1080/10705511.2024.2429544.

Abstract: Many of the advancements reconciling individual- and group-level results have occurred in the context of a discrete-time modeling framework. Discrete-time models are intuitive and offer relatively simple interpretations for the resulting dynamic structures; however, they do not possess the flexibility of models fitted in the continuous-time framework. We introduce ct-gimme, a continuous-time extension of the group iterative multiple model estimation (GIMME) procedure which enables researchers to fit complex, high-dimensional dynamic networks in continuous time. Our results indicate that ct-gimme outperforms $N = 1$ model fitting in continuous time by pooling information across multiple subjects. Likewise, ct-gimme outperforms group-level model fitting in the presence of within-sample heterogeneity. We conclude with an empirical illustration and highlight the limitations of the approach relating to the identification of meaningful starting values.

Pesigan et al.: SEM-based methods to form confidence intervals for indirect effect: Still applicable given nonnormality, under certain conditions Pesigan-Cheung-2020

Ivan Jacob Agaloos Pesigan and Shu Fai Cheung. "SEM-based methods to form confidence intervals for indirect effect: Still applicable given nonnormality, under certain conditions". In: *Frontiers in Psychology* 11 (Dec. 2020). DOI: 10.3389/fpsyg.2020.571928.

Abstract: A SEM-based approach using likelihood-based confidence interval (LBCI) has been proposed to form confidence intervals for unstandardized and standardized indirect effect in mediation models. However, when used with the maximum likelihood estimation, this approach requires that the variables are multivariate normally distributed. This can affect the LBCIs of unstandardized and standardized effect differently. In the present study, the robustness of this approach when the predictor is not normally distributed but the error terms are conditionally normal, which does not violate the distributional assumption of ordinary least squares (OLS) estimation, is compared to four other approaches: nonparametric bootstrapping, two variants of LBCI, LBCI assuming the predictor is fixed (LBCI-Fixed-X) and LBCI based on ADF estimation (LBCI-ADF), and Monte Carlo. A simulation study was conducted using a simple mediation model and a serial mediation model, manipulating the distribution of the predictor. The Monte Carlo method performed worst among the methods. LBCI and LBCI-Fixed-X had suboptimal performance when the distributions had high kurtosis and the population indirect effects were medium to large. In some conditions, the problem was severe even when the sample size was large. LBCI-ADF and nonparametric bootstrapping had coverage probabilities close to the nominal value in nearly all conditions, although the coverage probabilities were still suboptimal for the serial mediation model when the sample size was small with respect to the model. Implications of these findings in the context of this special case of nonnormal data were discussed.

Pesigan et al.: Monte Carlo confidence intervals for the indirect effect with missing data **Pesigan-Cheung-2024**

Ivan Jacob Agaloos Pesigan and Shu Fai Cheung. "Monte Carlo confidence intervals for the indirect effect with missing data". In: *Behavior Research Methods* 56.3 (Mar. 2024), pp. 1678–1696. DOI: 10.3758/s13428-023-02114-4.

Abstract: Missing data is a common occurrence in mediation analysis. As a result, the methods used to construct confidence intervals around the indirect effect should consider missing data. Previous research has demonstrated that, for the indirect effect in data with complete cases, the Monte Carlo method performs as well as nonparametric bootstrap confidence intervals (see MacKinnon et al., Multivariate Behavioral Research, 39(1), 99-128, 2004; Preacher & Selig, Communication Methods and Measures, 6(2), 77-98, 2012; Tofighi & MacKinnon, Structural Equation Modeling: A Multidisciplinary Journal, 23(2), 194-205, 2015). In this manuscript, we propose a simple, fast, and accurate two-step approach for generating confidence intervals for the indirect effect, in the presence of missing data, based on the Monte Carlo method. In the first step, an appropriate method, for example, full-information maximum likelihood or multiple imputation, is used to estimate the parameters and their corresponding sampling variance-covariance matrix in a mediation model. In the second step, the sampling distribution of the indirect effect is simulated using estimates from the first step. A confidence interval is constructed from the resulting sampling distribution. A simulation study with various conditions is presented. Implications of the results for applied research are discussed.

Pesigan et al.: How plausible is my model? Assessing model plausibility of structural equation models using Bayesian posterior probabilities (BPP) **Pesigan-Cheung-Wu-etal-2026**

Ivan Jacob Agaloos Pesigan, Shu Fai Cheung, et al. "How plausible is my model? Assessing model

plausibility of structural equation models using Bayesian posterior probabilities (BPP)". In: *Behavior Research Methods* 58.3 (Feb. 2026), p. 73. ISSN: 1554-3528. DOI: 10.3758/s13428-025-02921-x.

Abstract: In structural equation modeling (SEM), one method to select the most plausible model from several candidates, or to compare one or more hypothesized models with similar alternatives on plausibility, is to compare the models using Bayesian posterior probability (BPP). BPP can be computed from the Bayesian information criterion (BIC) scores (Wu et al. Multivariate Behavioral Research, 55(1), 1–16, 2020). This approach complements conventional goodness-of-fit indices such as the Comparative Fit Index (CFI), the root mean square error of approximation (RMSEA), and the standardized root mean square residual (SRMR) in giving concise BPP for assessing uncertainties among all models considered. It can also reveal evidence against a model otherwise hidden by these indices. However, Wu et al. Multivariate Behavioral Research, 55(1), 1–16. (2020) did not provide guidelines on deciding the models that should be considered. To facilitate the use of BPP, we proposed a novel method for selecting this set of models, called neighboring models, to help researchers decide on the initial set. This novel method integrates seamlessly into the typical workflow for SEM analysis. Researchers can fit a model as usual and then use this method to assess whether it is the most plausible model compared with the neighboring models. We believe the proposed method will make it easier for researchers to make better-informed decisions when evaluating their models. We developed a user-friendly R package, modelbpp, to automate all the steps: generating the set of neighboring models, fitting them, and computing the BPPs, all in a single function.

Pesigan et al.: Common and unique latent transition analysis (CULTA) as a way to examine the trait-state dynamics of alcohol intoxication Pesigan-Russell-Chow-2025b

Ivan Jacob Agaloos Pesigan, Michael A. Russell, and Sy-Miin Chow. "Common and unique latent transition analysis (CULTA) as a way to examine the trait-state dynamics of alcohol intoxication". In: *Psychology of Addictive Behaviors* 39.8 (Dec. 2025), pp. 743–762. ISSN: 0893-164X. DOI: 10.1037/adb0001106.

Abstract: Objective: This article introduces the common and unique latent transition analysis (CULTA), a novel approach to studying alcohol intoxication dynamics in young adults engaged in heavy episodic drinking. CULTA merges the common and unique trait-state model with latent transition analysis to separate stable, traitlike intoxication components from transient fluctuations while modeling transitions between distinct drinking profiles. Method: A sample of 222 young adults wore transdermal alcohol concentration sensors for 6 days, capturing real-time alcohol levels. The CULTA model decomposed intoxication variability into common and unique influences across four transdermal alcohol concentration features—peak, rise rate, fall rate, and duration. Latent intoxication profiles were identified, and transition probabilities between profiles were estimated with a focus on the influence of alcohol use disorder risk measured by the Alcohol Use Disorders Identification Test. Results: Two latent intoxication profiles emerged. The first, chronic heavy episodic drinking, was characterized by persistently high intoxication without significant inertia, while the second, inertia-driven drinking, featured moderate episodic intoxication with a strong autoregressive effect, reflecting lingering intoxication that dissipates over time. Individuals with higher Alcohol Use Disorders Identification Test scores were more likely to remain in or transition to the chronic heavy episodic drinking profile. Although peak intoxication and rise rate showed limited individual variability, fall rate and duration varied substantially, marking them as potential targets for intervention. Conclusions: CULTA advances our understanding of alcohol intoxication by distinguishing stable from transient influences and modeling transitions between drinking states. These findings suggest that interventions should address both persistent and situational aspects of intoxication—especially by reducing duration and fall rate—and encourage research across longer periods and populations.

Pesigan et al.: Inferences and effect sizes for direct, indirect, and total effects in continuous-time mediation models **Pesigan-Russell-Chow-2025a**

Ivan Jacob Agaloos Pesigan, Michael A. Russell, and Sy-Miin Chow. “Inferences and effect sizes for

direct, indirect, and total effects in continuous-time mediation models". In: *Psychological Methods* (Oct. 2, 2025). ISSN: 1082-989X. DOI: 10.1037/met0000779.

Abstract: Mediation modeling using longitudinal data is an exciting field that captures the interrelations in dynamic changes, such as mediated changes, over time. Even though discrete-time vector autoregressive approaches are commonly used to estimate indirect effects in longitudinal data, they have known limitations due to the dependency of inferential results on the time intervals between successive occasions and the assumption of regular spacing between measurements. Continuous-time vector autoregressive models have been proposed as an alternative to address these issues. Previous work in the area (e.g., Deboeck & Preacher, 2015; Ryan & Hamaker, 2021) has shown how the direct, indirect, and total effects, for a range of time-interval values, can be calculated using parameters estimated from continuous-time vector autoregressive models for causal inferential purposes. However, both standardized effects size measures and methods for calculating the uncertainty around the direct, indirect, and total effects in continuous-time mediation have yet to be explored. Drawing from the mediation model literature, we present and compare results using the delta, Monte Carlo, and parametric bootstrap methods to calculate SEs and confidence intervals for the direct, indirect, and total effects in continuous-time mediation for inferential purposes. Options to automate these inferential procedures and facilitate interpretations are available in the `cTMed R` package.

Pesigan et al.: From person-specific dynamics to population inference: A computationally efficient meta-analytic structural equation modeling framework for integrating complex multilevel dynamic models **Pesigan-Russell-Chow-2026**

Ivan Jacob Agaloos Pesigan, Michael A. Russell, and Sy-Miin Chow. "From person-specific dynamics to population inference: A computationally efficient meta-analytic structural equation modeling framework for integrating complex multilevel dynamic models". Manuscript under review. 2026.

Pesigan et al.: betaDelta and betaSandwich: Confidence intervals for standardized regression coefficients in R **Pesigan-Sun-Cheung-2023**

Ivan Jacob Agaloos Pesigan, Rong Wei Sun, and Shu Fai Cheung. "betaDelta and betaSandwich: Confidence intervals for standardized regression coefficients in R". In: *Multivariate Behavioral Research* 58.6 (Apr. 2023), pp. 1183–1186. DOI: 10.1080/00273171.2023.2201277.

Abstract: The multivariate delta method was used by Yuan and Chan to estimate standard errors and confidence intervals for standardized regression coefficients. Jones and Waller extended the earlier work to situations where data are nonnormal by utilizing Browne's asymptotic distribution-free (ADF) theory. Furthermore, Dudgeon developed standard errors and confidence intervals, employing heteroskedasticity-consistent (HC) estimators, that are robust to nonnormality with better performance in smaller sample sizes compared to Jones and Waller's ADF technique. Despite these advancements, empirical research has been slow to adopt these methodologies. This can be a result of the dearth of user-friendly software programs to put these techniques to use. We present the betaDelta and the betaSandwich packages in the R statistical software environment in this manuscript. Both the normal-theory approach and the ADF approach put forth by Yuan and Chan and Jones and Waller are implemented by the betaDelta package. The HC approach proposed by Dudgeon is implemented by the betaSandwich package. The use of the packages is demonstrated with an empirical example. We think the packages will enable applied researchers to accurately assess the sampling variability of standardized regression coefficients.

Pick et al.: Describing posterior distributions of variance components: Problems and the use of null distributions to aid interpretation **Pick-Kasper-Allegue-etal-2023**

Joel L. Pick et al. "Describing posterior distributions of variance components: Problems and the use of null distributions to aid interpretation". In: *Methods in Ecology and Evolution* 14.10 (Oct. 2023), pp. 2557–2574. ISSN: 2041-210X. DOI: 10.1111/2041-210x.14200.

Abstract: Assessing the biological relevance of variance components estimated using Markov chain Monte Carlo (MCMC)-based mixed-effects models is not straightforward. Variance estimates are constrained to be greater than zero and their posterior distributions are often asymmetric. Different measures of central tendency for these distributions can therefore vary widely, and credible intervals cannot overlap zero, making it difficult to assess the size and statistical support for among-group variance. Statistical support is often assessed through visual inspection of the whole posterior distribution and so relies on subjective decisions for interpretation. We use simulations to demonstrate the difficulties of summarizing the posterior distributions of variance estimates from MCMC-based models. We then describe different methods for generating the expected null distribution (i.e. a distribution of effect sizes that would be obtained if there was no among-group variance) that can be used to aid in the interpretation of variance estimates. Through comparing commonly used summary statistics of posterior distributions of variance components, we show that the posterior median is predominantly the least biased. We further show how null distributions can be used to derive a p -value that provides complementary information to the commonly presented measures of central tendency and uncertainty. Finally, we show how these p -values facilitate the implementation of power analyses within an MCMC framework. The use of null distributions for variance components can aid study design and the interpretation of results from MCMC-based models. We hope that this manuscript will make empiricists using mixed models think more carefully about their results, what descriptive statistics they present and what inference they can make.

Rajh-Weber et al.: A practice-oriented guide to statistical inference in linear modeling for non-normal or heteroskedastic error distributions

RajhWeber-Huber-Arendasy-2025

Hanna Rajh-Weber, Stefan Ernest Huber, and Martin Arendasy. "A practice-oriented guide to statistical inference in linear modeling for non-normal or heteroskedastic error distributions". In: *Behavior Research Methods* 57.12 (Nov. 2025). ISSN: 1554-3528. DOI: 10.3758/s13428-025-02801-4.

Abstract: Selecting an appropriate statistical method is a challenge frequently encountered by applied researchers, especially if assumptions for classical, parametric approaches are violated. To provide some guidelines and support, we compared classical hypothesis tests with their typical distributional assumptions of normality and homoskedasticity with common and easily accessible alternative inference methods (HC3, HC4, and six bootstrap methods) in the framework of ordinary least squares (OLS) regression. The method's performance was assessed for four different regression models with varying levels of non-normality and heteroskedasticity of errors, and for five different sample sizes ranging from 25 to 500 cases. For each scenario, 10,000 samples of observations were generated. Type I error and coverage rates, power, and standard error bias were examined to assess the methods' performance. No method considered here performed satisfactorily on all accounts. Using HC3 or HC4 standard errors, or a wild bootstrap procedure with percentile confidence intervals, could yield reliable results in many, but not all, scenarios. We suppose that, in the case of assumption violations, researchers might refer to a method that performed best in a scenario most similar to their data situation. To aid the selection of an appropriate method, we provide tables comparing relative performances in all considered scenarios.

Rajh-Weber et al.: Using heteroskedasticity-consistent standard errors and the bootstrap for linear regression analysis available in SPSS: A tutorial

RajhWeber-Huber-Arendasy-2026

Hanna Rajh-Weber, Stefan Ernest Huber, and Martin Arendasy. "Using heteroskedasticity-consistent standard errors and the bootstrap for linear regression analysis available in SPSS: A tutorial". In: *Advances in Methods and Practices in Psychological Science* 9.1 (Jan. 2026). ISSN: 2515-2459. DOI: 10.1177/25152459251408046.

Abstract: In the landscape of statistical software, from customizable programming-language-based to point-and-click systems, SPSS remains a popular choice among researchers. In SPSS, analyses with conventional methods, such as ordinary least squares regression, can be easily performed. However, violated assumptions, such as homoskedasticity or normality of the errors, can lead to altered

Type I error rates or a reduction in statistical power. SPSS provides a multitude of alternative inference methods associated with linear regression, but accessing them is not always straightforward. To facilitate data analysis when assumptions for conventional inference methods are not met, in this tutorial, we aim to provide applied researchers, particularly SPSS users, with a guide for performing linear regression analyses using heteroskedasticity-consistent (HC) standard errors (HC3 and HC4) and two different bootstrap resampling methods (pairs bootstrap and wild bootstrap). Each bootstrap method can further be combined with a bootstrap p value, a percentile confidence interval, or a bias-corrected and accelerated confidence interval. For illustration, the methods are then compared using a computer-generated data set. Although the focus of this article is on applied researchers who use mainly SPSS for their analyses, a tutorial on how to do everything shown here in R (with custom functions) is included in the supplementary materials.

Ray et al.: Capturing habitualness of drinking and smoking behavior in humans

Ray-Du-Grodin-etal-2020

Lara A. Ray et al. "Capturing habitualness of drinking and smoking behavior in humans". In: *Drug and Alcohol Dependence* 207 (Feb. 2020), p. 107738. ISSN: 0376-8716. DOI: 10.1016/j.drugalcdep.2019.107738.

Abstract: Background: Recent findings suggest that overreliance on habit may be common in individuals diagnosed with addiction. To advance our understanding of habit in clinical samples and from behavioral measures, this study examines the interrelations between self-reported habit index for smoking and drinking as well as behavioral measures of intraindividual variability in smoking and drinking. Methods: Treatment-seeking heavy drinking smokers ($N = 416$) completed the Self-Report Habit Index (SRHI) adapted for both smoking and drinking. "Behavioral habitualness" was computed from the degree of intraindividual variability in patterns of smoking and drinking over the past month. Using the 28-day Timeline-Follow Back (TLFB) interview, we derived two measures of intraindividual variability: interclass correlation (ICC) and autocorrelation [AR(7) coefficients]. Results: Self-report measures of habit were robustly associated with clinical severity of drinking and

smoking with higher habit scores indicating greater severity of drinking and smoking, respectively. ICC and AR(7) coefficients, the behavioral measure of "patterness" and putative habit, were not associated with SRHI scores. While ICC for smoking was associated with higher nicotine dependence scores, this pattern was not found for drinking ICC and alcohol problem severity. Conclusions: These results support the construct validity of the self-report measures of habit for smoking and drinking, as well an initial evaluation of behavioral measure of smoking "patterness" as a potential proxy for habit smoking. Because habit represents a complex phenotype with limited clinical translation, additional studies capturing a wider range of substance use severity and coupled with brain-based validation methods are warranted.

Revol et al.: Episode-contingent experience-sampling designs for accurate estimates of autoregressive dynamics **Revol-Ariens-Lafit-etal-2025**

Jordan Revol et al. "Episode-contingent experience-sampling designs for accurate estimates of autoregressive dynamics". In: *Psychological Methods* (May 2025). ISSN: 1082-989X. DOI: 10.1037/met0000758.

Abstract: Affect dynamics are often studied by means of first-order autoregressive (AR) modeling applied to intensive longitudinal data. A key target in these studies is the AR parameter, which is often tied conceptually to regulatory behavior in the affective process. The data are typically gathered using experience sampling methods, which are designed to pick up on fluctuations in affective variables as they evolve over time in naturalistic settings. In this article, we compare classical time-contingent sampling designs to episode-contingent sampling designs, which initiate sampling when an emotional episode has been signaled. We define emotional episodes as periods where an affective process strays relatively far away from its mean. Compared to time-contingent designs, episode-contingent designs leverage on increased affective variability, which can have beneficial implications for the precision of the ordinary least squares AR effect estimator. Using an extensive simulation study, we attempt to delineate which characteristics of an episode-contingent design are important to consider, and how these characteristics are related to estimation benefits. We then

turn to an empirical illustration, showing how an episode-contingent design can be implemented in practice. We also show that various patterns we expect based on the theoretical parts of the article are recovered in the data. We conclude that episode-contingent designs can have marked benefits for the precision of the AR effect estimator, and discuss a number of challenges when it comes to implementing episode-contingent designs in practice.

Rhemtulla et al.: Worse than measurement error: Consequences of inappropriate latent variable measurement models **Rhemtulla-vanBork-Borsboom-2020**

Mijke Rhemtulla, Riet van Bork, and Denny Borsboom. "Worse than measurement error: Consequences of inappropriate latent variable measurement models". In: *Psychological Methods* 25.1 (Feb. 2020), pp. 30–45. ISSN: 1082-989X. DOI: 10.1037/met0000220.

Abstract: Previous research and methodological advice has focused on the importance of accounting for measurement error in psychological data. That perspective assumes that psychological variables conform to a common factor model. We explore what happens when data that are not generated from a common factor model are nonetheless modeled as reflecting a common factor. Through a series of hypothetical examples and an empirical reanalysis, we show that when a common factor model is misused, structural parameter estimates that indicate the relations among psychological constructs can be severely biased. Moreover, this bias can arise even when model fit is perfect. In some situations, composite models perform better than common factor models. These demonstrations point to a need for models to be justified on substantive, theoretical bases in addition to statistical ones.

Richards et al.: Correspondence between alcohol use measured by a wrist-worn alcohol biosensor and self-report via ecological momentary assessment over a 2-week period **Richards-Barnett-Cook-etal-2023**

Veronica L. Richards, Nancy P. Barnett, et al. "Correspondence between alcohol use measured

by a wrist-worn alcohol biosensor and self-report via ecological momentary assessment over a 2-week period". In: *Alcohol: Clinical and Experimental Research* 47.2 (Feb. 2023), pp. 308–318. ISSN: 2993-7175. DOI: 10.1111/acer.14995.

Abstract: Background: Transdermal alcohol biosensors measure alcohol use continuously, passively, and non-invasively. There is little field research on the Skyn biosensor, a new-generation, wrist-worn transdermal alcohol biosensor, and little evaluation of its sensitivity and specificity and the day-level correspondence between transdermal alcohol concentration (TAC) and number of self-reported drinks. Methods: Participants ($N = 36$; 61% male, $M_{age} = 34.3$) wore the Skyn biosensor and completed ecological momentary assessment (EMA) surveys about their alcohol use over 2 weeks. A total of 497 days of biosensor and EMA data were collected. Skyn-measured drinking episodes were defined by TAC ¿ $5\mu g/L$. Skyn data were compared to self-reported drinking to calculate sensitivity and specificity (for drinking day vs. nondrinking day). Generalized estimating equations models were used to evaluate the correspondence between TAC features (peak TAC and TAC-area under the curve (AUC)) and number of drinks. Individual-level factors (sex, age, race/ethnicity, body mass index, human immunodeficiency virus status, and hazardous drinking) were examined to explore associations with TAC controlling for number of drinks. Results: Using a minimum TAC threshold of $5\mu g/L$ plus coder review, the biosensor had sensitivity of 54.7% and specificity of 94.6% for distinguishing drinking from nondrinking days. Without coder review, the sensitivity was 78.1% and the specificity was 55.2%. Peak TAC ($\beta = 0.92, p < 0.0001$) and TAC-AUC ($\beta = 1.60, p < 0.0001$) were significantly associated with number of drinks. Females had significantly higher TAC levels than males for the same number of drinks. Conclusions: Skyn-derived TAC can be used to measure alcohol use under naturalistic drinking conditions, additional research is needed to accurately identify drinking episodes based on Skyn TAC readings.

Richards et al.: Transdermal alcohol concentration features predict alcohol-induced blackouts in college students **Richards-Glenn-Turrisi-etal-2024**

Veronica L. Richards, Shannon D. Glenn, et al. "Transdermal alcohol concentration features predict

alcohol-induced blackouts in college students". In: *Alcohol, Clinical and Experimental Research* 48.5 (Apr. 2024), pp. 880–888. ISSN: 2993-7175. DOI: 10.1111/acer.15290.

Abstract: Background: Alcohol-induced blackouts (AIBs) are common in college students. Individuals with AIBs also experience acute and chronic alcohol-related consequences. Research suggests that how students drink is an important predictor of AIBs. We used transdermal alcohol concentration (TAC) sensors to measure biomarkers of increasing alcohol intoxication (rise rate, peak, and rise duration) in a sample of college students. We hypothesized that the TAC biomarkers would be positively associated with AIBs. Methods: Students were eligible to participate if they were aged 18-22 years, in their second or third year of college, reported drinking 4+ drinks on a typical Friday or Saturday, experienced ≥ 1 AIB in the past semester, owned an iPhone, and were willing to wear a sensor for 3 days each weekend. Students ($N = 79$, 55.7% female, 86.1% White, $M_{age} = 20.1$) wore TAC sensors and completed daily diaries over four consecutive weekends (89.9% completion rate). AIBs were assessed using the Alcohol-Induced Blackout Measure-2. Logistic multilevel models were conducted to test for main effects. Results: Days with faster TAC rise rates (OR = 2.69, 95% CI: 1.56, 5.90), higher peak TACs (OR = 2.93, 95% CI: 1.64, 7.11), and longer rise TAC durations (OR = 4.16, 95% CI: 2.08, 10.62) were associated with greater odds of experiencing an AIB. Conclusions: In a sample of "risky" drinking college students, three TAC drinking features identified as being related to rising intoxication independently predicted the risk for daily AIBs. Our findings suggest that considering how an individual drinks (assessed using TAC biomarkers), rather than quantity alone, is important for assessing risk and has implications for efforts to reduce risk. Not only is speed of intoxication important for predicting AIBs, but the height of the peak intoxication and the time spent reaching the peak are important predictors, each with different implications for prevention.

Richards et al.: Profiles of transdermal alcohol concentration and their prediction of negative and positive alcohol-related consequences in young adults' natural settings

Richards-Mallett-Turrisi-etal-2025

Veronica L. Richards, Kimberly A. Mallett, et al. "Profiles of transdermal alcohol concentration and their prediction of negative and positive alcohol-related consequences in young adults' natural settings". In: *Psychology of Addictive Behaviors* 39.2 (Jan. 2025), pp. 163–172. ISSN: 0893-164X. DOI: 10.1037/adb0001054.

Abstract: Objective: Transdermal alcohol concentration (TAC) sensors provide a multidimensional characterization of drinking events that self-reports cannot. These profiles may differ in their associated day-level alcohol-related consequences, but no research has tested this. We address this using multilevel latent profile analysis. Method: Two hundred twenty-two young adults who regularly engage in heavy drinking ($M_{\text{age}} = 22.3$, 64% female, 79% non-Hispanic White) responded to surveys and wore TAC sensors for 6 consecutive days. We tested whether four previously identified TAC profiles: (1) high-fast (8.5% of days), (2) moderate-fast (12.8%), (3) low-slow (20.4%), and (4) little-to-no-drinking days (58.2%) differed in numbers of negative and positive consequences and in the odds that both consequence types occurred on the same day. Results: High-fast (incident rate ratio [$\text{IRR}_{\text{low-slow}}$] = 6.18; $\text{IRR}_{\text{little-to-no-drinking}} = 9.47$) and moderate-fast ($\text{IRR}_{\text{low-slow}} = 3.71$; $\text{IRR}_{\text{little-to-no-drinking}} = 5.68$) days contained more negative consequences compared to low-slow and little-to-no-drinking days. High-fast (IRR = 2.05), moderate-fast (IRR = 1.88), and low-slow (IRR = 1.43) days contained more positive consequences than little-to-no-drinking days. The odds of having only positive consequences were highest on low-slow, $\chi^2(3) = 9.10$, $p < .05$, days but the odds of experiencing both consequence types increased on moderate-fast and high-fast days, $\chi^2(3) = 39.63$, $p < .001$. Conclusions: Compared to little-to-no-drinking days, TAC profiles indicative of drinking (high-fast, moderate-fast, and low-slow) contained more negative and positive consequences. However, the odds of experiencing only positive consequences were highest among low-slow days and decreased on moderate-fast and high-fast days as the odds of negative consequences rose. These findings provide novel evidence reinforcing harm reduction approaches that

seek to maximize positives and minimize negatives of alcohol consumption through emphasis on slow-paced, low-volume drinking.

Richards et al.: Subjective intoxication predicts alcohol-related consequences at equivalent alcohol concentrations in young adults using ecological momentary assessment and alcohol sensors **Richards-Turrisi-Russell-2024**

Veronica L. Richards, Robert J. Turrisi, and Michael A. Russell. "Subjective intoxication predicts alcohol-related consequences at equivalent alcohol concentrations in young adults using ecological momentary assessment and alcohol sensors". In: *Psychology of Addictive Behaviors* 38.3 (May 2024), pp. 334–346. ISSN: 0893-164X. DOI: 10.1037/adb0000993.

Abstract: Objective: Subjective intoxication (SI) when drinking may serve as an internal barometer of whether to continue drinking or engage in potentially unsafe behavior. Mobile assessments offer the potential to use SI as a prospective risk indicator during drinking episodes; little evidence exists for the validity of real-time SI measures. We test the correspondence of SI with estimated blood alcohol concentration and transdermal alcohol concentration (TAC) in young adults' natural settings. We provide a novel test of whether SI features (peak and mean SI) uniquely predict consequences adjusting for alcohol concentration. Method: Two hundred twenty-two heavy-drinking young adults (M_{age} = 22.3, 64% female, 79% non-Hispanic White, 84% undergraduates) participated in a 6-day study that used ecological momentary assessment of drinking and TAC sensors. SI was assessed every 30 min during drinking episodes. Multilevel modeling was used to test hypotheses. Results: Momentary SI and estimated blood alcohol concentration had moderate associations at the moment and day levels (standardized βs = 0.5-0.6); SI was moderately associated with TAC at the day level (βs = 0.5). Associations between SI and alcohol concentration varied widely between persons and across days. Day-level SI features predicted consequences when adjusting for alcohol concentration (incidence rate ratios, IRRs = 1.29-1.70). Conclusions: Our two-item SI measure shows evidence of validity in real-world settings with heavy-drinking young adults. SI was significantly correlated with alcohol concentration and was a unique predictor of consequences. The strength of these as-

sociations varied greatly across persons and days. Real-time SI measurement may be useful in preventive interventions, but continued research is needed into when and for whom momentary SI is most predictive of risk.

Ringwald et al.: Characterizing stress processes by linking big five personality states, traits, and day-to-day stressors **Ringwald-Nielsen-Mostajabi-etal-2024**

Whitney R. Ringwald et al. "Characterizing stress processes by linking big five personality states, traits, and day-to-day stressors". In: *Journal of Research in Personality* 110 (June 2024), p. 104487. ISSN: 0092-6566. DOI: 10.1016/j.jrp.2024.104487.

Abstract: The accumulation of day-to-day stressors can impact mental and physical health. How people respond to stressful events is a key mechanism responsible for the effects of stress, and individual differences in stress responses can either perpetuate or prevent negative consequences. Most research on daily stress processes has focused on affective responses to stressors, but stress responses can involve more than just affect (e.g., behavior, cognitions). Additionally, most research has studied the role of neuroticism in shaping those responses, but many other individual differences are associated with stress. In this study, we more broadly characterized daily stress processes by expanding the nomological networks of stress responses to include Big Five personality states. We also linked those stress responses to all Big Five traits, as well as individual differences in stress variety, severity, and controllability. We studied a sample of participants (N = 1,090) who reported on stressful events, their appraisal of events in terms of severity and controllability, and their Big Five personality states daily for 8–10 days (N = 8,870 observations). Multi-level structural equation models were used to separate how characteristics of the perceived stressful situation and characteristics of the person play into daily stress processes. Results showed that (1) all Big Five personality states shift in response to perceived stress, (2) all Big Five personality traits relate to average levels of perceived stress variety, severity, and controllability, and (3) individual differences in personality and average perceived stress variety and perceived severity relate to the strength of personality state responses to daily stress, albeit in a more limited fashion. Our results point to

new pathways by which stressors affect people in everyday life and begin to clarify processes that may explain individual differences in risk or resilience to the harmful effects of stress.

Rousselet et al.: The percentile bootstrap: A primer with step-by-step instructions in R **Rousselet-Pernet-Wilcox-2021**

Guillaume A. Rousselet, Cyril R. Pernet, and Rand R. Wilcox. "The percentile bootstrap: A primer with step-by-step instructions in R". In: *Advances in Methods and Practices in Psychological Science* 4.1 (Jan. 2021), pp. 1–10. DOI: 10.1177/2515245920911881.

Abstract: The percentile bootstrap is the Swiss Army knife of statistics: It is a nonparametric method based on data-driven simulations. It can be applied to many statistical problems, as a substitute to standard parametric approaches, or in situations for which parametric methods do not exist. In this Tutorial, we cover `R` code to implement the percentile bootstrap to make inferences about central tendency (e.g., means and trimmed means) and spread in a one-sample example and in an example comparing two independent groups. For each example, we explain how to derive a bootstrap distribution and how to get a confidence interval and a p value from that distribution. We also demonstrate how to run a simulation to assess the behavior of the bootstrap. For some purposes, such as making inferences about the mean, the bootstrap performs poorly. But for other purposes, it is the only known method that works well over a broad range of situations. More broadly, combining the percentile bootstrap with robust estimators (i.e., estimators that are not overly sensitive to outliers) can help users gain a deeper understanding of their data than they would using conventional methods.

Rowland et al.: Mindfulness and affect-network density: Does mindfulness facilitate disengagement from affective experiences in daily life? **Rowland-Wenzel-2020**

Zarah Rowland and Mario Wenzel. "Mindfulness and affect-network density: Does mindfulness

facilitate disengagement from affective experiences in daily life?" In: *Mindfulness* 11.5 (Feb. 2020), pp. 1253–1266. ISSN: 1868-8535. DOI: 10.1007/s12671-020-01335-4.

Abstract: Objectives: Mindfulness has been positively associated with affective well-being, which may be explained by improved affective disengagement from prior experiences. Methods: We used a dynamic network approach to represent temporal interconnections between specific affective states (i.e., network density)—with less interconnected states indicating better affective disengagement—to investigate whether mindfulness predicted a lower network density. One-hundred twenty-five undergraduate students (Mage $= 22.87; SD = 5.06$ years; 77.6% female) completed a 40-day ambulatory assessment, reporting on their affective experiences and state mindfulness six times a day, and either practiced mindfulness or were on a wait-list (control condition). Results: We found that neither the mindfulness training, $\beta = -.11, p = .222$, nor state mindfulness, $\beta = -.06, p = .484$, were associated with affective network density. However, given prior research indicating the importance of prior meditation experience, we found that participants with prior meditation experience reported lower affective network density when they reported higher state mindfulness than participants without meditation experiences, $\beta = -.20, p = .048$. Conclusion: The present study found that neither high levels of mindfulness nor a mindfulness training automatically improved affective disengagement in the form of lower network density. Rather, it seems that only those who were already familiar with practicing mindfulness and who were mindful in their daily life experienced easier disengagement from previous affective experiences. These findings may point toward a different, more complex, understanding of mindfulness in experienced meditators.

Rush et al.: Changes in daily stress reactivity and changes in physical health across 18 years of adulthood **Rush-Charles-Willroth-etal-2025**

Jonathan Rush et al. "Changes in daily stress reactivity and changes in physical health across 18 years of adulthood". In: *Annals of Behavioral Medicine* 59.1 (Jan. 2025). ISSN: 1532-4796. DOI: 10.1093/abm/kaae086.

Abstract: Background: Stress plays a pivotal role in physical health. Although many studies have linked stress reactivity (daily within-person associations between stress exposure and negative affect) to physical health outcomes, we know surprisingly little about how changes in stress reactivity are related to changes in physical health. Purpose: The current study examines how change in stress reactivity over 18 years is related to changes in functional health and chronic health conditions. Methods: Three measurement bursts from the National Study of Daily Experiences ($N = 2880$; 55% female) each included daily measures of stressor exposure and negative affect across 8 consecutive days, yielding 33,944 days of data across 18 years of adulthood. At each wave, participants reported their functional health limitations (ie, basic activities of daily living [ADL] and instrumental activities of daily living [IADL]) and chronic health conditions. Multilevel structural equation models simultaneously modeled stress reactivity at Level 1, longitudinal changes in stress reactivity at Level 2, and the association between changes in stress reactivity and changes in functional limitations and chronic conditions at Level 3. Results: Higher levels of stress reactivity at baseline were associated with more functional health limitations 18 years later (ADLs: Est. = 0.90, $P = .001$; IADLs: Est. = 1.78, $P < .001$). Furthermore, individuals who increased more in their stress reactivity across the 18-year period also showed greater increases in their functional health limitations (ADLs: Est. = 4.02, $P = .017$; IADLs: Est. = 5.74, $P < .001$) and chronic conditions (Est. = 11.17, $P = .008$). Conclusions: These findings highlight the strong connection between health and stress in daily life, and how they travel together across adulthood.

Russell et al.: Affect relative to day-level drinking initiation: Analyzing ecological momentary assessment data with multilevel spline modeling

Russell-LindenCarmichael-Lanza-etal-2020

Michael A. Russell, Ashley N. Linden-Carmichael, et al. "Affect relative to day-level drinking initiation: Analyzing ecological momentary assessment data with multilevel spline modeling". In: *Psychology of Addictive Behaviors* 34.3 (May 2020), pp. 434–446. ISSN: 0893-164X. DOI: 10.1037/adb0000550.

Abstract: Affect regulation models state that affect both motivates and reinforces alcohol use. We aimed to examine whether affect levels and rates of change differed across drinking versus nondrinking days in a manner consistent with affect regulation models. Four hundred four regularly drinking adults, aged 18-70 years, completed ecological momentary assessments over 3 weeks. Participants provided positive affect (PA; enthusiastic, excited, happy) and negative affect (NA; distressed, sad) reports during all prompts; alcohol consumption reports were also provided. Multilevel spline models revealed that on drinking days, PA was higher and NA was lower both before and after drinking compared to matched times on nondrinking days. PA and NA were also higher and lower, respectively, both before and after drinking, when heavy drinking days were compared to moderate drinking days. Examination of affect rates of change revealed that (a) accelerating increases in PA and accelerating decreases in NA preceded drinking initiation, (b) PA increases and NA decreases were seen up to 2 hr after drinking initiation, and (c) pre- and postdrinking PA increases were larger on heavy versus moderate drinking days, whereas only postdrinking NA decreases were larger on heavy drinking days. Results supported affect regulation models while adding nuance, showing accelerating changes in predrinking affect on drinking days and pre- and postdrinking differences in affect levels and rates of change across days of varying drinking intensity. Beyond theory, our results suggest that accelerating changes in affect may provide a clue to future commencement of heavy drinking, which may aid momentary intervention development.

Russell et al.: Adolescents' subjective social status predicts day-to-day mental health and future substance use **Russell-Odgers-2020**

Michael A. Russell and Candice L. Odgers. "Adolescents' subjective social status predicts day-to-day mental health and future substance use". In: *Journal of Research on Adolescence* 30.S2 (Feb. 2020), pp. 532–544. ISSN: 1532-7795. DOI: 10.1111/jora.12496.

Abstract: Adolescents' subjective social status (SSS) is associated with mental and behavioral health outcomes, independent of socioeconomic status (SES). Many previous findings, however, come from cross-sectional studies. We report results from a longitudinal study with 151 adolescents identified

as at risk for early substance use and behavioral problems sampled from low-SES neighborhoods. We examined whether adolescent's SSS predicted mental health (depression, anxiety, and inattention/impulsivity) measured over 30 days via ecological momentary assessment and risk for substance use at an 18-month follow-up. Results showed that with each perceived step "up" the SSS ladder, adolescents experienced fewer mental health symptoms in daily life and lower future substance use risk after adjusting for objective SES and previous psychopathology. Implications of these findings are discussed.

Russell et al.: Profiles of alcohol intoxication and their associated risks in young adults' natural settings: A multilevel latent profile analysis applied to daily transdermal alcohol concentration data **Russell-Richards-Turrisi-etal-2025**

Michael A. Russell, Veronica L. Richards, et al. "Profiles of alcohol intoxication and their associated risks in young adults' natural settings: A multilevel latent profile analysis applied to daily transdermal alcohol concentration data". In: *Psychology of Addictive Behaviors* 39.2 (Mar. 2025), pp. 173–185. ISSN: 0893-164X. DOI: 10.1037/adb0001022.

Abstract: Objective: Transdermal alcohol concentration (TAC) sensors capture aspects of drinking events that self-reports cannot. The multidimensional nature of TAC data allows novel classification of drinking days and identification of associated behavioral and contextual risks. We used multilevel latent profile analysis (MLPA) to create day-level profiles of TAC features and test their associations with (a) daily behaviors and contexts and (b) risk for alcohol use disorders at baseline. Method: Two hundred twenty-two regularly heavy-drinking young adults (Mage = 22.3) completed the Alcohol Use Disorders Identification Test (AUDIT) at baseline and then responded to mobile phone surveys and wore TAC sensors for six consecutive days. MLPA identified day-level profiles using four TAC features (peak, rise rate, fall rate, and duration). TAC profiles were tested as correlates of daily drinking behaviors, contexts, and baseline AUDIT. Results: Four profiles emerged: (a) high-fast (8.5% of days), (b) moderate-fast (12.8%), (c) low-slow (20.4%), and (d) little-to-no drinking days (58.2%). Profiles differed in the odds of risky drinking behaviors and contexts. The highest risk

occurred on high-fast days, followed by moderate-fast, low-slow, and little-to-no drinking days. Higher baseline AUDIT predicted higher odds of high-fast and moderate-fast days. Conclusions: Days with high and fast intoxication are reflective of high-risk drinking behaviors and were most frequent among those at risk for alcohol use disorders. TAC research using MLPA may offer novel and important insights to intervention efforts.

Russell et al.: Baseline protective behavioral strategy use predicts more moderate transdermal alcohol concentration dynamics and fewer negative consequences of drinking in young adults' natural settings. Russell-Smyth-Turrisi-etal-2024

Michael A. Russell, Joshua M. Smyth, et al. "Baseline protective behavioral strategy use predicts more moderate transdermal alcohol concentration dynamics and fewer negative consequences of drinking in young adults' natural settings." In: *Psychology of Addictive Behaviors* 38.3 (May 2024), pp. 347–359. ISSN: 0893-164X. DOI: 10.1037/adb0000941.

Abstract: Objective: Test whether frequent protective behavioral strategies (PBS) users report (a) fewer alcohol-related consequences and (b) less risky alcohol intoxication dynamics (measured via transdermal alcohol concentration [TAC] sensor "features") in daily life. Method: Two hundred twenty-two frequently heavy-drinking young adults (M_{age} = 22.3 years) wore TAC sensors for 6 consecutive days. TAC features peak (maximum TAC), rise rate (speed of TAC increase), and area under the curve (AUC) were derived for each day. Negative alcohol-related consequences were measured in the morning after each self-reported drinking day. Past-year PBS use was measured at baseline. Results: Young adults reporting more frequent baseline PBS use showed (a) fewer alcohol-related consequences and (b) lower intoxication dynamics on average (less AUC, lower peaks, and slower rise rates). Limiting/stopping and manner of drinking PBS showed the same pattern of findings as the total score. Serious harm reduction PBS predicted fewer negative alcohol-related consequences, but not TAC features. Multilevel path models showed that TAC features peak and rise rate partially explained associations between PBS (total, limiting/stopping, and manner of drinking) and consequences. Independent contributions of PBS subscales were small and

nonsignificant, suggesting that total PBS use was a more important predictor of risk/protection than the specific types of PBS used. Conclusions: Young adults using more total PBS may experience fewer alcohol-related consequences during real-world drinking episodes in part through less risky intoxication dynamics (TAC features). Future research measuring PBS at the daily level is needed to formally test TAC features as day-level mechanisms of protection from acute alcohol-related consequences.

Russell et al.: Transdermal sensor features correlate with ecological momentary assessment drinking reports and predict alcohol-related consequences in young adults' natural settings **Russell-Turrisi-Smyth-2022**

Michael A. Russell, Robert J. Turrisi, and Joshua M. Smyth. "Transdermal sensor features correlate with ecological momentary assessment drinking reports and predict alcohol-related consequences in young adults' natural settings". In: *Alcoholism: Clinical and Experimental Research* 46.1 (Jan. 2022), pp. 100–113. ISSN: 1530-0277. DOI: 10.1111/acer.14739.

Abstract: Background: Wearable transdermal alcohol concentration (TAC) sensors allow passive monitoring of alcohol concentration in natural settings and measurement of multiple features from drinking episodes, including peak intoxication level, speed of intoxication (absorption rate) and elimination, and duration. These passively collected features extend commonly used self-reported drink counts and may facilitate the prediction of alcohol-related consequences in natural settings, aiding risk stratification and prevention efforts. Method: A total of 222 young adults aged 21-29 ($M_{\text{age}} = 22.3$, 64 female, 79% non-Hispanic white, 84% undergraduates) who regularly drink heavily participated in a 5-day study that included the ecological momentary assessment (EMA) of alcohol consumption (daily morning reports and participant-initiated episodic EMA sequences) and the wearing of TAC sensors (SCRAM-CAM anklets). The analytic sample contained 218 participants and 1274 days (including 554 self-reported drinking days). Five features—area under the curve (AUC), peak TAC, rise rate (rate of absorption), fall rate (rate of elimination), and duration—were extracted from TAC-positive trajectories for each drinking day. Day- and person-level associations

of TAC features with drink counts (morning and episodic EMA) and alcohol-related consequences were tested using multilevel modeling. Results: TAC features were strongly associated with morning drink reports (r = 0.60.7) but only moderately associated with episodic EMA drink counts (r = 0.30.5) at both day and person levels. Higher peaks, larger AUCs, faster rise rates, and faster fall rates were significantly predictive of day-level alcohol-related consequences after adjusting for both morning and episodic EMA drink counts in separate models. Person means of TAC features added little above daily scores to the prediction of alcohol-related consequences. Conclusions: These results support the utility of TAC sensors in studies of alcohol misuse among young adults in natural settings and outline the specific TAC features that contribute to the day-level prediction of alcohol-related consequences. TAC sensors provide a passive option for obtaining valid and unique information predictive of drinking risk in natural settings.

Ryan et al.: Time to intervene: A continuous-time approach to network analysis and centrality **Ryan-Hamaker-2022**

Oisin Ryan and Ellen L. Hamaker. "Time to intervene: A continuous-time approach to network analysis and centrality". In: *Psychometrika* 87.1 (Mar. 2022), pp. 214–252. DOI: 10.1007/s11336-021-09767-0.

Abstract: Network analysis of ESM data has become popular in clinical psychology. In this approach, discrete-time (DT) vector auto-regressive (VAR) models define the network structure with centrality measures used to identify intervention targets. However, VAR models suffer from time-interval dependency. Continuous-time (CT) models have been suggested as an alternative but require a conceptual shift, implying that DT-VAR parameters reflect total rather than direct effects. In this paper, we propose and illustrate a CT network approach using CT-VAR models. We define a new network representation and develop centrality measures which inform intervention targeting. This methodology is illustrated with an ESM dataset.

Savalei et al.: Computational options for standard errors and test statistics with incomplete normal and nonnormal data in SEM **Savalei-Rosseel-2022**

Victoria Savalei and Yves Rosseel. "Computational options for standard errors and test statistics with incomplete normal and nonnormal data in SEM". In: *Structural Equation Modeling: A Multidisciplinary Journal* 29.2 (Mar. 2022), pp. 163–181. DOI: 10.1080/10705511.2021.1877548.

Abstract: This article provides an overview of different computational options for inference following normal theory maximum likelihood (ML) estimation in structural equation modeling (SEM) with incomplete normal and nonnormal data. Complete data are covered as a special case. These computational options include whether the information matrix is observed or expected, whether the observed information matrix is estimated numerically or using an analytic asymptotic approximation, and whether the information matrix and the outer product matrix of the score vector are evaluated at the saturated or at the structured estimates. A variety of different standard errors and robust test statistics become possible by varying these options. We review the asymptotic properties of these computational variations, and we show how to obtain them using lavaan in R. We hope that this article will encourage methodologists to study the impact of the available computational options on the performance of standard errors and test statistics in SEM.

Schneider et al.: Estimating reliabilities and correcting for sampling error in indices of within-person dynamics derived from intensive longitudinal data **Schneider-Junghaenel-2023**

Stefan Schneider and Doerte U. Junghaenel. "Estimating reliabilities and correcting for sampling error in indices of within-person dynamics derived from intensive longitudinal data". In: *Behavior Research Methods* 55.7 (Oct. 2023), pp. 3872–3891. ISSN: 1554-3528. DOI: 10.3758/s13428-022-01995-1.

Shao et al.: Seven challenges in affective inertia research **Shao-Ong-2026**

Sijing Shao and Anthony D. Ong. "Seven challenges in affective inertia research". In: *Emotion* 26.5 (Aug. 2026), pp. 1118–1132. DOI: 10.1037/emo0001630.

Abstract: Affective inertia describes the persistence of emotional states over time and has important implications for well-being and emotion regulation. Progress in this area has nevertheless been constrained by conceptual ambiguity, measurement problems, and statistical limitations. The authors identify seven related challenges involving distinctions between inertia and emotional stability, the treatment of ordinal observations with continuous-data models, and assumptions such as stationarity. They review developments in research design, measurement, model estimation, and diagnostic testing that may help address these problems. Greater conceptual and methodological precision should improve the interpretation of affective inertia and strengthen its use in basic and clinically oriented emotion research.

Shaygan et al.: Chronic pain in adolescents: Predicting role of emotional intelligence, self-esteem and parenting style **Shaygan-Karami-2020**

Maryam Shaygan and Zainab Karami. "Chronic pain in adolescents: Predicting role of emotional intelligence, self-esteem and parenting style". In: *International Journal of Community Based Nursing & Midwifery* 8 (2020). DOI: 10.30476/ijcbnm.2020.83153.1129.

Abstract: Background: Pediatric chronic pain is prevalent and disabling. The present study aimed to assess the prevalence of chronic pain among adolescents in Shiraz, Iran. We also compared emotional intelligence, self-esteem and parenting style between adolescents with chronic pain and healthy adolescents. Finally, we examined the predicting role of these variables regarding chronic pain in adolescents. Methods: This cross-sectional study, from January to June 2018, was conducted on 734 adolescents in Shiraz. A clustering sampling method was used. Adolescents with chronic pain were identified by affirmative answers to screening questions based on the International Classification of

Diseases 11th Revision (ICD-11) criteria. Participants completed three validated self-report questionnaires: Trait Emotional Intelligence Questionnaire, Rosenberg self-esteem scale and Baumrind parenting style questionnaire. The data were analyzed through SPSS v.22 software using Mann-Whitney and binary logistic regression tests. P¡0.05 was considered significant. Results: There were 221(30.1%) adolescents who met the ICD-11 criteria of chronic pain. Mann-Whitney tests showed that emotional intelligence (P¡0.001), self-esteem (P¡0.001), authoritative parenting style (P=0.004), and authoritarian parenting style (P=0.006) were significantly different in adolescents with chronic pain compared to healthy adolescents. Binary logistic regression revealed that emotional intelligence (P¡0.001), self-esteem (P¡0.001), authoritarian parenting style (P=0.04) and authoritative parenting style (P=0.01) were significantly correlated with chronic pain after controlling for demographic variables. Conclusion: Our findings indicate that emotional intelligence, self-esteem and parenting styles could be important factors in development or maintenance of chronic pain in adolescents. The results have potential to be extended to future interventions for adolescents with chronic pain.

Sladekova et al.: Sources of bias in general linear models: Evaluating the analytic practice in psychological research **Sladekova-Poupa-Field-2026**

Martina Sladekova, Veronika Poupa, and Andy Field. "Sources of bias in general linear models: Evaluating the analytic practice in psychological research". In: *Royal Society Open Science* 13.2 (Feb. 2026). ISSN: 2054-5703. DOI: 10.1098/rsos.250076.

Abstract: Unbiased estimation and hypothesis testing in general linear models (GLMs) require several conditions to be met, such as adherence to statistical assumptions and absence of influential data points. This study examined the extent to which researchers in psychology attend to these sources of bias when presented with analytic tasks. We recruited 188 psychology researchers who completed two data analytic scenarios. Researchers' analytic scripts and descriptions were subsequently coded based on their level of attentiveness to issues with non-normal model errors, heteroscedasticity, outliers, and influential cases. We found that for all categories, researchers either performed no checks for potential issues or performed checks that were insufficient to detect the

issue at hand. Researchers who teach research methods were more likely to perform the correct checks than non-teaching faculty, post-doctoral researchers and PhD students; however, this difference was small and the highest posterior density intervals overlapped substantially. We discuss the implications of routinely neglecting violated assumptions and influential cases and present the case for more frequent application of robust statistical methods to supplement or replace ordinary least squares GLMs.

Sun et al.: betaselectr: Selective (and proper) standardization in structural equation models **Sun-Chang-Yang-etal-2026**

Rong Wei Sun et al. "betaselectr: Selective (and proper) standardization in structural equation models". In: *Multivariate Behavioral Research* 61.5 (Sept. 2026), pp. 961–966. ISSN: 1532-7906. DOI: 10.1080/00273171.2026.2672692.

Tofighi et al.: Improved inference in mediation analysis: Introducing the model-based constrained optimization procedure **Tofighi-Kelley-2020b**

Davood Tofighi and Ken Kelley. "Improved inference in mediation analysis: Introducing the model-based constrained optimization procedure". In: *Psychological Methods* 25.4 (Aug. 2020), pp. 496–515. DOI: 10.1037/met0000259.

Abstract: Mediation analysis is an important approach for investigating causal pathways. One approach used in mediation analysis is the test of an indirect effect, which seeks to measure how the effect of an independent variable impacts an outcome variable through one or more mediators. However, in many situations the proposed tests of indirect effects, including popular confidence interval-based methods, tend to produce poor Type I error rates when mediation does not occur and, more generally, only allow dichotomous decisions of "not significant" or "significant" with regards to the statistical conclusion. To remedy these issues, we propose a new method, a likelihood ratio test (LRT), that uses non-linear constraints in what we term the model-based constrained

optimization (MBCO) procedure. The MBCO procedure (a) offers a more robust Type I error rate than existing methods; (b) provides a p-value, which serves as a continuous measure of compatibility of data with the hypothesized null model (not just a dichotomous reject or fail-to-reject decision rule); (c) allows simple and complex hypotheses about mediation (i.e., one or more mediators; different mediational pathways), and (d) allows the mediation model to use observed or latent variables. The MBCO procedure is based on a structural equation modeling framework (even if latent variables are not specified) with specialized fitting routines, namely with the use of nonlinear constraints. We advocate using the MBCO procedure to test hypotheses about an indirect effect in addition to reporting a confidence interval to capture uncertainty about the indirect effect because this combination transcends existing methods.

Tofighi et al.: Indirect effects in sequential mediation models: Evaluating methods for hypothesis testing and confidence interval formation **Tofighi-Kelley-2020a**

Davood Tofighi and Ken Kelley. "Indirect effects in sequential mediation models: Evaluating methods for hypothesis testing and confidence interval formation". In: *Multivariate Behavioral Research* 55.2 (Mar. 2020), pp. 188–210. DOI: 10.1080/00273171.2019.1618545.

Abstract: Complex mediation models, such as a two-mediator sequential model, have become more prevalent in the literature. To test an indirect effect in a two-mediator model, we conducted a large-scale Monte Carlo simulation study of the Type I error, statistical power, and confidence interval coverage rates of 10 frequentist and Bayesian confidence/credible intervals (CIs) for normally and nonnormally distributed data. The simulation included never-studied methods and conditions (e.g., Bayesian CI with flat and weakly informative prior methods, two model-based bootstrap methods, and two nonnormality conditions) as well as understudied methods (e.g., profile-likelihood, Monte Carlo with maximum likelihood standard error [MC-ML] and robust standard error [MC-Robust]). The popular BC bootstrap showed inflated Type I error rates and CI under-coverage. We recommend different methods depending on the purpose of the analysis. For testing the null hypothesis of no mediation, we recommend MC-ML, profile-likelihood, and two Bayesian methods. To report

a CI, if data has a multivariate normal distribution, we recommend MC-ML, profile-likelihood, and the two Bayesian methods; otherwise, for multivariate nonnormal data we recommend the percentile bootstrap. We argue that the best method for testing hypotheses is not necessarily the best method for CI construction, which is consistent with the findings we present.

Usami: On the differences between general cross-lagged panel model and random-intercept cross-lagged panel model: Interpretation of cross-lagged parameters and model choice **Usami-2021**

Satoshi Usami. "On the differences between general cross-lagged panel model and random-intercept cross-lagged panel model: Interpretation of cross-lagged parameters and model choice". In: *Structural Equation Modeling: A Multidisciplinary Journal* 28.3 (May 2021), pp. 331–344. ISSN: 1532-8007. DOI: 10.1080/10705511.2020.1821690.

Abstract: Many methods have been developed to infer reciprocal relations between longitudinally observed variables. Among them, the general cross-lagged panel model (GCLM) is the most recent development as a variant of the cross-lagged panel model (CLPM), while the random-intercept CLPM (RI-CLPM) has rapidly become a popular approach. In this article, we describe how common factors and cross-lagged parameters included in these models can be interpreted, using a unified framework that was recently developed. Because common factors are modeled with lagged effects in the GCLM, they have both direct and indirect influences on observed scores, unlike stable trait factors included in the RI-CLPM. This indicates that the GCLM does not control for stable traits as the RI-CLPM does, and that there are interpretative differences in cross-lagged parameters between these models. We also explain that including such common factors as well as moving-average terms in the GCLM makes this interpretation very complicated.

Usami: Within-person variability score-based causal inference: A two-step estimation for joint effects of time-varying treatments **Usami-2023**

Satoshi Usami. "Within-person variability score-based causal inference: A two-step estimation for joint effects of time-varying treatments". In: *Psychometrika* 88.4 (Dec. 2023), pp. 1466–1494. ISSN: 1860-0980. DOI: 10.1007/s11336-022-09879-1.

Abstract: Behavioral science researchers have shown strong interest in disaggregating within-person relations from between-person differences (stable traits) using longitudinal data. In this paper, we propose a method of within-person variability score-based causal inference for estimating joint effects of time-varying continuous treatments by controlling for stable traits of persons. After explaining the assumed data-generating process and providing formal definitions of stable trait factors, within-person variability scores, and joint effects of time-varying treatments at the within-person level, we introduce the proposed method, which consists of a two-step analysis. Within-person variability scores for each person, which are disaggregated from stable traits of that person, are first calculated using weights based on a best linear correlation preserving predictor through structural equation modeling (SEM). Causal parameters are then estimated via a potential outcome approach, either marginal structural models (MSMs) or structural nested mean models (SNMMs), using calculated within-person variability scores. Unlike the approach that relies entirely on SEM, the present method does not assume linearity for observed time-varying confounders at the within-person level. We emphasize the use of SNMMs with G-estimation because of its property of being doubly robust to model misspecifications in how observed time-varying confounders are functionally related to treatments/predictors and outcomes at the within-person level. Through simulation, we show that the proposed method can recover causal parameters well and that causal estimates might be severely biased if one does not properly account for stable traits. An empirical application using data regarding sleep habits and mental health status from the Tokyo Teen Cohort study is also provided.

Van Cappellen et al.: A new micro-intervention to increase the enjoyment and continued practice of meditation **VanCappellen-Catalino-Fredrickson-2020**

Patty Van Cappellen, Lahnna I. Catalino, and Barbara L. Fredrickson. "A new micro-intervention to increase the enjoyment and continued practice of meditation". In: *Emotion* 20.8 (Dec. 2020), pp. 1332–1343. DOI: 10.1037/emo0000684.

Abstract: Maintaining a newly adopted health behavior can be difficult, including the continued practice of meditation. Drawing on the upward spiral theory of lifestyle change, this experiment examined whether positive emotions experienced during an initial meditation session encouraged later practice. Midlife adults were assigned in a two-by-two design to receive either instruction about adaptive ways of prioritizing positivity or control information and then to complete either loving-kindness or mindfulness meditation. Participants subsequently reported their meditation behavior for 21 days. The positivity intervention enhanced enjoyment particularly during loving-kindness meditation, and greater positive emotion during the initial session predicted more frequent and longer subsequent practice. The findings highlight enjoyment as a potentially modifiable contributor to sustained meditation.

van Egmond et al.: Wearable transdermal alcohol monitors: A systematic review of detection validity, and relationship between transdermal and breath alcohol concentration and influencing factors **vanEgmond-Wright-Livingston-etal-2020**

Kelly van Egmond et al. "Wearable transdermal alcohol monitors: A systematic review of detection validity, and relationship between transdermal and breath alcohol concentration and influencing factors". In: *Alcoholism: Clinical and Experimental Research* 44.10 (Oct. 2020), pp. 1918–1932. ISSN: 1530-0277. DOI: 10.1111/acer.14432.

Abstract: Background Research on alcohol consumption mostly relies on self-reported data, which are subject to recall bias. Wearable transdermal alcohol concentration (TAC) monitors address this limitation by continuously measuring the ethanol excreted via the skin. This systematic review aims

to provide an overview of TAC monitors' reliability to detect alcohol consumption and methods to estimate breath alcohol concentration (BrAC) and number of standard drinks consumed in a given time frame. Methods The databases MEDLINE, PsycINFO, SCOPUS, Engineering Village, and CINAHL were systematically searched to identify 1,048 empirical research papers published from 2013 onwards, of which 13 were included after full-text screening. The selected studies included 3 TAC monitors: SCRAM, WristTAS, and Skyn. Results TAC measures of SCRAM, WrisTAS, and Skyn are found to be positively correlated with BrAC ($r = 0.56$ to 0.79) and/or self-reports ($r = 0.62$). Using the AMS criteria for detection results in low sensitivity, adjusted criteria can increase the sensitivity of the SCRAM from 39.9 to 68.5%. The WrisTAS and an early prototype of the Skyn showed high failure rates (17 to 38%). Recent advances toward transforming the TAC data into more clinically relevant measures have led to the development of mathematical models and the BrAC Estimator Software. Using TAC data, both approaches produce estimates explaining 70 to 82% of actual BrAC and self-reported drinking or to highly correlate with the actual BrAC measures ($\beta = 0.90$ to 0.91). Conclusions Transdermal alcohol monitors offer an opportunity to measure alcohol consumption in a valid and continuous way with mathematical models and software estimating BrAC values improving interpretation of TAC data. However, the SCRAM seems unable to detect low-to-moderate drinking levels using the thresholds and criteria set by the manufacturer. Moreover, the WrisTAS and the Skyn prototype show a high failure rate, raising questions about reliability. Future studies will assess the validity of new-generation wristbands, including the next Skyn generations.

van Ginkel: Standardized regression coefficients and newly proposed estimators for R^2 in multiply imputed data **vanGinkel-2020**

Joost R. van Ginkel. "Standardized regression coefficients and newly proposed estimators for R^2 in multiply imputed data". In: *Psychometrika* 85.1 (Mar. 2020), pp. 185–205. ISSN: 1860-0980. DOI: 10.1007/s11336-020-09696-4.

Abstract: Whenever statistical analyses are applied to multiply imputed datasets, specific formulas are needed to combine the results into one overall analysis, also called combination rules. In the context of regression analysis, combination rules for the unstandardized regression coefficients, the t-tests of the regression coefficients, and the F-tests for testing R^2 for significance have long been established. However, there is still no general agreement on how to combine the point estimators of R^2 in multiple regression applied to multiply imputed datasets. Additionally, no combination rules for standardized regression coefficients and their confidence intervals seem to have been developed at all. In the current article, two sets of combination rules for the standardized regression coefficients and their confidence intervals are proposed, and their statistical properties are discussed. Additionally, two improved point estimators of R^2 in multiply imputed data are proposed, which in their computation use the pooled standardized regression coefficients. Simulations show that the proposed pooled standardized coefficients produce only small bias and that their 95% confidence intervals produce coverage close to the theoretical 95%. Furthermore, the simulations show that the newly proposed pooled estimates for R^2 are less biased than two earlier proposed pooled estimates.

Vandekar et al.: A robust effect size index **Vandekar-Tao-Blume-2020**

Simon Vandekar, Ran Tao, and Jeffrey Blume. "A robust effect size index". In: *Psychometrika* 85.1 (Mar. 2020), pp. 232–246. ISSN: 1860-0980. DOI: 10.1007/s11336-020-09698-2.

Abstract: Effect size indices are useful tools in study design and reporting because they are unitless measures of association strength that do not depend on sample size. Existing effect size indices are developed for particular parametric models or population parameters. Here, we propose a robust effect size index based on M-estimators. This approach yields an index that is very generalizable because it is unitless across a wide range of models. We demonstrate that the new index is a function of Cohen's d , R^2 , and standardized log odds ratio when each of the parametric models is correctly specified. We show that existing effect size estimators are biased when the parametric models are incorrect (e.g., under unknown heteroskedasticity). We provide simple formulas to compute power and sample size and use simulations to assess the bias and standard error of the effect size estimator

in finite samples. Because the new index is invariant across models, it has the potential to make communication and comprehension of effect size uniform across the behavioral sciences.

Vehtari et al.: Rank-normalization, folding, and localization: An improved $\widehat{R}$ for assessing convergence of MCMC (with discussion) Vehtari-Gelman-Simpson-etal-2021

Aki Vehtari et al. "Rank-normalization, folding, and localization: An improved $\widehat{R}$ for assessing convergence of MCMC (with discussion)". In: *Bayesian Analysis* 16.2 (June 2021), pp. 667–718. ISSN: 1936-0975. DOI: 10.1214/20-ba1221.

Abstract: Markov chain Monte Carlo is a key computational tool in Bayesian statistics, but it can be challenging to monitor the convergence of an iterative stochastic algorithm. In this paper we show that the convergence diagnostic $\widehat{R}$ of Gelman and Rubin (1992) has serious flaws. Traditional $\widehat{R}$ will fail to correctly diagnose convergence failures when the chain has a heavy tail or when the variance varies across the chains. In this paper we propose an alternative rank-based diagnostic that fixes these problems. We also introduce a collection of quantile-based local efficiency measures, along with a practical approach for computing Monte Carlo error estimates for quantiles. We suggest that common trace plots should be replaced with rank plots from multiple chains. Finally, we give recommendations for how these methods should be used in practice.

L. Wang et al.: Dynamic modeling with intensive longitudinal data: One-step and two-step DSEM approaches Wang-Fang-Bergeman-2026

Lijuan Wang, Yuan Fang, and Cindy S. Bergeman. "Dynamic modeling with intensive longitudinal data: One-step and two-step DSEM approaches". In: *Structural Equation Modeling: A Multidisciplinary Journal* 33.2 (Jan. 2026), pp. 235–251. ISSN: 1532-8007. DOI: 10.1080/10705511.2025.2597284.

Abstract: This study evaluated and compared one-step and two-step dynamic structural equation modeling (DSEM) approaches for analyzing intensive longitudinal data. In one-step DSEM,

the within-person dynamic model and the between-person model are estimated simultaneously. In two-step DSEM, dynamic component estimates are first extracted from the within-person model, followed by estimation of between-person relations in a second step. We further examined whether including person-level variables as auxiliary variables in the first step affects the performance of two-step DSEM. The modeling approaches were evaluated by a simulation study and illustrated with a real data example. Results showed that two-step DSEM without auxiliary variables exhibited estimation bias, low coverage rates, and deflated Type I error rates. In contrast, one-step DSEM and two-step DSEM with auxiliary variables yielded satisfactory results when sufficient data were available (e.g., ≥ 30 time points and ≥ 100 individuals). Implications, modeling recommendations, and future research directions were discussed.

L. Wang et al.: Modeling intraindividual variability as a predictor with intensive longitudinal data **Wang-Liu-2026**

Lijuan Wang and Xiao Liu. "Modeling intraindividual variability as a predictor with intensive longitudinal data". In: *Psychological Methods* 31.3 (Apr. 2026), pp. 516–533. ISSN: 1082-989X. DOI: 10.1037/met0000818.

Abstract: In many areas of psychology, researchers are interested in studying whether intraindividual variability (IIV) is predictive of behavioral and health outcomes after controlling for the intraindividual mean. To the end, IIV indicators such as the observed intraindividual variance (OIVAR) or the observed intraindividual standard deviation (OISD) are often modeled as a predictor in regular regression analysis. However, OIVAR and OISD have been found to have a low-reliability problem, especially when the number of occasions is small. In this study, we analytically examined statistical features (mean and variance) of OIVAR and OISD. The results revealed that when measurement errors exist, regular regression can yield (a) more accurate results for the coefficient of intraindividual variance (IVAR) but (b) worse results for the coefficient of intraindividual standard deviation (ISD) when the number of occasions increases. Furthermore, we compared the performance of alternative modeling approaches, including the time-parceling, single indicator latent variable, and Bayesian

variability modeling approaches, to that of regular regression for modeling IVAR or ISD as a predictor. Simulation results were consistent with the analytical results and further suggested that our proposed (a) time-parceling with bootstrapping and (b) Bayesian variability modeling approaches performed well and better than regression for modeling IVAR as a predictor. When measurement errors exist, only the proposed Bayesian variability modeling approach performed well for modeling ISD as a predictor. Implications of the results and recommendations were discussed.

L. Wang et al.: Investigating the impact of the time interval selection on autoregressive mediation modeling: Result interpretations, effect reporting, and temporal designs

Wang-Zhang-2020

Lijuan Wang and Qian Zhang. "Investigating the impact of the time interval selection on autoregressive mediation modeling: Result interpretations, effect reporting, and temporal designs". In: *Psychological Methods* 25.3 (June 2020), pp. 271–291. DOI: 10.1037/met0000235.

Abstract: This study investigates the impact of the time interval (the time passed between 2 consecutive measurements) selection on autoregressive mediation modeling (AMM). For a widely used autoregressive mediation model, via analytical derivations, we explained why and how the conventionally reported time-specific coefficient estimates (e.g., $\hat{a}\hat{b}$ and $\hat{c}'$) and inference results in AMM provide limited information and can arrive in even misleading conclusions about direct and indirect effects over time. Furthermore, under the stationarity assumption, we proposed an approach to calculate the overall direct and indirect effect estimates over time and the time lag lengths at which they reach maxima, using AMM results. The derivation results revealed that the overall direct and indirect effect curves are asymptotically invariant to the time interval selection, under stationarity. With finite samples and thus sampling errors and potential computing problems, however, our simulation results revealed that the overall indirect effect curves were better recovered when the time interval is selected to be closer to half of the time lag length at which the overall indirect effect reaches its maximum. An R function and an R Shiny app were developed to obtain the overall direct and indirect effect curves over time and facilitate the time interval selection using AMM

results. Our findings provide another look at the connections between AMM and continuous time mediation modeling and the connections are discussed.

Wrzus et al.: Longitudinal coupling of momentary stress reactivity and trait neuroticism: Specificity of states, traits, and age period **Wrzus-Luong-Wagner-etal-2021**

Cornelia Wrzus et al. "Longitudinal coupling of momentary stress reactivity and trait neuroticism: Specificity of states, traits, and age period". In: *Journal of Personality and Social Psychology* 121.3 (Sept. 2021), pp. 691–706. ISSN: 0022-3514. DOI: 10.1037/pspp0000308.

Abstract: Personality traits like neuroticism show both continuity and change across adolescence and adulthood, with most pronounced changes occurring in young adulthood. It has been assumed, but insufficiently examined, that trait changes occur gradually over the years through the accumulation of daily experiences. The current longitudinal measurement burst study examined (a) how changes in average momentary stress reactivity are coupled with changes in trait neuroticism, (b) the extent to which this coupling is specific to stress reactivity and neuroticism, and (c) the extent to which there are age differences in the association between changes in stress reactivity and changes in neuroticism. Participants ($N = 581$; 50% male) between 14 and 86 years of age completed up to 3 waves (T1-T3) of Big Five trait questionnaires and experience-sampling assessments during 6 years. During each three-week experience-sampling period, participants reported their momentary affect and occurrences of hassles on average 55 times. Latent change models showed that increases over time in affective reactivity to daily hassles were associated with increases in neuroticism. This effect was consistent from T1 to T2 as well as from T2 to T3, and most pronounced in young adulthood. Importantly, the results were specific to associations between stress reactivity and neuroticism because changes in frequency of hassles in daily life did not predict changes in neuroticism, and stress reactivity did not consistently predict changes in the other Big Five traits. The findings help to inform theoretical models that outline how short-term states might contribute to gradual longer-term changes in traits like neuroticism.

Xie et al.: Is informal practice associated with outcomes in loving-kindness and compassion training? Evidence from pre-post and daily diary assessments

Xie-Riordan-Baldwin-etal-2024

Qiang Xie et al. "Is informal practice associated with outcomes in loving-kindness and compassion training? Evidence from pre-post and daily diary assessments". In: *Behaviour Research and Therapy* 177 (June 2024), p. 104537. ISSN: 0005-7967. DOI: 10.1016/j.brat.2024.104537.

Abstract: This study examined whether informal meditation practice, defined as applying meditative techniques outside periods of formal meditation, was associated with outcomes during smartphone-based loving-kindness and compassion training. Meditation-naive participants with clinically elevated symptoms completed a two-week intervention, with 351 participants providing measures of psychological distress, loneliness, empathy, and prosociality before and after training. Informal practice, distress, and loneliness were also assessed daily. Greater increases in informal practice were associated with larger pre-post reductions in psychological distress and loneliness but not with changes in empathy or prosociality. Longitudinal models separating within-person fluctuations from longer-term trajectories showed that greater informal practice on a given day predicted lower psychological distress on the following day, whereas a comparable prospective association was not found for loneliness. The findings suggest that informal practice may contribute to some outcomes of app-based loving-kindness and compassion training, particularly psychological distress.

You et al.: A diagnostic procedure for detecting outliers in linear state-space models

You-Hunter-Chen-etal-2020

Dongjun You et al. "A diagnostic procedure for detecting outliers in linear state-space models". In: *Multivariate Behavioral Research* 55.2 (Mar. 2020), pp. 231–255. ISSN: 1532-7906. DOI: 10.1080/00273171.2019.1627659.

Abstract: Outliers can be more problematic in longitudinal data than in independent observations due to the correlated nature of such data. It is common practice to discard outliers as they are

typically regarded as a nuisance or an aberration in the data. However, outliers can also convey meaningful information concerning potential model misspecification, and ways to modify and improve the model. Moreover, outliers that occur among the latent variables (innovative outliers) have distinct characteristics compared to those impacting the observed variables (additive outliers), and are best evaluated with different test statistics and detection procedures. We demonstrate and evaluate the performance of an outlier detection approach for multi-subject state-space models in a Monte Carlo simulation study, with corresponding adaptations to improve power and reduce false detection rates. Furthermore, we demonstrate the empirical utility of the proposed approach using data from an ecological momentary assessment study of emotion regulation together with an open-source software implementation of the procedures.

Zeileis et al.: Various versatile variances: An object-oriented implementation of clustered covariances in R **Zeileis-Koll-Graham-2020**

Achim Zeileis, Susanne Köll, and Nathaniel Graham. "Various versatile variances: An object-oriented implementation of clustered covariances in R". In: *Journal of Statistical Software* 95.1 (Oct. 2020). DOI: 10.18637/jss.v095.i01.

Abstract: Clustered covariances or clustered standard errors are very widely used to account for correlated or clustered data, especially in economics, political sciences, and other social sciences. They are employed to adjust the inference following estimation of a standard least-squares regression or generalized linear model estimated by maximum likelihood. Although many publications just refer to "the" clustered standard errors, there is a surprisingly wide variety of clustered covariances, particularly due to different flavors of bias corrections. Furthermore, while the linear regression model is certainly the most important application case, the same strategies can be employed in more general models (e.g., for zero-inflated, censored, or limited responses). In R, functions for covariances in clustered or panel models have been somewhat scattered or available only for certain modeling functions, notably the (generalized) linear regression model. In contrast, an object-oriented approach to "robust"' covariance matrix estimation - applicable beyond lm() and glm() - is available in the

sandwich package but has been limited to the case of cross-section or time series data. Starting with sandwich 2.4.0, this shortcoming has been corrected: Based on methods for two generic functions (estfun() and bread()), clustered and panel covariances are provided in vcovCL(), vcovPL(), and vcovPC(). Moreover, clustered bootstrap covariances are provided in vcovBS(), using model update() on bootstrap samples. These are directly applicable to models from packages including MASS, pscl, countreg, and betareg, among many others. Some empirical illustrations are provided as well as an assessment of the methods' performance in a simulation study.

G. Zhang et al.: Dynamic factor analysis with multivariate time series of multiple individuals: An error-corrected estimation method. **Zhang-Lee-Li-etal-2025**

Guangjian Zhang et al. "Dynamic factor analysis with multivariate time series of multiple individuals: An error-corrected estimation method." In: *Psychological Methods* (Jan. 2025). ISSN: 1082-989X. DOI: 10.1037/met0000722.

Abstract: Intensive longitudinal data, increasingly common in social and behavioral sciences, often consist of multivariate time series from multiple individuals. Dynamic factor analysis, combining factor analysis and time series analysis, has been used to uncover individual-specific processes from single-individual time series. However, integrating these processes across individuals is challenging due to estimation errors in individual-specific parameter estimates. We propose a method that integrates individual-specific processes while accommodating the corresponding estimation error. This method is computationally efficient and robust against model specification errors and nonnormal data. We compare our method with a Naive approach that ignores estimation error using both empirical and simulated data. The two methods produced similar estimates for fixed effect parameters, but the proposed method produced more satisfactory estimates for random effects than the Naive method. The relative advantage of the proposed method was more substantial for short to moderately long time series (T = 56-200).

M. X. Zhang et al.: Effects of social and outcome expectancies on hazardous drinking among Chinese university students: The mediating role of drinking motivations

Zhang-Ku-Wu-etal-2020

Meng Xuan Zhang et al. "Effects of social and outcome expectancies on hazardous drinking among Chinese university students: The mediating role of drinking motivations". In: *Substance Use & Misuse* 55.1 (Jan. 2020), pp. 156–166. ISSN: 1532-2491. DOI: 10.1080/10826084.2019.1658784.

Abstract: Background and Objectives: Based on the theory of reasoned action, the present study investigated the relative effects of drinking outcome expectancies and parental norms, as well as the mediating effect of drinking motivations, on hazardous drinking in Chinese university students. Method: A sample of Chinese university students in Hong Kong and Macao ($N = 973$, $M = 19.82$, $SD = 1.57$, 48.9% males), who reported drinking in the past 3 months, voluntarily completed an anonymous questionnaire. Path analysis was used to test the effects of the variables on hazardous drinking. Results: All the psychosocial variables showed positive correlations with hazardous drinking. In the path model, controlling for sex, parental norms had both direct and indirect effects on hazardous drinking through social and enhancement motivations. Courage had the strongest indirect effect on drinking behavior through social, enhancement, and coping motivations, whereas the relationship between tension reduction and hazardous drinking was mediated by enhancement and coping motivations. Sociality and sexuality only had indirect effect through social and coping motivations respectively. Negative outcome expectancies had no direct nor indirect effects on hazardous drinking. Conclusions: Perceived approval from parents and positive alcohol outcome expectancies may enhance individuals' tendency to engage in hazardous drinking by increasing their motivation to drink to be social, for enjoyment, and to cope with problems. Parents should explicitly show their disapproval of their children's drinking, and education efforts should focus on decreasing positive outcome expectancies and associated motivations for drinking among Chinese university students.

X. Zhang et al.: Asymptotic distribution of robust effect size index

Zhang-Muscatello-Jones-etal-2026

Xinyu Zhang et al. "Asymptotic distribution of robust effect size index". Jan. 2026. DOI: 10.48550/arXiv.2601.19004. arXiv: 2601.19004 [stat.ME]. URL: https://arxiv.org/abs/2601.19004.

Abstract: The Robust Effect Size Index (RESI) is a recently proposed standardized effect size to quantify association strength across models. However, its confidence interval construction has relied on computationally intensive bootstrap procedures. We establish a general theorem for the asymptotic distribution of the RESI using a Taylor expansion that accommodates a broad class of models. Simulations under various linear and logistic regression settings show that RESI and its CI have smaller bias and more reliable coverage than commonly used effect sizes such as Cohen's d and f. Combining with robust covariance estimation yields valid inference under model misspecification. We use the methods to investigate associations of depression and behavioral problems with sex and diagnosis in Autism spectrum disorders, and demonstrate that the asymptotic approach achieves up to a 50-fold speedup over the bootstrap. Our work provides a scalable and reliable alternative to bootstrap inference, greatly enhancing the applicability of RESI to high-dimensional studies.

Zyphur et al.: From data to causes I: Building a general cross-lagged panel model (GCLM)

Zyphur-Allison-Tay-etal-2020

Michael J. Zyphur, Paul D. Allison, et al. "From data to causes I: Building a general cross-lagged panel model (GCLM)". In: *Organizational Research Methods* 23.4 (Oct. 2020), pp. 651–687. ISSN: 1552-7425. DOI: 10.1177/1094428119847278.

Abstract: This is the first paper in a series of two that synthesizes, compares, and extends methods for causal inference with longitudinal panel data in a structural equation modeling (SEM) framework. Starting with a cross-lagged approach, this paper builds a general cross-lagged panel model (GCLM) with parameters to account for stable factors while increasing the range of dy-

namic processes that can be modeled. We illustrate the GCLM by examining the relationship between national income and subjective well-being (SWB), showing how to examine hypotheses about short-run (via Granger-Sims tests) versus long-run effects (via impulse responses). When controlling for stable factors, we find no short-run or long-run effects among these variables, showing national SWB to be relatively stable, whereas income is less so. Our second paper addresses the differences between the GCLM and other methods. Online Supplementary Materials offer an Excel file automating GCLM input for Mplus (with an example also for Lavaan in R) and analyses using additional data sets and all program input/output. We also offer an introductory GCLM presentation at https://youtu.be/tHnnaRNPbXs. We conclude with a discussion of issues surrounding causal inference.

Zyphur et al.: From data to causes II: Comparing approaches to panel data analysis

Zyphur-Voelkle-Tay-etal-2020

Michael J. Zyphur, Manuel C. Voelkle, et al. "From data to causes II: Comparing approaches to panel data analysis". In: *Organizational Research Methods* 23.4 (Oct. 2020), pp. 688–716. ISSN: 1552-7425. DOI: 10.1177/1094428119847280.

Abstract: This article compares a general cross-lagged model (GCLM) to other panel data methods based on their coherence with a causal logic and pragmatic concerns regarding modeled dynamics and hypothesis testing. We examine three "static" models that do not incorporate temporal dynamics: random- and fixed-effects models that estimate contemporaneous relationships; and latent curve models. We then describe "dynamic" models that incorporate temporal dynamics in the form of lagged effects: cross-lagged models estimated in a structural equation model (SEM) or multilevel model (MLM) framework; Arellano-Bond dynamic panel data methods; and autoregressive latent trajectory models. We describe the implications of overlooking temporal dynamics in static models and show how even popular cross-lagged models fail to control for stable factors over time. We also show that Arellano-Bond and autoregressive latent trajectory models have various shortcomings. By contrasting these approaches, we clarify the benefits and drawbacks of common methods for

modeling panel data, including the GCLM approach we propose. We conclude with a discussion of issues regarding causal inference, including difficulties in separating different types of time-invariant and time-varying effects over time.